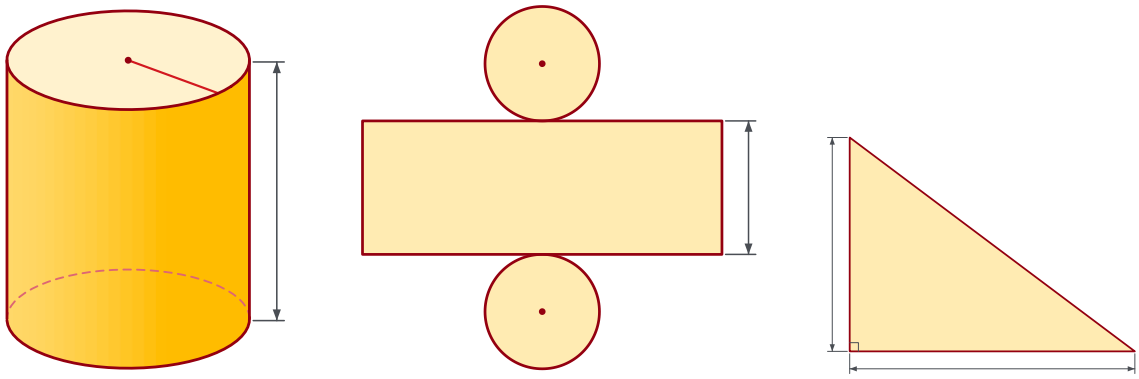
ACCESS TRADES

Trades Numeracy Textbook

Whole numbers to trigonometry, with key-by-key calculator steps for the TI-30Xa and the Sharp EL-531XT. For fabrication, pipe trades and millwright pre-apprentices.



ACCESS

Skills for Success: Numeracy
Metric and imperial

About this book

This textbook covers the mathematics a pre-apprentice uses every day in the shop: whole numbers, fractions and decimals, percent and ratio, measurement and unit conversion, formulas and how to rearrange them, perimeter, area, volume, right triangles and trigonometry.

Each chapter opens with what you will learn and the key words to watch for. The ideas are explained in plain language, then shown in worked examples. Short Try it sets let you practise straight away. Each chapter ends with Shape Canvas worksheets that use the same drawings and practice ideas as the ACCESS Shape Canvas app.

Two calculators are used throughout: the Texas Instruments TI-30Xa and the Sharp EL-531XT. Every calculator box shows the keys for both, side by side, and what each display reads after the last key.

Sizes are given in metric first and imperial second. Answers to every Try it set and worksheet are in the separate Answer Key. The companion Exercise Book has more practice for each chapter.

HOW TO READ A SIZE

In this book 1,250 mm means one thousand two hundred fifty millimetres. $5' \; 3\frac{1}{2}''$ means five feet three and one half inches. The symbol $\approx$ means “about equal to” and is used when an answer is rounded.

Contents

1 Whole Numbers 9

1.1 Whole numbers and place value 10
1.2 Rounding 11
1.3 Estimating 12
1.4 Adding 12
1.5 Subtracting 13
1.6 Multiplying 15
1.7 Dividing 16
1.8 Averages and multi-step problems 19

2 Your Calculator: TI-30Xa and Sharp EL-531XT 27

2.1 Meet the two calculators 28
2.2 Turning on, clearing and fixing mistakes 30
2.3 Second functions: the 2nd and 2ndF keys 31
2.4 Two ways to take keys 31
2.5 Negative numbers 32
2.6 Brackets, powers and π 33
2.7 The fraction key 34
2.8 Percent 35
2.9 Memory: STO and RCL 36
2.10 Angles: DEG and the trig keys 37
2.11 Checking a result 39

3 Order of Operations 43

3.1 Why order matters 44
3.2 BEDMAS 44
3.3 Brackets 45
3.4 Exponents, squares and square roots 47
3.5 Left to right 48
3.6 The fraction bar groups 49
3.7 Your calculator and BEDMAS 51

4 Fractions 55

4.1 What a fraction is 56
4.2 Reading an inch tape and rule 57
4.3 Equivalent fractions and lowest terms 59
4.4 Improper fractions and mixed numbers 61
4.5 The fraction key 62
4.6 Adding and subtracting fractions 64
4.7 Multiplying and dividing fractions 67

5 Decimals 75

5.1 Decimals and place value 76

5.2 Comparing and rounding decimals 77
5.3 Adding and subtracting decimals 78
5.4 Multiplying decimals 80
5.5 Dividing decimals 81
5.6 Fractions and decimals 82
5.7 Money 86

6 Percent, Ratio and Proportion 91
6.1 What percent means 92
6.2 Finding a percent of a number 93
6.3 Finding the percent and the whole 94
6.4 Waste allowance, increase and decrease 96
6.5 Ratio 98
6.6 Rate 100
6.7 Proportion 100
6.8 Scale drawings 102

7 Measurement and Unit Conversion 107
7.1 Units and the metric system 108
7.2 Imperial units, feet and inches 109
7.3 Converting with exact factors 111
7.4 Units of area and volume 113
7.5 Litres and gallons 116
7.6 Temperature 117

8 Formulas and Transposing 123
8.1 Reading a formula 124
8.2 Substituting values 124
8.3 Keeping an equation balanced 126
8.4 Transposing one-step formulas 127
8.5 Transposing two-step formulas 129
8.6 Formulas with squares and square roots 131
8.7 Checking by substituting back 134

9 Perimeter and Circumference 139
9.1 Perimeter of straight-sided shapes 140
9.2 The parts of a circle and π 141
9.3 Circumference 142
9.4 Stretch-out of a pipe or duct 144
9.5 Arcs 146
9.6 Composite perimeters 147

10 Area 153
10.1 Area and square units 154
10.2 Rectangles and squares 154
10.3 Triangles 156
10.4 Parallelograms and trapezoids 158

10.5 Circles and rings 159
10.6 Composite shapes 161
10.7 Surface area and nets 163

11 Volume and Capacity 171
11.1 Volume and cubic units 172
11.2 Prisms 172
11.3 Capacity: litres and US gallons 175
11.4 Cylinders: tanks and pipes 177
11.5 Cones and pyramids 179
11.6 Spheres 180
11.7 Composite solids 181

12 Angles and Right Triangles 187
12.1 Angles and degrees 188
12.2 Types of angles 189
12.3 Angles in a triangle 190
12.4 Right triangles and the Pythagorean theorem 191
12.5 The 3-4-5 check for square 194
12.6 Diagonals of rectangles 195

13 Trigonometry 201
13.1 Naming the sides 202
13.2 Sine, cosine and tangent 203
13.3 Finding a side 204
13.4 Finding an angle 207
13.5 Slopes and ramps 208
13.6 Pipe offsets 210
13.7 Bevel and mitre angles 213

Formula sheet 219

Conversion factors 220

Glossary 222

How to use this book

Worked examples

Examples are numbered by chapter. Read the question, try it yourself, then check each step.

Example 1.8 Counting bolts

A job needs 24 brackets. Each bracket is held by 4 bolts. How many bolts are needed?

SOLUTION

1. 24 groups of 4: 24 × 4.
2. 20 × 4 = 80 and 4 × 4 = 16.
   80 + 16 = 96.

Answer **96 bolts**

Calculator boxes

The left column shows the TI-30Xa keys, the right column the Sharp EL-531XT keys. Grey keys are numbers, white keys are operations and functions, and a gold key is the second-function key (2nd or 2ndF) pressed before the key after it. The display shows what that calculator reads after the last key.

CALCULATOR

TI-30Xa	Sharp EL-531XT
π × 152.4 =	π × 152.4 =
Display 478.7787204	Display 478.7787204

Rounded to two decimal places: **478.78**

Try it

Quick practice at the end of a section. Do these before you move on. Answers are in the Answer Key.

Boxes

TIP
A shortcut or a good habit.

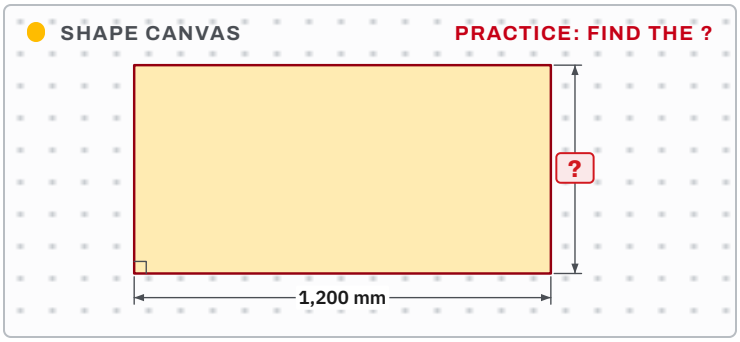
WATCH OUT
A common mistake and how to avoid it.

ON THE JOB
How the idea shows up in the shop.

KEY IDEA
The one thing to remember from a section.

Shape Canvas worksheets

The worksheets use the drawings and practice ideas of the ACCESS Shape Canvas app. A red ? marks a size that is hidden: find it. Dashed lines **break a part down** into simple shapes. A solid shown beside its flat pattern is **unfolded** into its net. Callouts point to the part a question is about. Show your work in the working box and write the answer on the answer line.



CHAPTER

1

Whole Numbers

Every job in the shop starts with whole numbers. You count bolts and brackets, read sizes off a drawing, add up a cut list and work out how many pieces you can get from one length of stock.

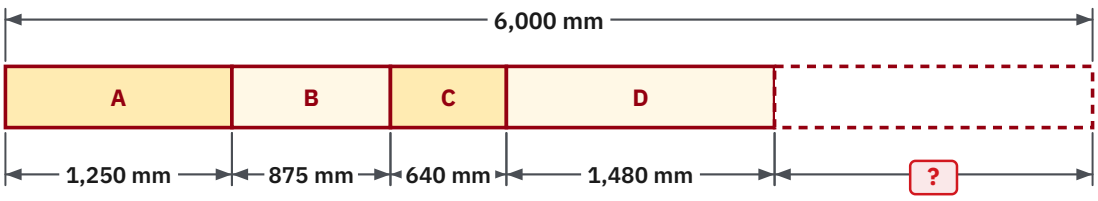
This chapter reviews place value, rounding and estimating, and the four operations: adding, subtracting, multiplying and dividing. Each method is shown by hand first and then on the TI-30Xa and the Sharp EL-531XT, so you can check your work two ways.

In this chapter you will

- Read and write whole numbers using place value
- Round whole numbers and use rounding to estimate
- Add and subtract lengths on a cut list
- Multiply and divide to find totals, pieces and spacing
- Work out averages and multi-step shop problems
- Check answers with an estimate and with your calculator

Key words

- whole number
- digit
- place value
- rounding
- estimate
- sum and difference
- factors and product
- quotient and remainder
- kerf
- average



1.1 Whole numbers and place value

A **whole number** is a number with no fraction or decimal part: 0, 1, 2, 3 and so on. Whole numbers are written with ten **digits**, 0 to 9.

The value of a digit depends on where it sits in the number. This is called **place value**. Each place is worth ten times the place to its right.

Number	Ten thousands	Thousands	Hundreds	Tens	Ones
6,096		6	0	9	6
12,450	1	2	4	5	0
305			3	0	5

Table 1.1 Place value of three numbers.

A zero holds a place. In 305 the zero shows there are no tens. Without it the number would read 35, which is a very different length.

Example 1.1 Reading a size off a drawing

A drawing gives the length of a beam as 12,450 mm. Write the number in words and say what the 2 is worth.

SOLUTION

1. Split the number at the comma: 12 thousand and 450.
2. Read each part: twelve thousand, four hundred fifty.
3. The 2 is in the thousands place, so it is worth 2 × 1,000 = 2,000.

Answer **Twelve thousand four hundred fifty millimetres. The 2 is worth 2,000.**

COMMAS AND SPACES

This book puts a comma between each group of three digits in numbers with four or more digits, for example 12,450. You may also see a small space used instead, as in 12 450. Both mean the same number. The conversion table at the back of this book groups long decimals with spaces, as in 0.453 592 37.

TRY IT 1.1

1. Write 3,048 in words.
2. Write in digits: seven thousand six hundred two.
3. What is the 5 worth in 25,410?
4. What is the 9 worth in 6,096?
5. Which is longer, 2,450 mm or 2,405 mm?
6. Write in digits: twenty thousand fifty.

Answers are in the Answer Key.

1.2 Rounding

Rounding replaces a number with a nearby number that is easier to work with. You round to a place: the nearest ten, the nearest hundred or the nearest thousand.

1. Find the digit in the place you are rounding to.
2. Look at the digit just to its right.
3. If that digit is 5 or more, add one to the rounding digit. If it is 4 or less, leave the rounding digit as it is.
4. Change every digit to the right of the rounding place to zero.

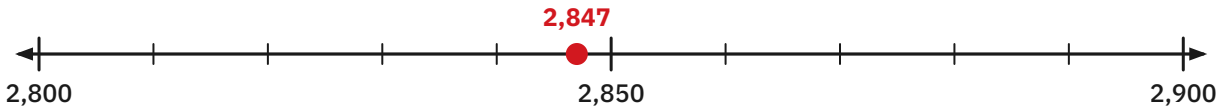


Figure 1.1 2,847 is closer to 2,800 than to 2,900, so it rounds to 2,800 to the nearest hundred.

Example 1.2 Rounding a length

Round 2,847 mm to the nearest ten and to the nearest hundred.

SOLUTION

1. Nearest ten: the tens digit is 4. The digit to its right is 7, which is 5 or more, so the 4 becomes 5: 2,850 mm.
2. Nearest hundred: the hundreds digit is 8. The digit to its right is 4, which is 4 or less, so the 8 stays: 2,800 mm.

Answer **2,850 mm to the nearest ten, 2,800 mm to the nearest hundred.**

Example 1.3 Rounding up past a nine

Round 6,960 to the nearest hundred.

SOLUTION

1. The hundreds digit is 9. The digit to its right is 6, so the 9 goes up by one.
2. Nine plus one is ten, so write 0 in the hundreds place and carry 1 into the thousands: 6 thousands become 7 thousands.

Answer **7,000**

ROUND AT THE END

Do the whole calculation with the exact numbers and round only the final answer. Rounding each step adds small errors together and the result can be a cut that is too short.

TRY IT 1.2

1. Round 1,374 to the nearest ten.
2. Round 1,375 to the nearest ten.
3. Round 6,096 to the nearest hundred.
4. Round 6,096 to the nearest thousand.
5. Round 49,950 to the nearest hundred.
6. Round 845 to the nearest hundred.

Answers are in the Answer Key.

1.3 Estimating

An **estimate** is a quick answer made with rounded numbers. It tells you roughly what the exact answer should be. If your calculator gives an answer far from your estimate, a key was probably missed or pressed twice.

Example 1.4 Estimating a cut list

A cut list has four pieces: 1,250 mm, 875 mm, 640 mm, 1,480 mm. Estimate the total length.

SOLUTION

1. Round each length to the nearest hundred: 1,300, 900, 600, 1,500.
2. Add the rounded numbers: 1,300 + 900 + 600 + 1,500 = 4,300.

Answer **About 4,300 mm. The exact total, found in Example 1.5, is 4,245 mm, which is close to the estimate.**

KEY IDEA

Estimate first, calculate second, compare third. An estimate does not replace the exact answer. It catches mistakes before they reach the saw.

1.4 Adding

Adding puts amounts together. The answer to an addition is the **sum**. To add by hand, line the numbers up by place value, ones under ones and tens under tens, and add each column from the right. When a column adds to 10 or more, write the ones digit and carry the tens digit to the next column.

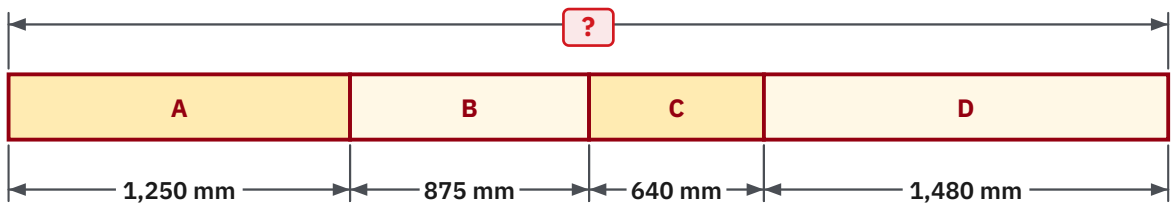


Figure 1.2 The cut list from Example 1.4. The total length is hidden.

Example 1.5 Total length of a cut list

Find the exact total length of the four pieces: 1,250 mm, 875 mm, 640 mm, 1,480 mm.

SOLUTION

1. Ones: 0 + 5 + 0 + 0 = 5.
2. Tens: 5 + 7 + 4 + 8 = 24. Write 4, carry 2.
3. Hundreds: 2 + 8 + 6 + 4 + 2 (carried) = 22. Write 2, carry 2.
4. Thousands: 1 + 1 + 2 (carried) = 4.

CALCULATOR

TI-30Xa

1250 + 875 + 640 + 1480 =

Display 4245.

Sharp EL-531XT

1250 + 875 + 640 + 1480 =

Display 4’245.

The TI-30Xa shows 4245. The Sharp EL-531XT shows 4’245, with a small mark between the thousands and the hundreds. The total is **4,245 mm**.

Answer **4,245 mm, close to the estimate of 4,300 mm.**

The TI-30Xa shows no separator between groups of digits. Read its display from the right in groups of three to put the commas back: 4245 is 4,245. The Sharp EL-531XT marks each group of three with a small raised mark, as in 4’245.

TRY IT 1.4

1. 2,450 + 1,375
2. 608 + 1,295 + 47
3. Pieces of 915, 915 and 1,220 mm. Total length?
4. 12,450 + 3,048 + 9,960

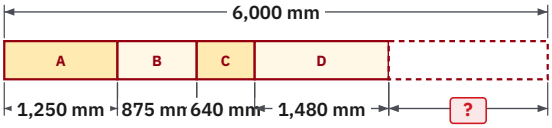
Answers are in the Answer Key.

1.5 Subtracting

Subtracting takes one amount away from another. The answer is the **difference**. Line the numbers up by place value with the larger number on top. When the top digit in a column is smaller than the bottom digit, borrow 1 from the next column to the left. That 1 is worth 10 in the column you are working on.

Example 1.6 What is left of the bar?

The four pieces from Example 1.5 (4,245 mm in total) are cut from a 6,000 mm bar. Ignore the width of the saw cuts. How long is the offcut?



SOLUTION

1. Write 6,000 on top and 4,245 below it.
2. Ones: 0 − 5 needs a borrow. The tens and hundreds are 0, so borrow through them: 6,000 becomes 5 thousands, 9 hundreds, 9 tens and 10 ones.
3. 10 − 5 = 5, 9 − 4 = 5, 9 − 2 = 7, 5 − 4 = 1.

CALCULATOR

TI-30Xa

6000 − 4245 =

Display 1755.

Sharp EL-531XT

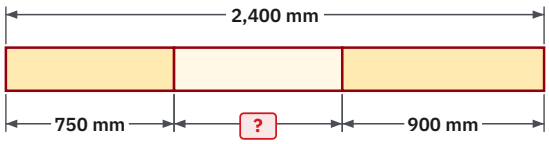
6000 − 4245 =

Display 1′755.

Answer **1,755 mm**

Example 1.7 Finding a hidden dimension

A drawing gives the overall length of a part as 2,400 mm and two of its three sections as 750 mm and 900 mm. Find the middle section.



SOLUTION

1. The three sections add up to the overall length.
2. Add the sections you know: 750 + 900 = 1,650.
3. Subtract from the overall length: 2,400 − 1,650 = 750.

CALCULATOR

TI-30Xa

2400 − 750 − 900 =

Display 750.

Sharp EL-531XT

2400 − 750 − 900 =

Display 750.

Answer **The middle section is 750 mm.**

CHECK A SUBTRACTION BY ADDING

Add the answer back to the number you took away. You should get the number you started with: 1,755 + 4,245 = 6,000.

TRY IT 1.5

1. 5,000 − 1,875
2. 12,450 − 9,960
3. A 3,600 mm bar has 2,475 mm cut off. How much is left?
4. Overall 1,800 mm. Sections of 600 mm and ? mm. Find the ?

Answers are in the Answer Key.

1.6 Multiplying

Multiplying is a fast way to add the same number many times. 24 × 4 means 24 groups of 4. The numbers you multiply are **factors**, and the answer is the **product**.

Example 1.8 Counting bolts

A job needs 24 brackets. Each bracket is held by 4 bolts. How many bolts are needed?

SOLUTION

1. 24 groups of 4: 24 × 4.
2. 20 × 4 = 80 and 4 × 4 = 16.
3. 80 + 16 = 96.

CALCULATOR

TI-30Xa

24 × 4 =

Display 96.

Sharp EL-531XT

24 × 4 =

Display 96.

Answer **96 bolts**

Example 1.9 Multiplying by a two-digit number

You need 36 pieces, each 125 mm long. What total length of stock do the pieces use? Ignore the saw cuts.

SOLUTION

1. Split 36 into 30 + 6.
2. 125 × 6 = 750.
3. 125 × 30 = 3,750.
4. Add the two parts: 750 + 3,750 = 4,500.

CALCULATOR

TI-30Xa

36 × 125 =

Display 4500.

Sharp EL-531XT

36 × 125 =

Display 4'500.

Answer **4,500 mm**

SAW CUTS TAKE MATERIAL

A saw blade turns a strip of metal into chips. The width of that strip is the **kerf**. When you cut many pieces, the kerf adds up. With a kerf of 3 mm, 12 cuts remove 12 × 3 = 36 mm of stock. The problems in this book give the kerf whenever it counts.

TRY IT 1.6

1. 15 × 8
2. 48 × 25
3. 125 × 64
4. 18 pieces at 450 mm. Total length?
5. 7 cuts with a 3 mm kerf. Stock lost?
6. 6 rows of 14 bolts. How many bolts?

Answers are in the Answer Key.

1.7 Dividing

Dividing splits an amount into equal groups or equal parts. In 6,000 ÷ 450, the number being divided (6,000) is the dividend, the number you divide by (450) is the divisor, and the answer is the **quotient**. When the division does not come out even, the amount left over is the **remainder**.

Example 1.10 Pieces from one bar

How many 450 mm pieces can you cut from a 6,000 mm bar? Ignore the saw cuts. How long is the offcut?

SOLUTION

1. Divide: 6,000 ÷ 450. The calculator shows 13.33333333.
2. Only whole pieces count, so you get 13 pieces.
3. The 13 pieces use 13 × 450 = 5,850 mm.
4. The offcut is the remainder: 6,000 − 5,850 = 150 mm.

CALCULATOR

TI-30Xa

Pieces

6000 ÷ 450 =

Display 13.33333333

Offcut

6000 − 13 × 450 =

Display 150.

Sharp EL-531XT

Pieces

6000 ÷ 450 =

Display 13.33333333

Offcut

6000 − 13 × 450 =

Display 150.

Answer **13 pieces, with a 150 mm offcut.**

WHOLE PIECES ONLY

The calculator shows 13.33333333 pieces. Only whole pieces count, so the answer is 13. Even 13.9 would still be 13 pieces, because the 14th piece does not fit. Think about what the answer means before you round.

Example 1.11 Long division by hand

Divide 4,500 by 36.

SOLUTION

1. 36 does not go into 4. It goes into 45 once: 1 × 36 = 36, and 45 − 36 = 9.
2. Bring down the 0 to make 90. 36 goes into 90 twice: 2 × 36 = 72, and 90 − 72 = 18.
3. Bring down the last 0 to make 180. 36 goes into 180 five times: 5 × 36 = 180, remainder 0.
4. Read the answer from the top: 125.

CALCULATOR

TI-30Xa

4500 ÷ 36 =

Display 125.

Sharp EL-531XT

4500 ÷ 36 =

Display 125.

Answer **125. This undoes Example 1.9: 36 × 125 = 4,500.**

Equal spacing is a division problem with one twist. When there is a hole at each end of a straight row, the number of spaces is one less than the number of holes.

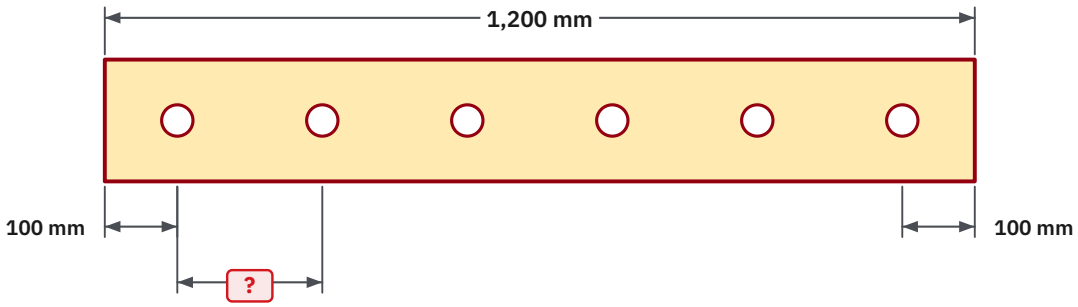


Figure 1.3 Six holes in a row, with the end holes 100 mm from the ends of a 1,200 mm plate.

Example 1.12 Equal hole spacing

Six holes are drilled in a straight row along a 1,200 mm plate. The first and last holes are 100 mm from the ends. The holes are evenly spaced. How far apart are the holes, centre to centre?

SOLUTION

1. Distance from the first hole to the last hole: 1,200 − 100 − 100 = 1,000 mm.
2. Six holes make 6 − 1 = 5 spaces.
3. Spacing: 1,000 ÷ 5 = 200 mm.

CALCULATOR

TI-30Xa

(1200 − 2 × 100) ÷ 5 =

Display 200.

Sharp EL-531XT

(1200 − 2 × 100) ÷ 5 =

Display 200.

Answer **200 mm centre to centre.**

KEY IDEA

In a straight row with a hole at each end, spaces = holes − 1. Draw a quick sketch and count the gaps if you are unsure.

TRY IT 1.7

1. 144 ÷ 12
2. 5,850 ÷ 450
3. 1,000 ÷ 8
4. How many 380 mm pieces from a 3,000 mm bar? Offcut?
5. 96 bolts shared equally among 24 brackets. Bolts each?
6. 6 holes evenly spaced, end holes 50 mm from the ends of a 650 mm bar. Spacing?

Answers are in the Answer Key.

1.8 Averages and multi-step problems

The **average** of a set of numbers is one number that represents the whole set. Add the numbers, then divide by how many numbers there are.

Example 1.13 Average welds per day

Over five days a welder completes 38, 42, 45, 40 and 35 joints. What is the average number of joints per day?

SOLUTION

1. Add: 38 + 42 + 45 + 40 + 35 = 200.
2. There are 5 days, so divide by 5: 200 ÷ 5 = 40.

CALCULATOR

TI-30Xa

(38 + 42 + 45 + 40 + 35) ÷ 5 =

Display 40.

Sharp EL-531XT

(38 + 42 + 45 + 40 + 35) ÷ 5 =

Display 40.

The brackets make the calculator add first and divide second. Chapter 3 explains why.

Answer **40 joints per day.**

Many shop problems take more than one step. Write down what you know, decide which operation each step needs, and check the final answer against an estimate.

Example 1.14 Ordering brackets

A job has 3 units. Each unit needs 14 brackets, and you want 6 spares. Brackets come in boxes of 10. How many boxes do you order?

SOLUTION

1. Brackets for the units: 3 × 14 = 42.
2. Add the spares: 42 + 6 = 48 brackets.
3. Boxes: 48 ÷ 10 = 4.8.
4. 4 boxes hold only 40 brackets, which is not enough. Round up to 5 boxes.

CALCULATOR

TI-30Xa

Brackets

3 × 14 + 6 =

Display 48.

Boxes

48 ÷ 10 =

Display 4.8

Sharp EL-531XT

Brackets

3 × 14 + 6 =

Display 48.

Boxes

48 ÷ 10 =

Display 4.8

Answer **5 boxes (50 brackets).**

ROUNDING UP OR ROUNDING DOWN

When you are counting pieces you can cut, round down: a part piece is scrap. When you are counting boxes, lengths or sheets you need to buy, round up: a part box still has to be bought.

WORKSHEET 1A

Cut lists on the canvas

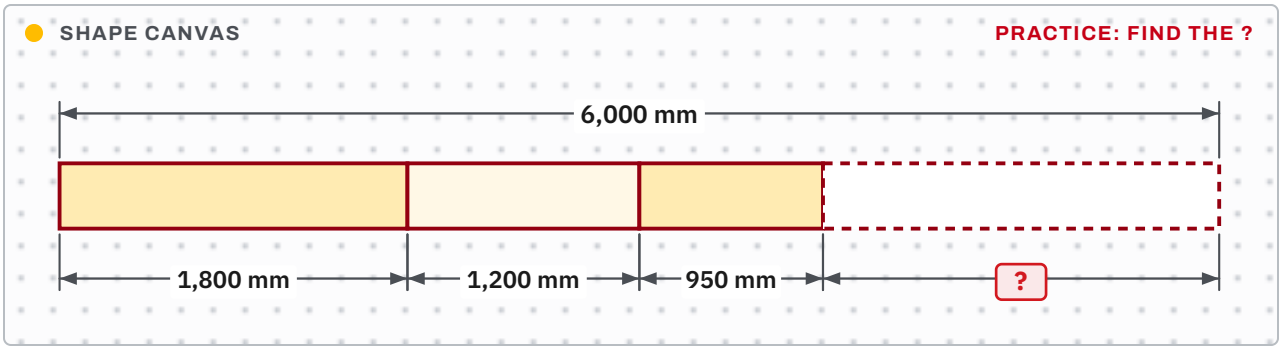
Each drawing is to scale. Find every red ?

Name _____

Date _____

Work each problem in the working box. Ignore saw cuts unless the problem gives a kerf. Estimate first, then check with your calculator.

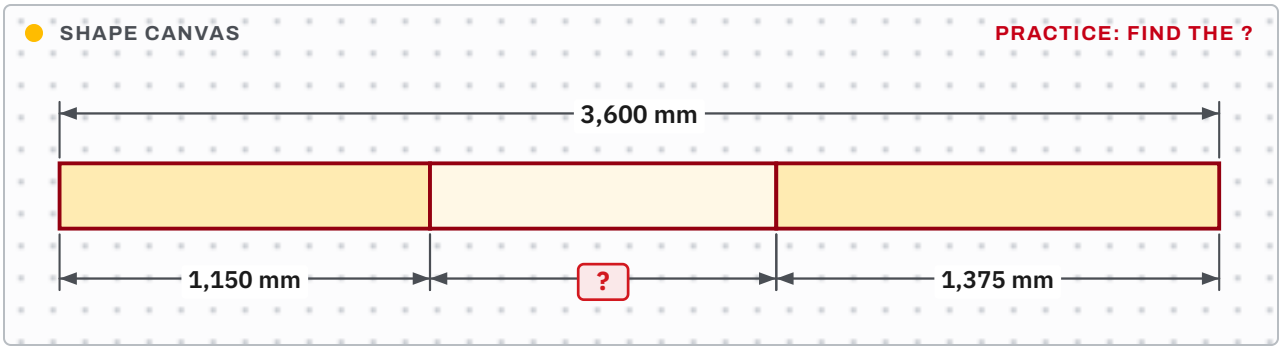
1 Three pieces are cut from a 6,000 mm bar. How long is the offcut?



WORKING

Answer _____

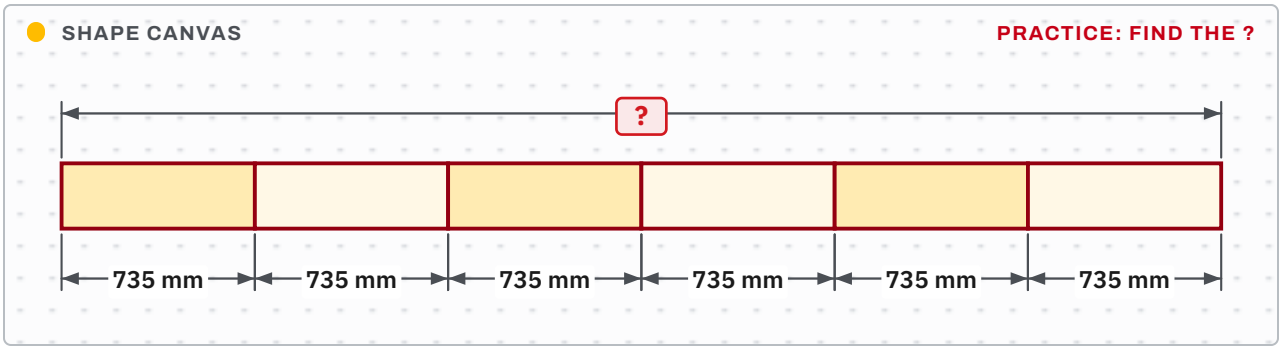
2 The overall length is 3,600 mm. Find the middle section.



WORKING

Answer _____

3 Six equal pieces of 735 mm. Find the total length.



WORKING

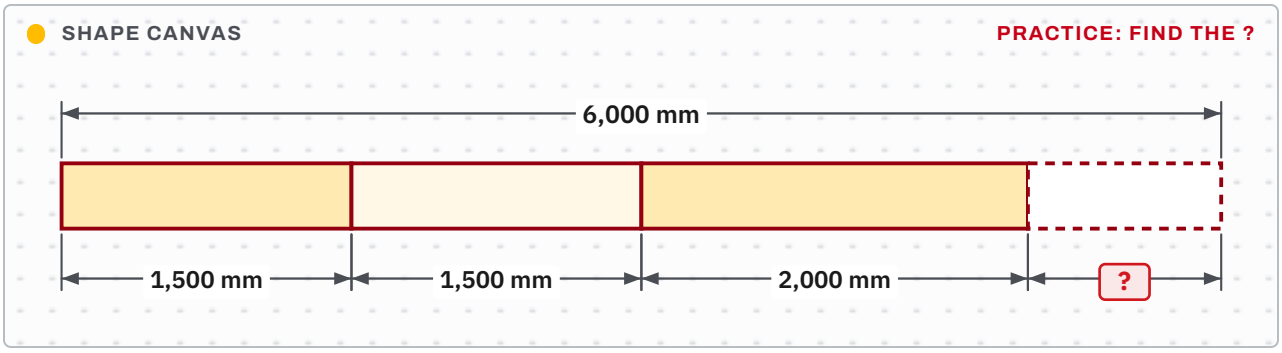
Answer _____

4 How many 380 mm pieces can be cut from a 6,000 mm bar? How long is the offcut?

WORKING

Answer _____

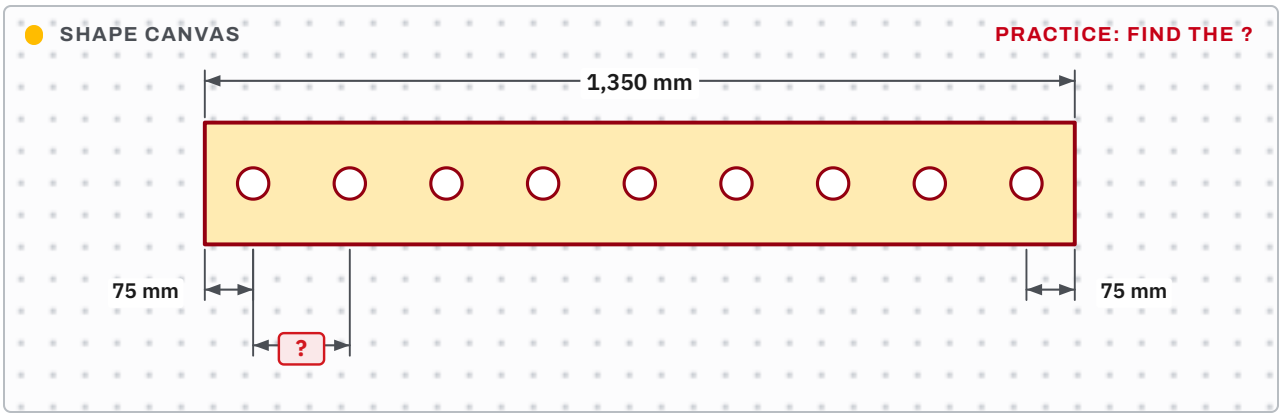
5 Two 1,500 mm pieces and one 2,000 mm piece are cut from a 6,000 mm bar. You make one cut for each piece and the kerf is 3 mm. How long is the offcut?



WORKING

Answer _____

6 Nine holes are evenly spaced along a 1,350 mm bar. The end holes are 75 mm from the ends. Find the spacing.



WORKING

Answer _____

WORKSHEET 1B

Break it down: plates and parts

Split each part into simple pieces to find the ?

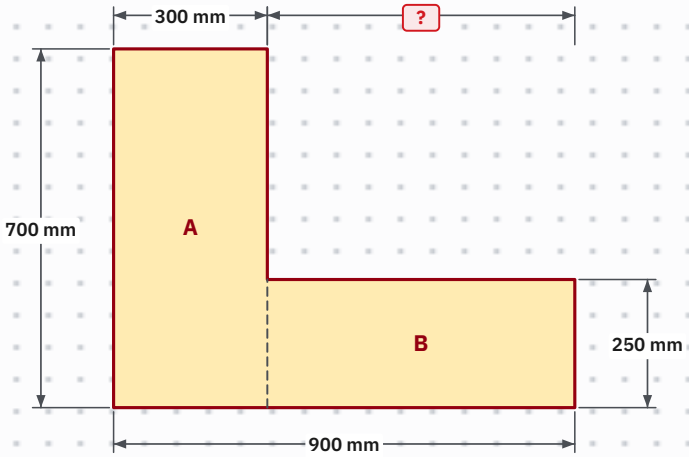
Name _____

Date _____

The dashed line splits the plate into a part A and a part B. Use the sizes you know to find the sizes you do not know.

1 Find the hidden width.

SHAPE CANVAS — PRACTICE: FIND THE ?

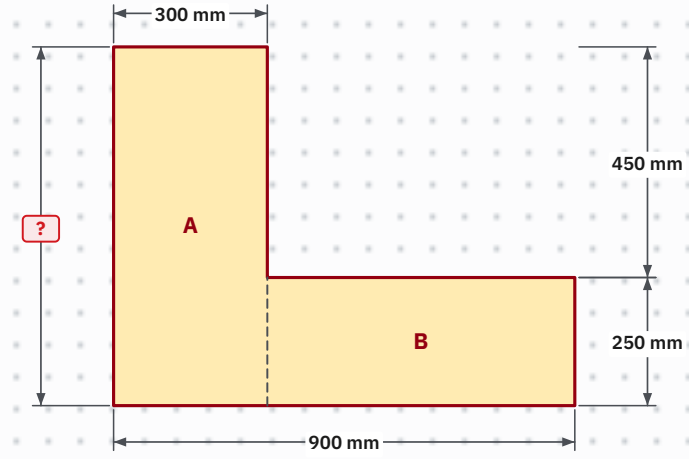


WORKING

Answer _____

2 Find the overall height.

SHAPE CANVAS — PRACTICE: FIND THE ?

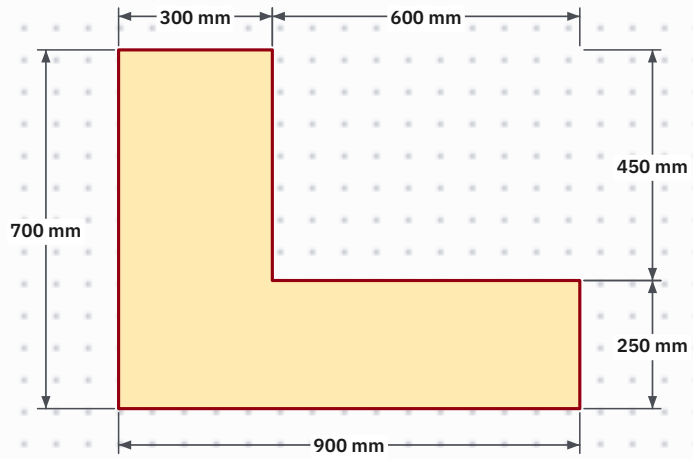


WORKING

Answer _____

3 Find the length of all six edges of the plate, then add them to find the distance around.

SHAPE CANVAS — READ THE DRAWING

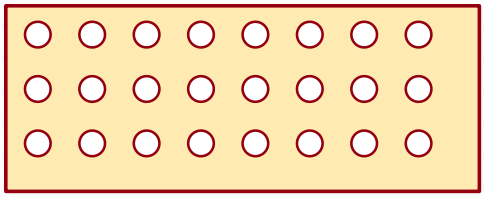


WORKING

Answer _____

4 A plate has holes in equal rows. Count them with one multiplication. Each hole takes 2 minutes to drill. How long does the plate take?

SHAPE CANVAS — READ THE DRAWING



WORKING

Answer _____

5 Estimate 4 × 1,180 by rounding 1,180 to the nearest hundred. Then find the exact answer.

WORKING

Answer _____

6 Four test plates measure 312, 298, 305 and 301 mm across. Find the average.

WORKING

Answer _____

CHAPTER

2

Your Calculator: TI-30Xa and Sharp EL-531XT

A scientific calculator is a shop tool like a tape measure or a square. This book shows every calculator step on two calculators, the TI-30Xa and the Sharp EL-531XT. Both give the same answers, but they do not always take the keys in the same order. This chapter walks you around both faceplates: which keys do what, how each calculator wants its keys pressed, and how to read the display and check a result.

In this chapter you will

- Find the main keys on both calculators
- Turn on, clear and fix a mistyped number
- Use the 2nd and 2ndF keys
- Key in negatives, brackets, squares, roots, powers and π
- Use the fraction, percent and memory keys
- Set DEG and use the trig keys, then check the answer

Key words

- key sequence
- display
- second function
- D.A.L.
- memory



TI-30Xa



Sharp EL-531XT

2.1 Meet the two calculators

Both calculators have the same kinds of keys: number keys, the four operations (+ − × ÷), an equals key, and function keys for squares, roots, powers, π, fractions, memory and angles. The keys are in different places, so take a few minutes with the drawings below and your own calculator.

The **display** is the screen. The TI-30Xa display shows a number of up to 10 digits, with small indicators such as DEG and 2nd. The Sharp EL-531XT shows two lines: the calculation you keyed in on the top line and the answer on the bottom line.

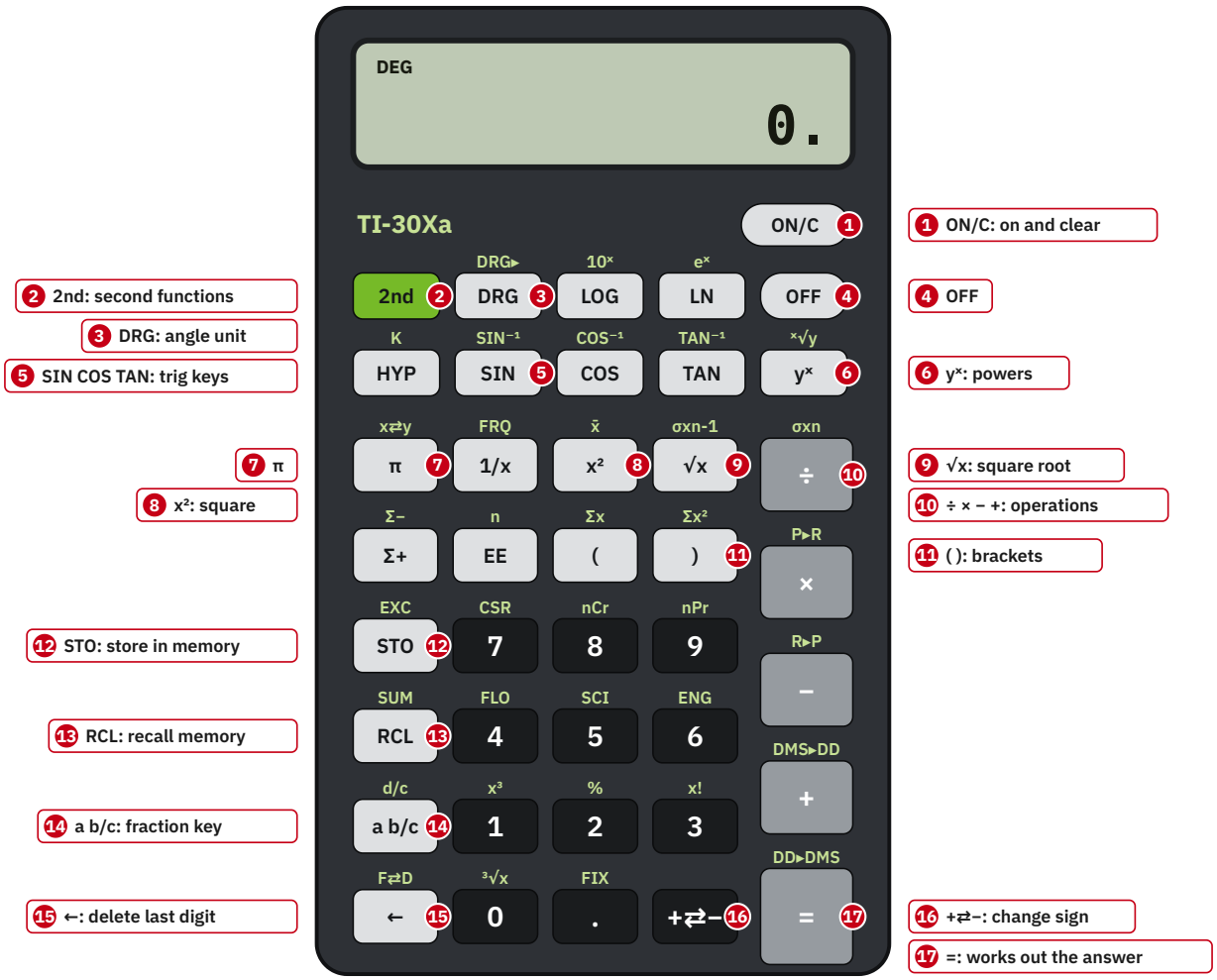


Figure 2.1 The TI-30Xa faceplate. The small labels above the keys are second functions, used with the 2nd key. The numbered notes name the keys used most in this book.

A **key sequence** is the list of keys you press, in order, to do a calculation. Every calculator box in this book shows the key sequence for both calculators and the display each one shows at the end. Table 2.1 lists the keys that look or work differently on the two calculators.

Job	TI-30Xa	Sharp EL-531XT
Turn on	ON/C	ON/C
Turn off	OFF	2ndF then ON/C (OFF is printed above it)
Second functions	2nd	2ndF
Change the sign of a number	+⇄− after the digits	+/− before the number
Square root	√x after the number	√ before the number, then =
Square	x² after the number	x² after the number, then =
Store in memory	STO then 1, 2 or 3	STO then a letter key (A to F, X, Y, M)
Fix a mistyped digit	←	DEL

Table 2.1 Keys that differ between the two calculators.

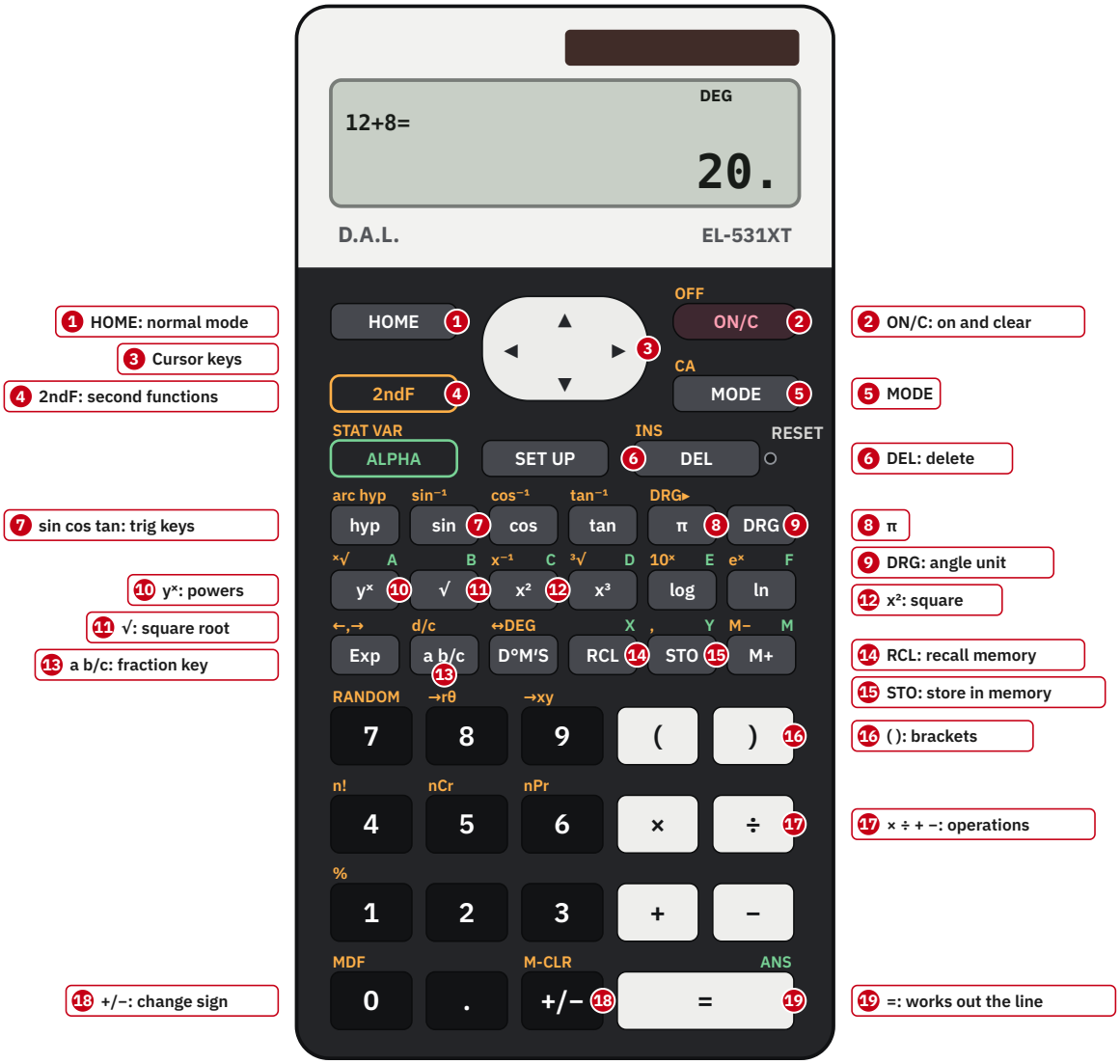


Figure 2.2 The Sharp EL-531XT faceplate. Orange labels above the keys are second functions, used with the 2ndF key. The letters at the upper right of some keys (green in this drawing) are memory names.

2.2 Turning on, clearing and fixing mistakes

Press ON/C to turn either calculator on. To turn the TI-30Xa off, press OFF. To turn the Sharp off, press 2ndF and then ON/C, because OFF is the second function printed in orange above the ON/C key. These steps are for the battery TI-30Xa, the model drawn in this book. The TI-30Xa Solar works differently: its guidebook says to expose the solar panel to light to turn it on and to cover the panel with the slide case to turn it off.

Start every calculation with a clear calculator. The calculator boxes in this book all start that way: press ON/C twice before the keys shown, and check that DEG is on the display.

Example 2.1 Fixing a wrong number

You meant to key 7 + 9 but pressed 8 instead of 9. Then you meant to key 125 × 4 but typed 1255. Fix each mistake without starting over.

SOLUTION

1. TI-30Xa: press ON/C once. One press clears only the number you were keying, so 7 + is still waiting. Key 9 and press =.
2. TI-30Xa: press ← to delete the last digit typed. 1255 becomes 125. Carry on with × 4 =.
3. Sharp: press DEL. When the cursor is at the right end of the line, DEL deletes the last character. Then key the right digit.

CALCULATOR

TI-30Xa

ON/C once

7 + 8 ON/C 9 =

Display 16.

Delete the last digit

1255 ← × 4 =

Display 500.

Sharp EL-531XT

ON/C once

7 + 9 =

Display 16.

Delete the last digit

125 × 4 =

Display 500.

The Sharp column shows each line keyed without the mistake. On the Sharp, use DEL to make the fix. ON/C on the Sharp clears the entry.

Answer **7 + 9 = 16 and 125 × 4 = 500.**

WHEN IN DOUBT, START AGAIN

On the TI-30Xa, pressing ON/C twice clears the display and every waiting operation. On the Sharp, ON/C clears the entry. If you are not sure what the calculator is holding, clear it and key the calculation again from the start.

2.3 Second functions: the 2nd and 2ndF keys

There are more jobs than keys, so most keys do two jobs. The job printed on the key is its first function. The job printed above the key is its **second function**. To use a second function, press 2nd on the TI-30Xa or 2ndF on the Sharp, then press the key.

On the TI-30Xa, 2nd shows on the display while it waits for the next key. On the Sharp, the second functions are printed in orange, and 2ndF shows on the display after you press 2ndF. Some functions are a second function on one calculator and a key of their own on the other. The cube, x^3, is an example.

Example 2.2 Cube and cube root

Find 5^3 (5 × 5 × 5) and the cube root of 27.

SOLUTION

1. On the TI-30Xa, x^3 and $\sqrt[3]{x}$ are second functions, so press 2nd before each one.
2. On the Sharp, x^3 has its own key. $\sqrt[3]{}$ is the second function printed above the x^3 key, so press 2ndF first.

CALCULATOR

TI-30Xa

5^3

[5] [2nd] [x^3]

Display: 125.

Cube root of 27

[27] [2nd] [$\sqrt[3]{x}$]

Display: 3.

Sharp EL-531XT

5^3

[5] [x^3] [=]

Display: 125.

Cube root of 27

[2ndF] [$\sqrt[3]{}$] [27] [=]

Display: 3.

Answer **5^3 = 125 and the cube root of 27 is 3.**

READ THE KEY BEFORE YOU PRESS IT

If the label you want is above the key, you need 2nd or 2ndF first.

2.4 Two ways to take keys

The two calculators work out the same answer but want some keys in a different order.

On the TI-30Xa, a function key such as x^2, $\sqrt{x}$ or SIN goes after the number, and it works at once. Key 144, press $\sqrt{x}$, and 12 shows straight away.

The Sharp has **D.A.L.** printed on its faceplate. For you, D.A.L. means this: press the keys in the same order the calculation is written. For √144 that is √ first, then 144, then =. The line is worked out when you press =. The top line of the display shows what you keyed, so you can read it back first.

Example 2.3 Square and square root

Find 12^2 and √144.

SOLUTION

1. 12^2 means 12 × 12. Both calculators take x^2 after the number. The Sharp needs = to finish.
2. √144 asks which number times itself makes 144. The TI-30Xa takes √x after the number. The Sharp takes √ first, as written.

CALCULATOR

TI-30Xa

12^2

[12] [x²]

Display 144.

√144

[144] [√x]

Display 12.

Sharp EL-531XT

12^2

[12] [x²] [=]

Display 144.

√144

[√] [144] [=]

Display 12.

Answer **12^2 = 144 and √144 = 12. Each answer checks the other.**

TRY IT 2.4

1. Write the TI-30Xa keys for √81.
2. Write the Sharp keys for √81.
3. Find 15^2.
4. Find √196.
5. Find 4^3.
6. Find the cube root of 64.

Answers are in the Answer Key.

2.5 Negative numbers

A negative number is less than zero, such as −5. Both calculators have a change-sign key that makes a number negative. It is not the minus key. The minus key subtracts one number from another.

On the TI-30Xa, key the digits first and then press +⇄−. On the Sharp, press +/− first and then the digits, the way the number is written.

Example 2.4 A calculation with a negative number

Find 42 × (−5) + 120.

SOLUTION

1. The −5 is a negative number, so it needs the change-sign key, not the minus key.
2. 42 × (−5) = −210, then −210 + 120 = −90. Both displays show the minus sign in front.

CALCULATOR

TI-30Xa

[42] [×] [5] [+⇄−] [+] [120] [=]

Display -90.

Sharp EL-531XT

[42] [×] [+/−] [5] [+] [120] [=]

Display -90.

Answer **−90**

CHANGE SIGN OR SUBTRACT?

Use the minus key between two numbers: 50 − 8. Use the change-sign key for a number that is negative on its own: 6 × (−4).

TRY IT 2.5

1. −8 + 15
2. 6 × (−4)
3. −20 − 12
4. (−36) ÷ 9

Answers are in the Answer Key.

2.6 Brackets, powers and π

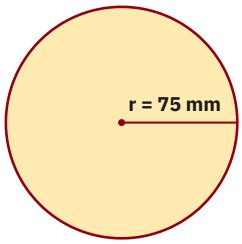
Brackets tell the calculator which part to work out first. Both calculators have (and) keys and take them in the order written: for (12 + 8) × 3, key (12 + 8) × 3 = and both show 60. Chapter 3 explains when you need them.

An **exponent** says how many of that number are multiplied together. $2^5 = 2 \times 2 \times 2 \times 2 \times 2$. Use the y^x key: the number, then y^x, then the exponent, then =. For 2^5, key 2 y^x 5 = on either calculator. Both show 32, and the Sharp shows 2^5= on its top line.

The π key enters the number π, about 3.14. Use it instead of typing 3.14, so the answer keeps all the digits.

Example 2.5 Using the π key

The distance around a circle is π × D, where D is the diameter (across the circle through its centre). The area is π × r², where r is the radius (half the diameter). A round duct has D = 150 mm and r = 75 mm. Find both, to two decimal places.



SOLUTION

1. Distance around: π × 150 = 471.2388..., which rounds to 471.24 mm.
2. Area: 75² = 5,625, then π × 5,625 = 17,671.458..., which rounds to 17,671.46 mm². Key 75 x² × π = on both calculators.

CALCULATOR

TI-30Xa

π × D

[π] [×] [150] [=]

Display 471.238898

π × r²

[75] [x²] [×] [π] [=]

Display 17671.45868

Sharp EL-531XT

π × D

[π] [×] [150] [=]

Display 471.238898

π × r²

[75] [x²] [×] [π] [=]

Display 17'671.45868

Answer **About 471.24 mm around and 17,671.46 mm² of area.**

TRY IT 2.6

1. (25 + 15) × 4
2. 3^4
3. 10^3 using y^x
4. π × 200, to two decimal places
5. π × 40², to two decimal places
6. (18 − 6) ÷ 4

Answers are in the Answer Key.

2.7 The fraction key

Both calculators have an a b/c key for fractions. To key a mixed number such as 3¼, key the whole number, a b/c, the top number, a b/c, the bottom number. The answer comes back as a fraction. Chapter 4 covers fractions in full.

Example 2.6 Fractions of an inch

Add 3¼ in and 2⅝ in. Then change ⅜ in to a decimal.

SOLUTION

1. Key 3 a b/c 1 a b/c 4 for 3¼ and 2 a b/c 5 a b/c 8 for 2⅝.
2. The TI-30Xa shows 5⌟7⌟8 and the Sharp shows 5 ⌜7 ⌜8. Both mean 5 and 7/8: the whole number, then the top number, then the bottom number.
3. To change a fraction answer to a decimal, press 2nd F⇄D on the TI-30Xa. On the Sharp, press a b/c after the answer. The second row changes ⅜ to a decimal.

CALCULATOR

TI-30Xa

3¼ + 2⅝

3 | a b/c | 1 | a b/c | 4 | + | 2 | a b/c | 5 | a b/c | 8 | =

Display: 5⌟7⌟8

⅜ as a decimal

3 | a b/c | 8 | 2nd | F⇄D

Display: 0.375

Sharp EL-531XT

3¼ + 2⅝

3 | a b/c | 1 | a b/c | 4 | + | 2 | a b/c | 5 | a b/c | 8 | =

Display: 5 ⌜7 ⌜8

⅜ as a decimal

3 | a b/c | 8 | = | a b/c

Display: 0.375

Answer $5\frac{7}{8}$ **in, and ⅜ in = 0.375 in.**

TRY IT 2.7

1. 1½ + 2¾ with the fraction key
2. Change ⅝ to a decimal.
3. Change 2⅜ to a decimal.
4. ¾ × 120

Answers are in the Answer Key.

2.8 Percent

The percent key (%) is a second function on both calculators: 2nd % on the TI-30Xa and 2ndF % on the Sharp, where % is printed above the 1 key. On the TI-30Xa you finish with =. On the Sharp, 2ndF % works out the line itself, so you do not press = after it.

Example 2.7 Percent of a number, and adding a percent

An order is for 250 bolts. Find 15% of 250. Then find 250 plus 15% more.

SOLUTION

1. 15% of 250: on the TI-30Xa key 250 × 15 2nd % =. On the Sharp key 250 × 15 2ndF %.
2. 250 plus 15%: on the TI-30Xa key 250 + 15 2nd % =. On the Sharp key 250 + 15 2ndF %. The calculator adds 15% of 250 to 250.

CALCULATOR

TI-30Xa

15% of 250

250 × 15 2nd % =

Display 37.5

250 + 15%

250 + 15 2nd % =

Display 287.5

Sharp EL-531XT

15% of 250

250 × 15 2ndF %

Display 37.5

250 + 15%

250 + 15 2ndF %

Display 287.5

Answer **15% of 250 is 37.5. Adding 15% gives 287.5, so order 288 bolts.**

PERCENT WITHOUT THE % KEY

15% means 15 out of 100. You can always key it as × 15 ÷ 100: 250 × 15 ÷ 100 = 37.5. Chapter 6 covers percent in full.

2.9 Memory: STO and RCL

A **memory** holds a number so you do not have to write it down and key it again. STO (store) puts a number in a memory. RCL (recall) brings it back.

The TI-30Xa has three memories, 1, 2 and 3. Press STO then 1 to store the display in memory 1. The Sharp names its memories with letters: A to F, X, Y and M. The letters are printed at the upper right above some keys. A is above y^x, so STO then y^x stores in A, and the box shows that key as A. On the Sharp, STO finishes the line you keyed and stores the answer.

Example 2.8 Storing a cut list total

Two pieces, 1,250 mm and 375 mm, are cut from a 6,000 mm bar. Ignore the saw cuts. Store the total of the pieces, clear the display, then find the offcut.

SOLUTION

1. 1,250 + 375 = 1,625. Store it in memory 1 (A on the Sharp).
2. Press ON/C. The memory keeps the number.
3. 6,000 − RCL 1 = 6,000 − 1,625 = 4,375.

CALCULATOR

TI-30Xa

1250 + 375 = STO 1 ON/C 6000 − RCL 1 =

Display 4375.

Sharp EL-531XT

1250 + 375 STO A ON/C 6000 − RCL A =

Display 4’375.

The TI-30Xa shows 4375. and the Sharp shows 4’375. The memory kept 1,625 after ON/C on both calculators.

Answer **4,375 mm**

OLD NUMBERS STAY IN MEMORY

Clearing with ON/C does not empty the memories, so a memory can still hold a number from an earlier job. Store the number you need before you recall it.

2.10 Angles: DEG and the trig keys

The trig keys SIN, COS and TAN work with angles. Chapter 13 explains what they mean. Angles can be measured in degrees, radians or grads, and the calculator must be set to the unit you are using. This book uses degrees.

Each press of DRG changes the angle unit in the order DEG, RAD, GRAD. The unit shows at the top of the display. Before any trig calculation, check that DEG is showing. If RAD or GRAD shows, press DRG until DEG comes back.



Figure 2.3 DEG at the top of each display means the calculator is set to degrees.

The trig keys follow the same rule as the other functions. On the TI-30Xa, key the angle and then SIN. On the Sharp, press sin, the angle and =. The inverse keys, SIN^{-1}, COS^{-1} and TAN^{-1}, are second functions. They take a number and give back an angle.

Example 2.9 Trig keys and their inverses

Find sin 30°. Find the angle whose tangent is 0.75, to two decimal places. Then find 600 × tan 35°, to two decimal places.

SOLUTION

1. sin 30° = 0.5.
2. The angle whose tangent is 0.75 uses TAN^{-1}: 2nd TAN^{-1} on the TI-30Xa, 2ndF $\tan^{-1}$ on the Sharp. The display shows 36.86989765, which rounds to 36.87°.
3. 600 × tan 35°: on the TI-30Xa, 35 TAN gives tan 35°, then × 600. On the Sharp, key tan 35 × 600 = as written. 420.1245229 rounds to 420.12.

CALCULATOR

TI-30Xa

sin 30°

[30] [SIN]

Display 0.5

$\tan^{-1}$ 0.75

[0.75] [2nd] [TAN^{-1}]

Display 36.86989765

600 × tan 35°

[35] [TAN] [×] [600] [=]

Display 420.1245229

Sharp EL-531XT

sin 30°

[sin] [30] [=]

Display 0.5

$\tan^{-1}$ 0.75

[2ndF] [$\tan^{-1}$] [0.75] [=]

Display 36.86989765

600 × tan 35°

[tan] [35] [×] [600] [=]

Display 420.1245229

Answer **sin 30° = 0.5, the angle is 36.87°, and 600 × tan 35° = 420.12.**

TRY IT 2.10

1. cos 60°
2. sin 45°, to four decimal places
3. tan 20°, to four decimal places
4. The angle whose sine is 0.5
5. The angle whose cosine is 0.8, to two decimal places
6. 250 × cos 40°, to two decimal places

Answers are in the Answer Key.

2.11 Checking a result

A calculator does exactly what you key, including your mistakes. Before you cut or order, check the answer in at least one of these ways.

1. Estimate first. Round the numbers and work it out in your head, as in Chapter 1. The calculator answer should be close.
2. Work it backwards. If $\sqrt{144} = 12$, then 12^2 should give 144. If 6,000 − 1,625 = 4,375, then 4,375 + 1,625 should give 6,000.
3. Read the Sharp top line. It shows the calculation you keyed, so you can spot a wrong number before you press =.
4. Key it again. If the second answer differs from the first, key it a third time slowly.

An answer with more than 10 digits does not fit on the display. The calculator then shows it with a small number at the right, called an exponent of 10. For 123,456,789 × 1,000, both calculators show 1.23456789 with a small 11, which means $1.23456789 \times 10^{11}$, or 123,456,789,000.

> **WRITE THE ANSWER WITH ITS UNIT**
>
> The display shows a number, not a size. Write the unit next to every answer you copy down: 4,375 mm, not 4375. A number without a unit is a guess on the shop floor.

WORKSHEET 2A

Key it in

Write the keys for both calculators, then the answer

Name _____

Date _____

For each problem, write the key sequence for the TI-30Xa and for the Sharp EL-531XT. Then work it out on your own calculator. Start each one with ON/C ON/C and DEG on the display.

1 2,450 + 1,375 − 980

WORKING

Answer _____

2 18^2

WORKING

Answer _____

3 √529

WORKING

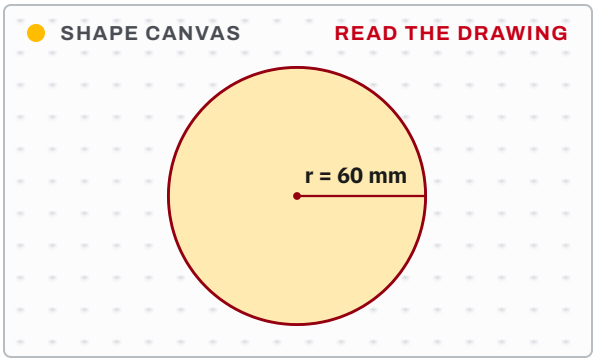
Answer _____

4 (−12) × 4 + 60

WORKING

Answer _____

5 The circle has the radius shown. Find D = 2 × r, then the distance around, π × D, to two decimal places.



WORKING

Answer _____

6 Add 2¾ in and 1⅝ in with the fraction key. Write the answer as each display shows it, then in the usual way.

WORKING

Answer _____

7 Find 20% of 340 kg.

WORKING

Answer _____

8 Find 800 × sin 25°, to two decimal places. Check that DEG is on the display first.

WORKING

Answer _____

WORKSHEET 2B

Know your keys, read the display

Name the lettered keys and read each display

Name _____

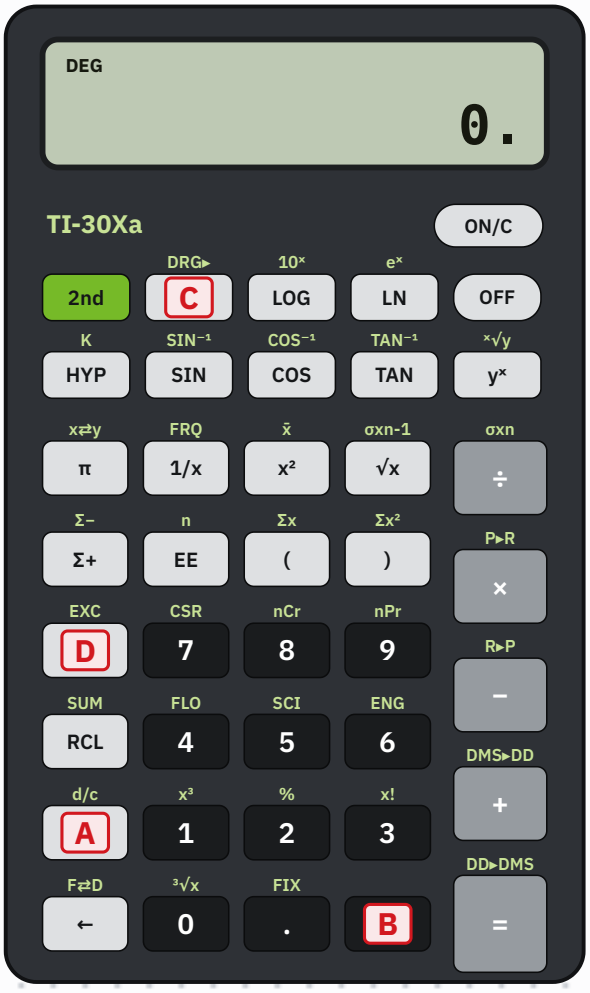
Date _____

The red letters mark keys on each faceplate. Name each key and say what it does. For the displays, write the number in the usual way.

1 Name keys A to H and say what each one does. A to D are on the TI-30Xa, E to H on the Sharp EL-531XT.

SHAPE CANVAS

NAME THE KEYS

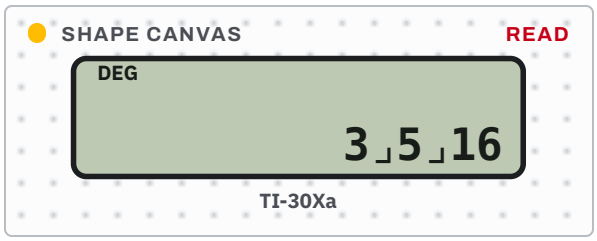




WORKING

Answer _____

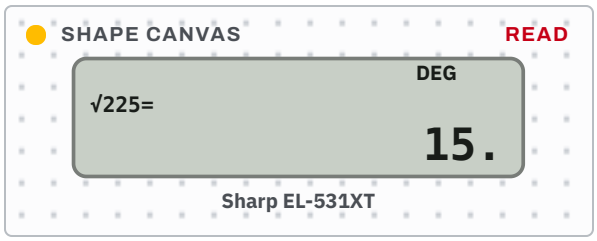
2 Write the number this display shows.



WORKING

Answer _____

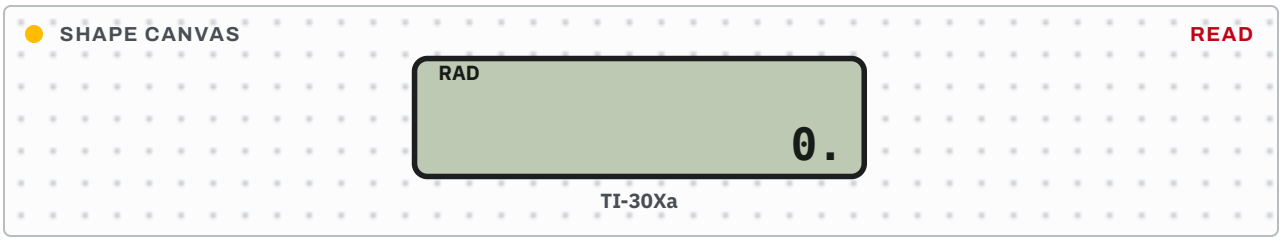
3 What calculation was keyed? What is the answer?



WORKING

Answer _____

4 This display shows RAD. What do you press before you use SIN?



WORKING

Answer _____

CHAPTER

3

Order of Operations

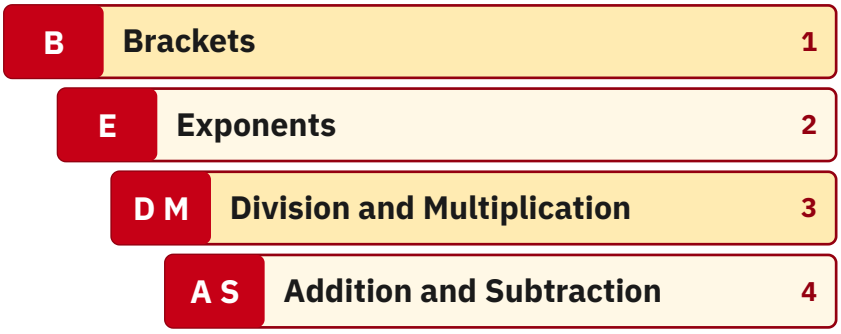
What is 3 + 4 × 5? Add first and you get 35. Multiply first and you get 23. Only one answer can be right, so everyone who works with numbers follows the same order, and so does your calculator. This chapter sets out that order, BEDMAS, and shows when the calculator needs brackets that are not written on the page.

In this chapter you will

- Use BEDMAS to work a calculation step by step
- Work out brackets, powers and roots in the right order
- Work left to right for × and ÷, and for + and −
- Treat a fraction bar as a grouping symbol
- Key brackets so the calculator follows the same order

Key words

- order of operations
- BEDMAS
- brackets
- exponent
- square
- square root



3.1 Why order matters

A calculation with more than one operation can be read in more than one way. The **order of operations** is the agreed order for working one out. When everyone uses the same order, a calculation written on a drawing or in a formula means the same thing to everyone who reads it.

Example 3.1 Bolts in a crate

A crate holds 3 loose bolts and 4 bags of 5 bolts. The count is written 3 + 4 × 5. How many bolts are in the crate?

SOLUTION

1. The 4 bags of 5 bolts are 4 × 5 = 20 bolts. That multiplication has to happen first.
2. Add the loose bolts: 3 + 20 = 23.
3. Adding first, 3 + 4 = 7 and 7 × 5 = 35, counts the 3 loose bolts as if they were in bags. That is wrong.

CALCULATOR

TI-30Xa

3 + 4 × 5 =

Display 23.

Sharp EL-531XT

3 + 4 × 5 =

Display 23.

Both calculators multiply before they add, so both show 23 even though the keys are pressed left to right.

Answer 23 bolts

3.2 BEDMAS

BEDMAS is a memory aid for the order of operations. Each letter stands for one kind of operation, from the first to be done to the last. The ladder on the first page of this chapter shows the four steps: start at the top step and work down.

Letter	Stands for	Example
B	Brackets ( ) and [ ]	(12 + 8) × 3: add 12 + 8 first
E	Exponents: squares, powers and roots	100 − 5^2: work out 5^2 first
D and M	Division and Multiplication, left to right	24 ÷ 4 × 3: divide first, because ÷ comes first
A and S	Addition and Subtraction, left to right	20 − 8 + 2: subtract first, because − comes first

Table 3.1 What each letter of BEDMAS means.

KEY IDEA

D and M are one step, not two. Neither one goes first. Work them in the order they appear, left to right. The same goes for A and S.

Example 3.2 Working through BEDMAS

Find 8 − 3 × 2 + 1.

SOLUTION

1. No brackets and no exponents.
2. Multiply: 3 × 2 = 6. The calculation is now 8 − 6 + 1.
3. Add and subtract left to right: 8 − 6 = 2, then 2 + 1 = 3.

CALCULATOR

TI-30Xa

8 − 3 × 2 + 1 =

Display 3.

Sharp EL-531XT

8 − 3 × 2 + 1 =

Display 3.

Answer 3

REWRITE THE LINE AFTER EACH STEP

Write the whole calculation again after each step, with the part you just worked out replaced by its answer. Each line gets shorter until one number is left.

TRY IT 3.2

1. 6 + 4 × 3
2. 20 − 12 ÷ 4
3. 5 × 6 − 4 × 3
4. 18 ÷ 3 + 2 × 7

Answers are in the Answer Key.

3.3 Brackets

Brackets group part of a calculation so it is worked out first. They come first in BEDMAS. Inside the brackets, use BEDMAS again.

Sometimes there are brackets inside brackets. Square brackets [ ] are often used for the outer pair so they are easy to tell apart. Work out the innermost brackets first, then work outward. On the calculator, every bracket is keyed with the (and) keys.

Example 3.3 Brackets change the answer

Find 12 + 8 × 3 and (12 + 8) × 3.

SOLUTION

1. 12 + 8 × 3: multiply first, 8 × 3 = 24, then 12 + 24 = 36.
2. (12 + 8) × 3: brackets first, 12 + 8 = 20, then 20 × 3 = 60.

CALCULATOR

TI-30Xa

No brackets

12 + 8 × 3 =

Display 36.

With brackets

(12 + 8) × 3 =

Display 60.

Sharp EL-531XT

No brackets

12 + 8 × 3 =

Display 36.

With brackets

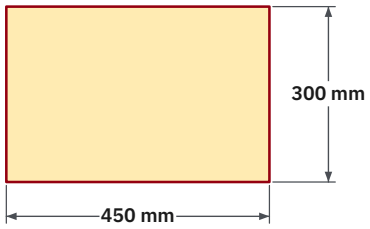
(12 + 8) × 3 =

Display 60.

Answer **36 and 60. The same numbers give different answers.**

Example 3.4 A frame and brackets inside brackets

A rectangular frame is 450 mm by 300 mm. The distance around it is 2 × (450 + 300). Find it. Then find 2 × [15 − (4 + 6)].



SOLUTION

1. Frame: brackets first, 450 + 300 = 750. Then 2 × 750 = 1,500 mm.
2. Inner brackets first: 4 + 6 = 10. The line is now 2 × [15 − 10].
3. Outer brackets: 15 − 10 = 5. Then 2 × 5 = 10.

On the calculator, square brackets are keyed with the same (and) keys: 2 × (15 − (4 + 6)) =. Close every bracket you open.

Answer **The frame is 1,500 mm around, and 2 × [15 − (4 + 6)] = 10.**

KEY THE × BEFORE A BRACKET

On paper, 3(4 + 5) means 3 × (4 + 5). The × is often left out next to a bracket. On the calculator, always key the × yourself: 3 × (4 + 5) = 27.

TRY IT 3.3

1. $(6 + 4) \times (3 - 2)$
2. $(25 - 5) \div 4$
3. $3 \times (7 + 8)$
4. $2 \times (1{,}200 + 800)$

Answers are in the Answer Key.

3.4 Exponents, squares and square roots

An **exponent** is the small raised number that says how many of that number are multiplied together. In 5^3 the exponent is 3: $5^3 = 5 \times 5 \times 5 = 125$.

The **square** of a number is the number multiplied by itself. The square of 5 is written 5^2 and read "5 squared": $5 \times 5 = 25$. The **square root** undoes a square. $\sqrt{25} = 5$ because $5 \times 5 = 25$.

Exponents and roots come right after brackets in BEDMAS. An exponent belongs only to the number or bracket it touches.

Example 3.5 What does the square belong to?

Find $100 - 5^2$, $(100 - 5)^2$, 5×3^2 and $(5 \times 3)^2$.

SOLUTION

1. $100 - 5^2$: the square belongs to the 5. $5^2 = 25$, then $100 - 25 = 75$.
2. $(100 - 5)^2$: brackets first, 95, then $95^2 = 9{,}025$.
3. 5×3^2: the square belongs to the 3. $3^2 = 9$, then $5 \times 9 = 45$.
4. $(5 \times 3)^2$: brackets first, 15, then $15^2 = 225$.

CALCULATOR

TI-30Xa

$100 - 5^2$

[100] [−] [5] [x²] [=]

Display: 75.

$(100 - 5)^2$

[(] [100] [−] [5] [)] [x²]

Display: 9025.

Sharp EL-531XT

$100 - 5^2$

[100] [−] [5] [x²] [=]

Display: 75.

$(100 - 5)^2$

[(] [100] [−] [5] [)] [x²] [=]

Display: 9'025.

Answer **75, 9,025, 45 and 225.**

A square root sign works like a bracket. The bar over the top covers everything under it, and all of that is worked out before the root is taken.

Example 3.6 A root over a sum

Find √(9 + 16) and √9 + √16.

SOLUTION

1. √(9 + 16): the root covers the sum. 9 + 16 = 25, then √25 = 5.
2. √9 + √16: two separate roots. √9 = 3 and √16 = 4, then 3 + 4 = 7.
3. On the Sharp, key √ and then the sum in brackets. On the TI-30Xa, key the sum in brackets, then √x after the closing bracket.

CALCULATOR √(9 + 16)

TI-30Xa

(9 + 16) √x

Display 5.

Sharp EL-531XT

√ (9 + 16) =

Display 5.

Answer **√(9 + 16) = 5 but √9 + √16 = 7. They are not the same.**

TRY IT 3.4

1. 4^2
2. 2×3^2
3. $(2 \times 3)^2$
4. $10 + 2^3$

Answers are in the Answer Key.

3.5 Left to right

Division and multiplication share one step of BEDMAS. Addition and subtraction share the last step. When a line has two operations from the same step, work them in the order you meet them, from left to right.

MULTIPLICATION DOES NOT ALWAYS COME FIRST

The M comes after the D in BEDMAS, but that does not mean divide before multiply. They share a step. 24 ÷ 4 × 3 = 18, not 24 ÷ 12 = 2. In the same way, 50 − 20 + 5 = 35, not 50 − 25 = 25.

A bar is 6,000 mm long. You cut off 1,250 mm, then 875 mm, and then weld a 400 mm piece back on. Ignoring saw cuts, its length is 6,000 − 1,250 − 875 + 400. Left to right, that is 4,750, then 3,875, then 4,275 mm.

Example 3.7 Left to right

Find 48 ÷ 6 ÷ 2, 24 ÷ 4 × 3 and 20 − 8 − 2.

SOLUTION

1. 48 ÷ 6 ÷ 2: left to right, 48 ÷ 6 = 8, then 8 ÷ 2 = 4. Working 6 ÷ 2 first would give 48 ÷ 3 = 16, which is wrong.
2. 24 ÷ 4 × 3: ÷ comes first, 24 ÷ 4 = 6, then 6 × 3 = 18.
3. 20 − 8 − 2: 20 − 8 = 12, then 12 − 2 = 10. Working 8 − 2 first would give 14, which is wrong.

CALCULATOR

TI-30Xa

48 ÷ 6 ÷ 2

[48] [÷] [6] [÷] [2] [=]

Display 4.

24 ÷ 4 × 3

[24] [÷] [4] [×] [3] [=]

Display 18.

Sharp EL-531XT

48 ÷ 6 ÷ 2

[48] [÷] [6] [÷] [2] [=]

Display 4.

24 ÷ 4 × 3

[24] [÷] [4] [×] [3] [=]

Display 18.

Answer 4, 18 and 10.

TRY IT 3.5

1. 48 ÷ 4 × 2
2. 90 ÷ 3 ÷ 5
3. 40 − 16 ÷ 4 × 2
4. 30 − 10 + 5

Answers are in the Answer Key.

3.6 The fraction bar groups

A fraction bar means divide. It also groups: everything above the bar is worked out as one number, everything below the bar is worked out as one number, and then the top is divided by the bottom. It acts like a pair of brackets around the top and another pair around the bottom.

On a calculator there is no fraction bar for a calculation like this, so you must key the brackets yourself: (top) ÷ (bottom).

Example 3.8 Brackets for a fraction bar

Find $\frac{18 + 6}{15 - 8}$ to two decimal places.

SOLUTION

1. Top: 18 + 6 = 24. Bottom: 15 − 8 = 7.
2. Divide: 24 ÷ 7 = 3.428571429..., which rounds to 3.43.
3. Keyed without brackets, 18 + 6 ÷ 15 − 8 = gives 10.4, because the calculator divides 6 ÷ 15 first. That answer is wrong.

CALCULATOR

TI-30Xa

With brackets

(18 + 6) ÷ (15 − 8) =

Display 3.428571429

Without brackets (wrong)

18 + 6 ÷ 15 − 8 =

Display 10.4

Sharp EL-531XT

With brackets

(18 + 6) ÷ (15 − 8) =

Display 3.428571429

Without brackets (wrong)

18 + 6 ÷ 15 − 8 =

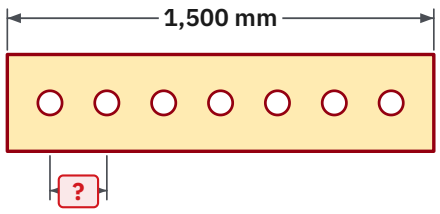
Display 10.4

This is the same calculation as an example in the Sharp manual, keyed with brackets.

Answer **3.43**

Example 3.9 Hole spacing

Seven holes are evenly spaced along a 1,500 mm plate, with the end holes 150 mm from each end. Seven holes make 6 spaces. The spacing is $\frac{1{,}500 - 2 \times 150}{6}$. Find it.



SOLUTION

1. Top first, using BEDMAS inside it: 2 × 150 = 300, then 1,500 − 300 = 1,200.
2. Divide by the bottom: 1,200 ÷ 6 = 200.
3. On either calculator, key (1500 − 2 × 150) ÷ 6 =.

Answer **200 mm centre to centre.**

TRY IT 3.6

1. $\frac{45 + 27}{14 - 6}$
2. $\frac{312 + 305 + 298 + 309}{4}$
3. $\frac{100}{2 + 3}$
4. $\frac{2{,}400 - 2 \times 200}{5}$

Answers are in the Answer Key.

3.7 Your calculator and BEDMAS

Both calculators follow BEDMAS. Key a calculation from left to right, as it is written, and the calculator holds back an addition until any multiplication after it is done. That is why 3 + 4 × 5 = gives 23 on both. The TI-30Xa guidebook and the Sharp manual both list this order, with anything in brackets worked out first.

The calculator only knows what you key. It cannot see a fraction bar, a root bar over a sum, or a square that belongs to a whole bracket. Before you key a formula from a drawing or a manual, find every grouping on paper, then key each one as brackets. On the Sharp, read the top line of the display before you press =.

Written	Key it as	Why
$\frac{18 + 6}{15 - 8}$	(18 + 6) ÷ (15 − 8) =	A fraction bar groups the top and the bottom
√(9 + 16)	TI-30Xa: (9 + 16) √x Sharp: √ (9 + 16) =	The root bar covers the whole sum
$(100 - 5)^2$	TI-30Xa: (100 − 5) x^2 Sharp: (100 − 5) x^2 =	The square belongs to the bracket
3(4 + 5)	3 × (4 + 5) =	Key the × that is not written

Table 3.2 When the calculator needs brackets.

WORKSHEET 3A

Order on the canvas

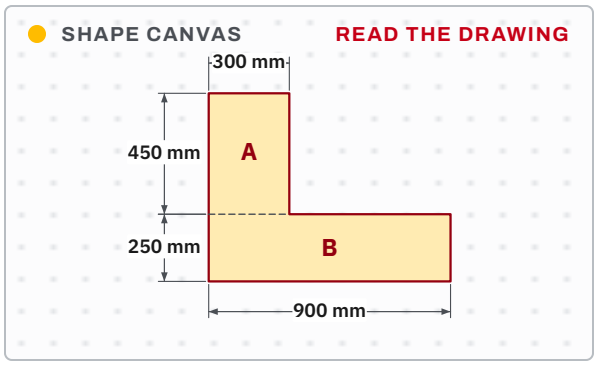
Write the calculation, then work it out with BEDMAS

Name _____

Date _____

Each drawing comes with a calculation. Work it out step by step on paper, rewriting the line after each step, then check it on your calculator.

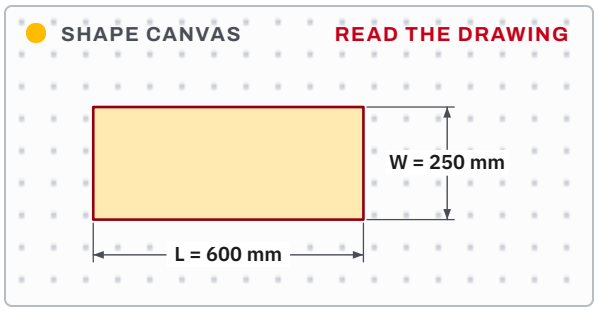
1 The plate is split into A and B. Its area in mm² is 300 × 450 + 900 × 250. Work it out.



WORKING

Answer _____

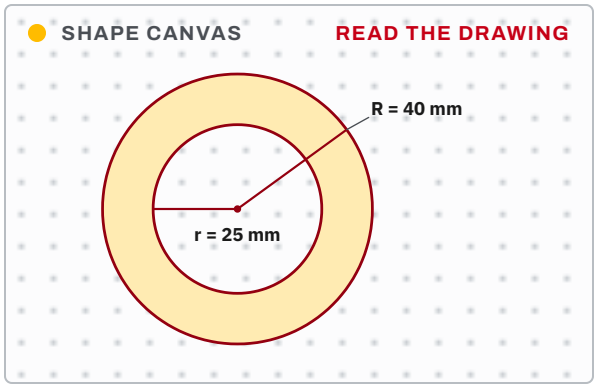
2 The distance around the frame is 2 × (L + W). Find it. If L doubles, what is the new distance around?



WORKING

Answer _____

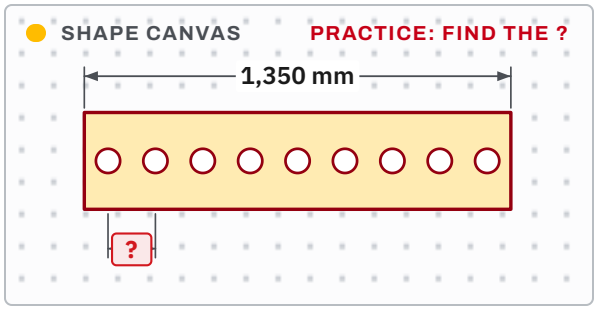
3 The ring has R = 40 mm and r = 25 mm. Its area is π × (R² − r²). Find it to two decimal places.



WORKING

Answer _____

4 Nine holes are evenly spaced on a 1,350 mm bar, with the end holes 75 mm from the ends. The spacing is (1,350 − 2 × 75) ÷ 8. Find it.



WORKING

Answer _____

WORKSHEET 3B

Brackets or not?

Work out, add brackets, and key it right

Name _____

Date _____

Use BEDMAS for each problem. Where you add brackets, check the new line gives the answer asked for.

1 Find $6 + 4 \times 3 - 2$.

WORKING

Answer _____

2 Find $9 + 36 \div 4 - 2$.

WORKING

Answer _____

3 Find 2×3^2 and $(2 \times 3)^2$. Why are they different?

WORKING

Answer _____

4 Find $40 - 16 \div 4 \times 2$.

WORKING

Answer _____

5 Add one pair of brackets to make this true: $12 + 8 \div 4 = 5$.

WORKING

Answer _____

6 Add one pair of brackets to make this true: $3 \times 5 - 2 + 4 = 13$.

WORKING

Answer _____

7 Find $\sqrt{(36 + 64)}$ and $\sqrt{36} + \sqrt{64}$.

WORKING

Answer _____

8 Write the keys for $\frac{45 + 27}{14 - 6}$ and find the answer.

WORKING

Answer _____

9 Three 1,500 mm pieces are cut from a 6,000 mm bar with three cuts. The kerf is 3 mm. The offcut is $6{,}000 - (3 \times 1{,}500 + 3 \times 3)$. Find it.

WORKING

Answer _____

10 In 5×3^2, change the 3 to a 4. What is the new answer? By how much did it grow?

WORKING

Answer _____

11 A crew splits 120 m and 60 m of cable equally among 3 jobs. A learner keys $120 + 60 \div 3 =$ and gets 140. What went wrong? Write the right keys and the right answer.

WORKING

Answer _____

CHAPTER

4

Fractions

Many inch sizes are fractions. A tape reads $2\frac{5}{16}$ in, a drawing calls for $\frac{3}{8}$ in plate, and a cut list adds up lengths like $36\frac{3}{8}$ in. To read, add and divide these sizes you need to work with fractions.

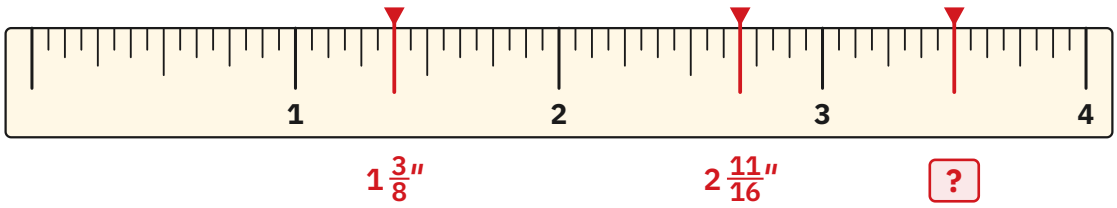
This chapter starts with what a fraction means and how to read an inch tape and a steel rule. It then covers equivalent fractions, lowest terms, mixed numbers and the four operations. Each method is shown by hand and then with the fraction key on the TI-30Xa and the Sharp EL-531XT.

In this chapter you will

- Read an inch tape to the nearest 1/16 in and a rule to 1/32 and 1/64 in
- Write equivalent fractions and put fractions in lowest terms
- Change between improper fractions and mixed numbers
- Compare fractions using a common denominator
- Add, subtract, multiply and divide fractions and mixed numbers
- Use the fraction key on both calculators

Key words

- fraction
- numerator and denominator
- proper and improper fraction
- mixed number
- equivalent fractions
- lowest terms
- common denominator
- reciprocal



4.1 What a fraction is

A **fraction** names part of a whole. The whole is split into equal parts, and the fraction tells you how many of those parts you have.

In $\frac{3}{8}$ the bottom number is the **denominator**. It tells how many equal parts the whole is split into: 8. The top number is the **numerator**. It tells how many of those parts you have: 3.

The line between the two numbers is the fraction bar. It means divide: $\frac{3}{8}$ is the same as $3 \div 8$. Chapter 5 uses this to change fractions into decimals.

A fraction whose numerator is smaller than its denominator is a **proper fraction**. It is worth less than one whole. $\frac{5}{8}$, $\frac{3}{4}$ and $\frac{15}{16}$ are proper fractions.

Example 4.1 Parts of a bar

A flat bar is marked into 8 equal lengths. Three lengths are cut off. What fraction of the bar is cut off, and what fraction is left?



SOLUTION

1. The bar is split into 8 equal parts, so the denominator is 8.
2. 3 parts are cut off, so $\frac{3}{8}$ of the bar is cut off.
3. $8 - 3 = 5$ parts are left, so $\frac{5}{8}$ of the bar is left.

Answer **$\frac{3}{8}$ is cut off and $\frac{5}{8}$ is left. Together they make $\frac{8}{8}$, the whole bar.**



Figure 4.1 One part of each bar is shaded. The bigger the denominator, the smaller each part: $\frac{1}{16}$ in is smaller than $\frac{1}{8}$ in.

TRY IT 4.1

1. A bar is split into 4 equal parts and 3 are shaded. What fraction is shaded?
2. In $\frac{7}{16}$, which number is the numerator?
3. In $\frac{7}{16}$, how many equal parts is the whole split into?
4. A box holds 12 couplings and 5 are used. What fraction is used? What fraction is left?
5. Which is smaller, $\frac{1}{4}$ in or $\frac{1}{8}$ in?
6. Is $\frac{9}{8}$ a proper fraction? Explain.

Answers are in the Answer Key.

4.2 Reading an inch tape and rule

Inch tapes and rules split each inch into equal parts. On the rule below, the longest lines mark whole inches. The next longest line marks the half inch. Shorter lines mark quarters, then eighths, and the shortest lines mark sixteenths. This book reads tapes to the nearest $\frac{1}{16}$ in, and steel rules to $\frac{1}{32}$ in or $\frac{1}{64}$ in.

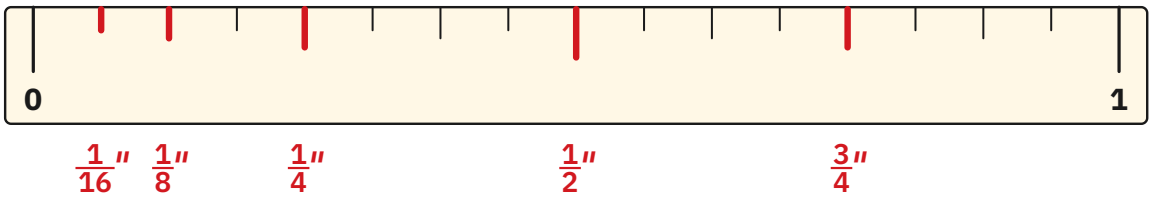


Figure 4.2 One inch split into sixteenths. $\frac{1}{2} = \frac{8}{16}$, $\frac{1}{4} = \frac{4}{16}$ and $\frac{1}{8} = \frac{2}{16}$. The length of each line shows which fraction it marks.

1. Find the last whole inch line before the mark. That gives the whole number.
2. Count the sixteenth spaces from that inch line to the mark.
3. Write the count over 16. That is the fraction part.
4. Put the fraction in lowest terms (Section 4.3 shows how).

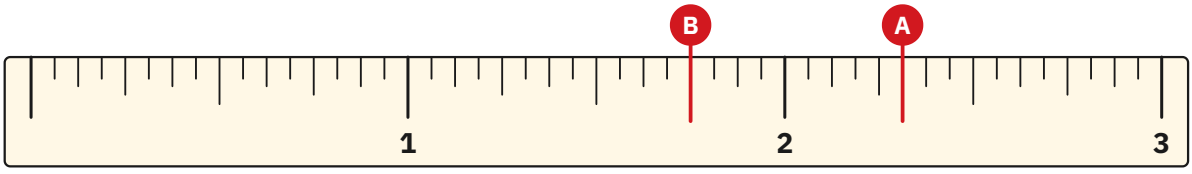


Figure 4.3 A rule marked in sixteenths. Marks A and B are read in Example 4.2.

Example 4.2 Reading marks A and B

Read marks A and B on the rule above.

SOLUTION

1. A: the last whole inch line before A is the 2. Count the spaces from the 2 line to A: 5 sixteenths. 5 is odd, so $\frac{5}{16}$ is in lowest terms. A reads $2\frac{5}{16}$ in.
2. B: the last whole inch line before B is the 1. Count 12 sixteenths past it. Divide the top and bottom of $\frac{12}{16}$ by 4 to get $\frac{3}{4}$. B reads $1\frac{3}{4}$ in.
3. Check: B sits on a quarter line, the line for $\frac{3}{4}$.

Answer **A is $2\frac{5}{16}$ in and B is $1\frac{3}{4}$ in.**

COUNT SPACES, NOT LINES

Start counting at the first space after the whole inch line. If you count the inch line itself as 1, every reading comes out $\frac{1}{16}$ in too long.

A rule marked in 32nds splits each inch into 32 equal parts, and a rule marked in 64ths splits each inch into 64. The lines are close together, so read them in two steps. First find the last eighth line before the mark, then count the small spaces past it.

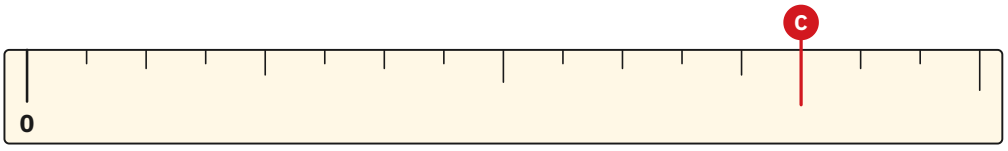


Figure 4.4 The first half inch of a rule marked in 32nds.

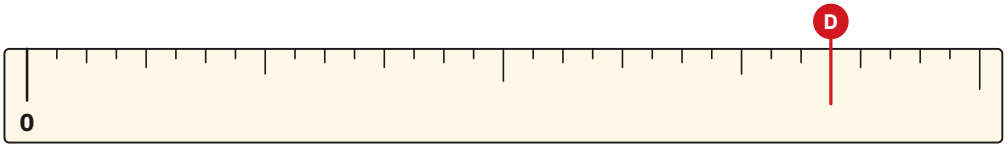


Figure 4.5 The first half inch of a rule marked in 64ths.

Example 4.3 Reading 32nds and 64ths

Read mark C on the rule marked in 32nds and mark D on the rule marked in 64ths.

SOLUTION

1. C: the last eighth line before C is the $\frac{3}{8}$ line, and $\frac{3}{8} = \frac{12}{32}$. C is 1 small space past it: 12 + 1 = 13. C reads $\frac{13}{32}$ in.
2. D: the last eighth line before D is also the $\frac{3}{8}$ line, and $\frac{3}{8} = \frac{24}{64}$. D is 3 small spaces past it: 24 + 3 = 27. D reads $\frac{27}{64}$ in.
3. 13 and 27 are odd, so both readings are in lowest terms.

Answer C is $\frac{13}{32}$ in and D is $\frac{27}{64}$ in.

Read each mark on the rule below for Try it 4.2. Give each reading to the nearest $\frac{1}{16}$ in, in lowest terms.

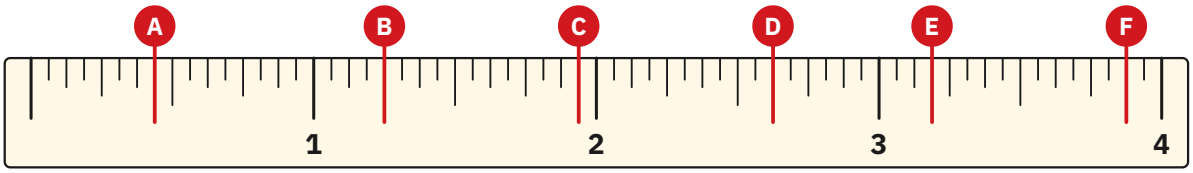


Figure 4.6 Marks A to F for Try it 4.2.

TRY IT 4.2

1. Mark A
2. Mark B
3. Mark C
4. Mark D
5. Mark E
6. Mark F

Answers are in the Answer Key.

4.3 Equivalent fractions and lowest terms

Equivalent fractions are written differently but have the same value. In the figure below, $\frac{1}{2}$, $\frac{2}{4}$, $\frac{4}{8}$ and $\frac{8}{16}$ of the bars are shaded. The four shaded parts are the same length.



Figure 4.7 $\frac{1}{2} = \frac{2}{4} = \frac{4}{8} = \frac{8}{16}$. The dashed line is at the half.

MAKING AN EQUIVALENT FRACTION

Multiply or divide the numerator and the denominator by the same number. The value of the fraction does not change. To write $\frac{3}{4}$ in sixteenths, multiply the top and bottom by 4: $\frac{3}{4} = \frac{12}{16}$.

A fraction is in **lowest terms** when no whole number bigger than 1 divides evenly into both the numerator and the denominator. To put a fraction in lowest terms, divide the top and the bottom by a number that goes into both, and repeat until no number bigger than 1 goes into both. This book writes tape readings in lowest terms: $1\frac{3}{4}$ in, not $1\frac{12}{16}$ in.

Example 4.4 Reducing a reading

Write $\frac{12}{16}$ and $\frac{24}{64}$ in lowest terms.

SOLUTION

1. $\frac{12}{16}$: both numbers are even. Divide both by 2 to get $\frac{6}{8}$, then by 2 again to get $\frac{3}{4}$. 3 is odd, so stop.
2. $\frac{24}{64}$: divide by 8 in one step. $24 \div 8 = 3$ and $64 \div 8 = 8$, so $\frac{24}{64} = \frac{3}{8}$.

CALCULATOR

TI-30Xa

12 | a b/c | 16 | =

Display: 3⌟4

Sharp EL-531XT

12 | a b/c | 16 | =

Display: 3⌜4

When you press =, the display shows the fraction in lowest terms. The TI-30Xa shows 3⌟4 and the Sharp shows 3⌜4. Read either one as $\frac{3}{4}$.

Answer $\frac{12}{16} = \frac{3}{4}$ **and** $\frac{24}{64} = \frac{3}{8}$

A QUICK TEST FOR TAPE FRACTIONS

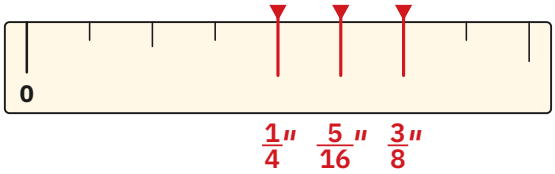
Tape and rule fractions have 2, 4, 8, 16, 32 or 64 as the denominator. If the numerator is odd, the fraction is already in lowest terms. If the numerator is even, divide the top and the bottom by 2 and test again.

Comparing fractions

When two fractions have the same denominator, the one with the bigger numerator is bigger: $\frac{9}{16}$ is bigger than $\frac{7}{16}$. When the denominators are different, rewrite the fractions with the same denominator first. A denominator that the fractions share is a **common denominator**. For tape fractions, 16 works for halves, quarters, eighths and sixteenths. Use 32 or 64 when a fraction has 32 or 64 on the bottom.

Example 4.5 Putting sizes in order

Three bolts have diameters of $\frac{3}{8}$ in, $\frac{5}{16}$ in and $\frac{1}{4}$ in. List them from smallest to largest.



SOLUTION

1. Change each to sixteenths: $\frac{3}{8} = \frac{6}{16}$, $\frac{5}{16}$ stays as it is, and $\frac{1}{4} = \frac{4}{16}$.
2. Put the numerators in order: 4, 5, 6.
3. The rule shows the same order from left to right.

Answer $\frac{1}{4}, \frac{5}{16}, \frac{3}{8}$

For fractions that are not tape fractions, use the smallest number that both denominators divide into evenly. For $\frac{2}{3}$ and $\frac{3}{4}$ that number is 12: $\frac{2}{3} = \frac{8}{12}$ and $\frac{3}{4} = \frac{9}{12}$, so $\frac{3}{4}$ is bigger.

TRY IT 4.3

1. Write $\frac{10}{16}$ in lowest terms.
2. Write $\frac{14}{16}$ in lowest terms.
3. Write $\frac{20}{32}$ in lowest terms.
4. Write $\frac{48}{64}$ in lowest terms.
5. Write $\frac{5}{8}$ in sixteenths.
6. Write $\frac{3}{4}$ in 32nds.
7. Which is bigger, $\frac{3}{8}$ or $\frac{7}{16}$?
8. Which is bigger, $\frac{5}{32}$ or $\frac{1}{8}$?
9. Put in order from smallest: $\frac{3}{4}, \frac{5}{8}, \frac{11}{16}$.
10. Which is bigger, $\frac{2}{3}$ or $\frac{5}{8}$? Use 24 as the common denominator.

Answers are in the Answer Key.

4.4 Improper fractions and mixed numbers

An **improper fraction** has a numerator equal to or bigger than its denominator, so it is worth one whole or more. A **mixed number** is a whole number and a proper fraction written together. Tape readings such as $2\frac{3}{4}$ in are mixed numbers. Improper fractions make multiplying and dividing easier.

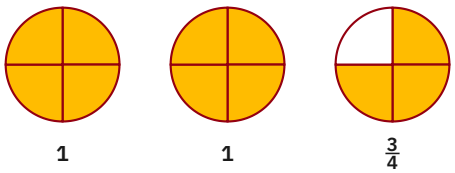


Figure 4.8 Eleven quarters make 2 whole circles and 3 quarters more: $\frac{11}{4} = 2\frac{3}{4}$.

Example 4.6 Improper to mixed

Change $\frac{43}{16}$ to a mixed number.

SOLUTION

1. Divide the numerator by the denominator: 43 ÷ 16 = 2 with a remainder. 2 × 16 = 32 and 43 − 32 = 11.
2. The whole number is 2. The remainder 11 goes over the same denominator: $\frac{11}{16}$.

Answer $\frac{43}{16} = 2\frac{11}{16}$

Example 4.7 Mixed to improper

Change $3\frac{5}{8}$ to an improper fraction.

SOLUTION

1. Multiply the whole number by the denominator: 3 whole inches hold 3 × 8 = 24 eighths.
2. Add the numerator: 24 + 5 = 29. Write it over the same denominator.

Answer $3\frac{5}{8} = \frac{29}{8}$

TRY IT 4.4

1. Change $\frac{7}{4}$ to a mixed number.
2. Change $\frac{29}{16}$ to a mixed number.
3. Change $\frac{21}{8}$ to a mixed number.
4. Change $2\frac{3}{8}$ to an improper fraction.
5. Change $5\frac{1}{2}$ to an improper fraction.
6. Change $1\frac{15}{16}$ to an improper fraction.

Answers are in the Answer Key.

4.5 The fraction key

The TI-30Xa and the Sharp EL-531XT each have a fraction key, marked a b/c. To key in a mixed number, press the whole number, the fraction key, the numerator, the fraction key again and then the denominator. For a fraction with no whole number, start with the numerator.

CALCULATOR Keying in $3\frac{5}{8}$

TI-30Xa

Key in $3\frac{5}{8}$ and press =

3 | a b/c | 5 | a b/c | 8 | =

Display: 3⌟5⌟8

Change it to an improper fraction

3 | a b/c | 5 | a b/c | 8 | = | 2nd | d/c

Display: 29⌟8

Change it to a decimal

3 | a b/c | 5 | a b/c | 8 | = | 2nd | F⇄D

Display: 3.625

Sharp EL-531XT

Key in $3\frac{5}{8}$ and press =

3 | a b/c | 5 | a b/c | 8 | =

Display: 3⌜5⌜8

Change it to an improper fraction

3 | a b/c | 5 | a b/c | 8 | = | 2ndF | d/c

Display: 29⌜8

Change it to a decimal

3 | a b/c | 5 | a b/c | 8 | = | a b/c

Display: 3.625

The TI-30Xa marks the parts of a fraction with ⌟ and the Sharp uses ⌜. Read 3⌟5⌟8 and 3⌜5⌜8 as $3\frac{5}{8}$, and 29⌟8 as $\frac{29}{8}$.

Job	TI-30Xa	Sharp EL-531XT
Key in a fraction	a b/c between the parts	a b/c between the parts
Change a result between a fraction and a decimal	2nd, then F⇄D	a b/c after the =
Change between a mixed number and an improper fraction	2nd, then d/c	2ndF, then d/c
Change a decimal such as 0.375 to a fraction	0.375, 2nd, F⇄D	0.375, =, a b/c

Table 4.1 Fraction keys on the two calculators. Both show 0.375 as $\frac{3}{8}$.

CALCULATOR A decimal as a fraction

TI-30Xa

0.375 | 2nd | F⇄D

Display: 3⌟8

Sharp EL-531XT

0.375 | = | a b/c

Display: 3⌜8

Both displays read $\frac{3}{8}$.

A DECIMAL IN A FRACTION CALCULATION

On the Sharp, if any number in a fraction calculation is keyed as a decimal, the answer comes out as a decimal: its manual shows 1.25 + 2/5 = 1.65. Key every number as a fraction when you want a fraction answer. To switch a fraction answer to a decimal and back, press 2nd F⇄D on the TI-30Xa or a b/c on the Sharp.

TRY IT 4.5

1. Write the keys to enter $2\frac{7}{16}$.
2. The TI-30Xa shows 1⌟3⌟8. What number is that?
3. The Sharp shows 15┌16. What number is that?
4. Use the fraction key to put $\frac{28}{32}$ in lowest terms.
5. Use d/c to change $4\frac{3}{4}$ to an improper fraction.
6. Use the calculator to change 0.625 to a fraction.

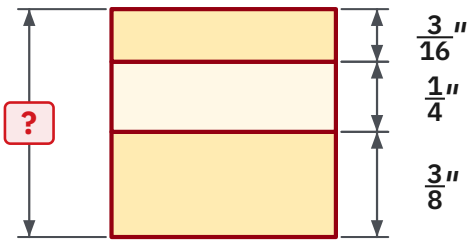
Answers are in the Answer Key.

4.6 Adding and subtracting fractions

To add or subtract fractions with the same denominator, add or subtract the numerators and keep the denominator: $\frac{3}{16} + \frac{5}{16} = \frac{8}{16} = \frac{1}{2}$. When the denominators are different, change to a common denominator first: $\frac{7}{8} - \frac{5}{16} = \frac{14}{16} - \frac{5}{16} = \frac{9}{16}$. Then put the answer in lowest terms.

Example 4.8 Stacking plates

Three plates are stacked. They are $\frac{3}{8}$ in, $\frac{1}{4}$ in and $\frac{3}{16}$ in thick. How thick is the stack?



SOLUTION

1. Change to sixteenths: $\frac{6}{16}$, $\frac{4}{16}$ and $\frac{3}{16}$.
2. Add the numerators: 6 + 4 + 3 = 13.
3. The total is $\frac{13}{16}$. 13 is odd, so it is in lowest terms.

CALCULATOR

TI-30Xa

3 | a b/c | 8 | + | 1 | a b/c | 4 | + | 3 | a b/c | 16 | =

Display: 13⌟16

Sharp EL-531XT

3 | a b/c | 8 | + | 1 | a b/c | 4 | + | 3 | a b/c | 16 | =

Display: 13┌16

The TI-30Xa shows 13⌟16 and the Sharp shows 13┌16. Both mean $\frac{13}{16}$.

Answer $\frac{13}{16}$ **in**

To add mixed numbers, add the whole numbers and add the fractions. If the fractions come to one whole or more, change that part to a mixed number and carry the whole number.

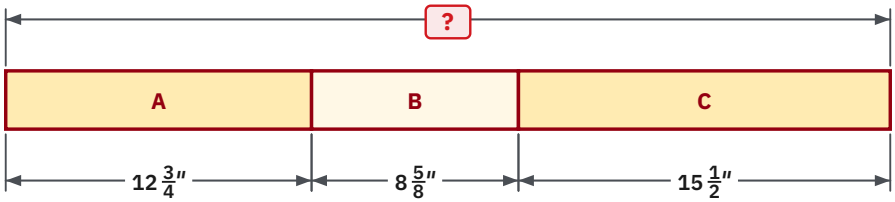


Figure 4.9 Three pieces on a cut list. The total length is hidden.

Example 4.9 Total of a cut list

Find the total length of the three pieces: $12\frac{3}{4}$ in, $8\frac{5}{8}$ in, $15\frac{1}{2}$ in.

SOLUTION

1. Add the whole numbers: 12 + 8 + 15 = 35.
2. Change the fractions to eighths and add them: $\frac{6}{8} + \frac{5}{8} + \frac{4}{8} = \frac{15}{8}$.
3. $\frac{15}{8} = 1\frac{7}{8}$. Carry the 1: 35 + 1 = 36. The total is $36\frac{7}{8}$ in.

CALCULATOR

TI-30Xa

12 | a b/c | 3 | a b/c | 4 | + | 8 | a b/c | 5 | a b/c | 8 | + | 15 | a b/c | 1 | a b/c | 2 | =

Display: 36⌟7⌟8

Sharp EL-531XT

12 | a b/c | 3 | a b/c | 4 | + | 8 | a b/c | 5 | a b/c | 8 | + | 15 | a b/c | 1 | a b/c | 2 | =

Display: 36 ⌜7 ⌜8

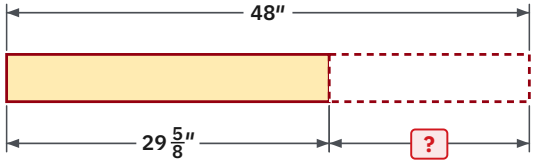
The TI-30Xa shows 36⌟7⌟8 and the Sharp shows 36 ⌜7 ⌜8. Both mean $36\frac{7}{8}$.

Answer $36\frac{7}{8}$ in

To subtract mixed numbers, subtract the fractions first and then the whole numbers. When the fraction on top is smaller than the fraction you take away, borrow 1 from the whole number. Write the borrowed 1 as a fraction with the same denominator, such as $\frac{8}{8}$ or $\frac{16}{16}$, and add it to the fraction. For example, $5\frac{1}{4} - 2\frac{7}{16} = 5\frac{4}{16} - 2\frac{7}{16} = 4\frac{20}{16} - 2\frac{7}{16} = 2\frac{13}{16}$.

Example 4.10 What is left of the bar?

A piece $29\frac{5}{8}$ in long is cut from a 48 in bar. Ignore the saw cut. How long is the offcut?



SOLUTION

1. There is no fraction in 48 to take $\frac{5}{8}$ from, so borrow 1: $48 = 47\ \frac{8}{8}$.
2. Fractions: $\frac{8}{8} - \frac{5}{8} = \frac{3}{8}$.
3. Whole numbers: $47 - 29 = 18$.

CALCULATOR

TI-30Xa

48 | – | 29 | a b/c | 5 | a b/c | 8 | =

Display: 18⌟3⌟8

Sharp EL-531XT

48 | – | 29 | a b/c | 5 | a b/c | 8 | =

Display: 18 ⌜3 ⌜8

The TI-30Xa shows 18⌟3⌟8 and the Sharp shows 18 ⌜3 ⌜8. Both mean $18\frac{3}{8}$.

Answer $18\frac{3}{8}$ **in**

CHECK BY ADDING

Add your answer to the number you took away. You should get the number you started with: $18\frac{3}{8} + 29\frac{5}{8} = 48$.

TRY IT 4.6

1. $\frac{3}{8} + \frac{7}{16}$
2. $\frac{5}{8} + \frac{3}{4}$
3. $10\frac{7}{8} + 6\frac{3}{4}$
4. Pieces of $18\frac{3}{8}$ in, $24\frac{1}{2}$ in and $9\frac{15}{16}$ in. What is the total length?
5. $\frac{15}{16} - \frac{3}{8}$
6. $6 - 2\frac{3}{8}$
7. $12\frac{3}{16} - 5\frac{5}{8}$
8. A piece $22\frac{11}{16}$ in long is cut from a 36 in bar. Ignore the saw cut. How long is the offcut?

Answers are in the Answer Key.

4.7 Multiplying and dividing fractions

To multiply fractions, multiply the numerators, multiply the denominators and put the answer in lowest terms: $\frac{2}{3} \times \frac{3}{4} = \frac{6}{12} = \frac{1}{2}$. A whole number can be written over 1: $6 = \frac{6}{1}$. Change mixed numbers to improper fractions before you multiply.

In word problems, the word "of" means multiply. $\frac{3}{4}$ of a 24 in bar is $\frac{3}{4} \times \frac{24}{1} = \frac{72}{4} = 18$ in.

SAW CUTS IN FRACTIONS

When a problem gives the **kerf** as a fraction of an inch, work with it like any other fraction. With a kerf of $\frac{1}{8}$ in, 4 cuts remove $4 \times \frac{1}{8} = \frac{4}{8} = \frac{1}{2}$ in of stock. The problems in this book give the kerf whenever it counts.

Example 4.11 Pieces and saw cuts

Four pieces, each $11\frac{3}{4}$ in long, are cut from a 48 in bar. You make 4 cuts and the kerf is $\frac{1}{8}$ in. How long is the offcut?

SOLUTION

1. Pieces: $4 \times 11\frac{3}{4} = 4 \times \frac{47}{4} = \frac{188}{4} = 47$ in.
2. Saw cuts: $4 \times \frac{1}{8} = \frac{1}{2}$ in.
3. Offcut: $48 - 47 - \frac{1}{2} = \frac{1}{2}$ in.

CALCULATOR

TI-30Xa

48 | − | 4 | × | 11 | a b/c | 3 | a b/c | 4 | − | 4 | × | 1 | a b/c | 8 | =

Display: 1⌟2

Sharp EL-531XT

48 | − | 4 | × | 11 | a b/c | 3 | a b/c | 4 | − | 4 | × | 1 | a b/c | 8 | =

Display: 1⌜2

Answer $\frac{1}{2}$ **in**

The **reciprocal** of a fraction is the fraction turned upside down. The reciprocal of $\frac{3}{4}$ is $\frac{4}{3}$, and the reciprocal of 5 (or $\frac{5}{1}$) is $\frac{1}{5}$. A number times its reciprocal is 1: $\frac{3}{4} \times \frac{4}{3} = \frac{12}{12} = 1$.

DIVIDING BY A FRACTION

To divide by a fraction, multiply by its reciprocal. Change mixed numbers to improper fractions first. Then turn the number you are dividing by upside down and multiply.

For example, a $30\frac{1}{2}$ in bar cut into 4 equal pieces, ignoring the saw cuts, gives $30\frac{1}{2} \div 4 = \frac{61}{2} \times \frac{1}{4} = \frac{61}{8} = 7\frac{5}{8}$ in for each piece.

Example 4.12 How many pieces?

How many spacers $1\frac{3}{4}$ in long can be cut from a 24 in bar? Ignore the saw cuts. How long is the offcut?

SOLUTION

1. $24 \div 1\frac{3}{4} = \frac{24}{1} \times \frac{4}{7} = \frac{96}{7} = 13\frac{5}{7}$.
2. Only whole spacers count: 13 spacers.
3. 13 spacers use $13 \times 1\frac{3}{4} = 22\frac{3}{4}$ in. The offcut is $24 - 22\frac{3}{4} = 1\frac{1}{4}$ in.

CALCULATOR

TI-30Xa

Spacers

24 ÷ 1 a b/c 3 a b/c 4 =

Display 13⌟5⌟7

Offcut

24 − 13 × 1 a b/c 3 a b/c 4 =

Display 1⌟1⌟4

Sharp EL-531XT

Spacers

24 ÷ 1 a b/c 3 a b/c 4 =

Display 13 ⌜5 ⌜7

Offcut

24 − 13 × 1 a b/c 3 a b/c 4 =

Display 1 ⌜1 ⌜4

The first display reads $13\frac{5}{7}$. Only the whole number of spacers counts.

Answer **13 spacers, with an offcut of $1\frac{1}{4}$ in.**

LENGTHS THAT ARE NOT ON THE TAPE

Some answers have denominators that are not on a tape, such as thirds. To mark them, multiply the fraction part by 16 and round to the nearest whole number of sixteenths. A 25 in bar cut into 3 equal pieces gives $\frac{25}{3} = 8\frac{1}{3}$ in. $\frac{1}{3} \times 16 = 5\frac{1}{3}$, which rounds to 5, so mark each piece at $8\frac{5}{16}$ in. Chapter 5 does the same job with decimal inches.

TRY IT 4.7

1. $\frac{3}{8} \times 4$
2. $\frac{2}{3}$ of 18
3. $1\frac{1}{2} \times 2\frac{1}{4}$
4. 6 pieces at $2\frac{3}{8}$ in. What is the total length, ignoring saw cuts?
5. What is the reciprocal of $\frac{5}{8}$?
6. $\frac{7}{8} \div 2$
7. $10\frac{1}{2} \div 1\frac{3}{4}$
8. A $29\frac{1}{4}$ in bar is cut into 3 equal pieces. Ignore the saw cuts. How long is each piece?

Answers are in the Answer Key.

WORKSHEET 4A

Read the rule

Read each red pointer, then find the hidden lengths

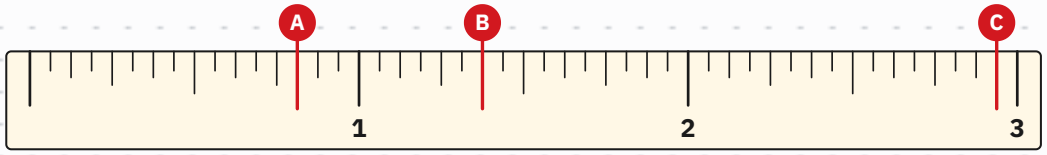
Name _____

Date _____

Give every reading in lowest terms. Ignore saw cuts unless the problem gives a kerf. Check your answers with the fraction key.

1 Read marks A, B and C to the nearest $\frac{1}{16}$ in.

SHAPE CANVAS — READ THE DRAWING

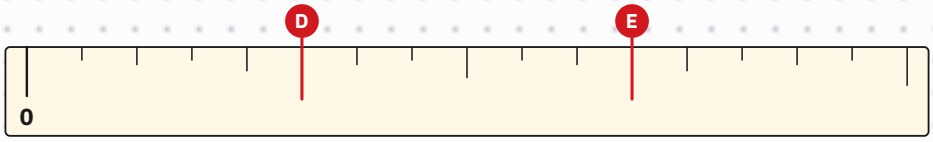


WORKING

Answer _____

2 This half inch of a rule is marked in 32nds. Read marks D and E.

SHAPE CANVAS — READ THE DRAWING

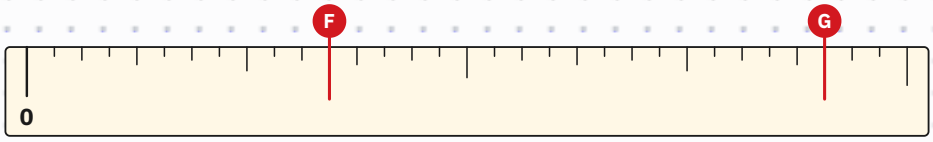


WORKING

Answer _____

3 This half inch of a rule is marked in 64ths. Read marks F and G.

SHAPE CANVAS — READ THE DRAWING

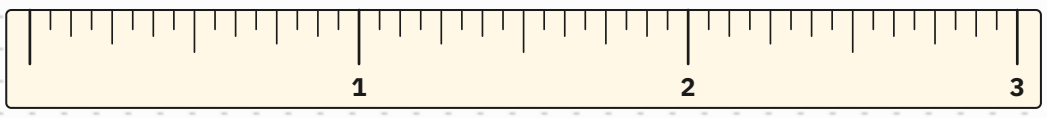


WORKING

Answer _____

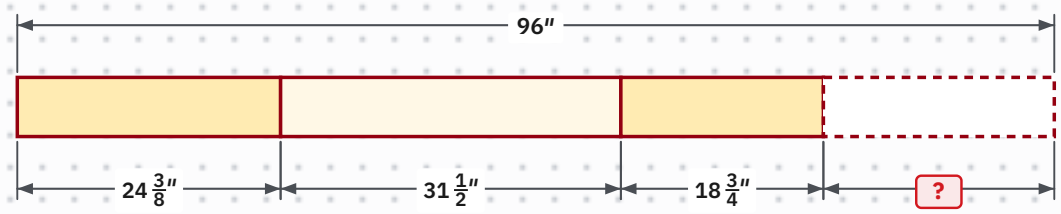
4 Draw a pointer on the rule at $1\frac{3}{8}$ in and another at $2\frac{11}{16}$ in. Label them.

SHAPE CANVAS — READ THE DRAWING



5 Three pieces are cut from a 96 in bar. How long is the offcut?

SHAPE CANVAS — PRACTICE: FIND THE ?

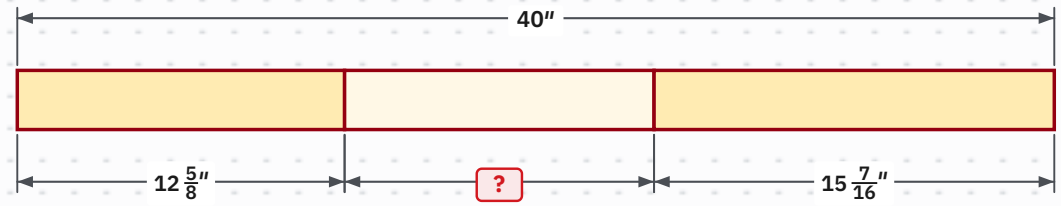


WORKING

Answer _____

6 The overall length is 40 in. Find the middle piece.

SHAPE CANVAS — PRACTICE: FIND THE ?



WORKING

Answer _____

7 A frame needs 2 pieces at $36\frac{3}{8}$ in and 2 pieces at $24\frac{5}{8}$ in, cut from a 144 in bar. You make 4 cuts and the kerf is $\frac{1}{8}$ in. How long is the offcut?

WORKING

Answer _____

8 Change one value: 8 pieces are each $5\frac{3}{8}$ in long. What is their total length? If each piece is made $\frac{1}{16}$ in longer, how much more stock is needed?

WORKING

Answer _____

9 A 72 in bar is cut into 5 equal pieces. Ignore the saw cuts. How long is each piece, to the nearest $\frac{1}{16}$ in?

WORKING

Answer _____

WORKSHEET 4B

Break it down: parts in inches

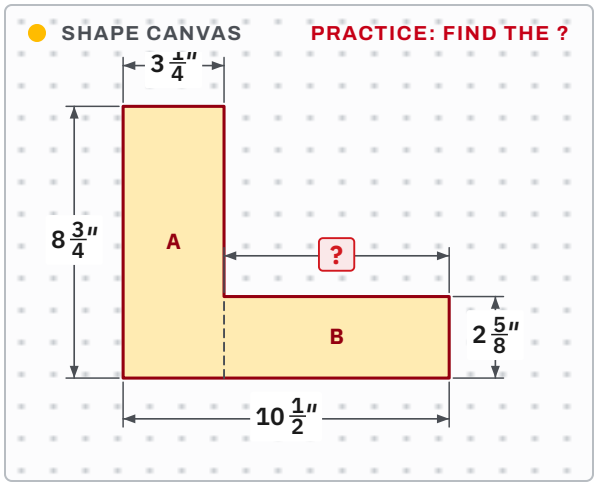
Split each part into pieces you can work with

Name _____

Date _____

Each drawing is to scale. Find every red ?. Show the fractions you add or take away in the working box.

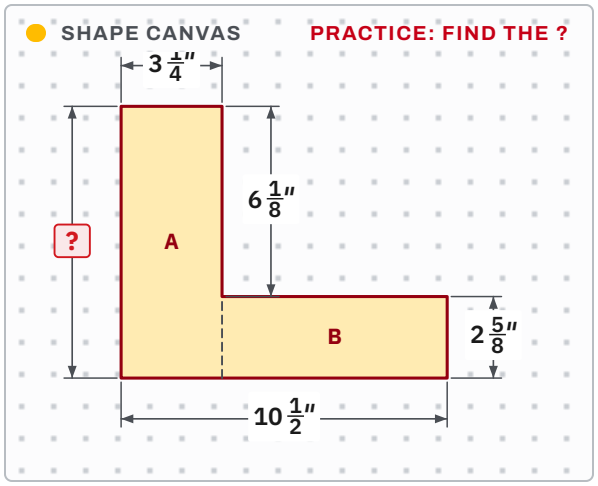
1 Find the hidden width.



WORKING

Answer _____

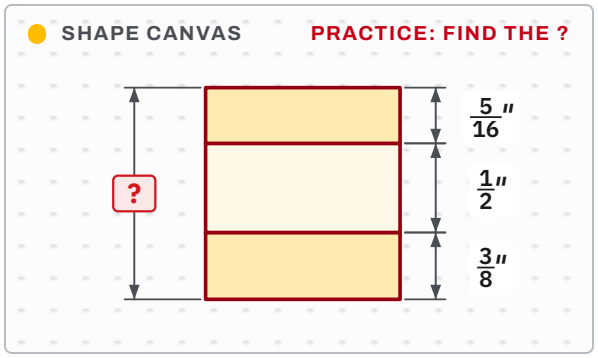
2 Find the overall height.



WORKING

Answer _____

3 Three plates are stacked. How thick is the stack?



WORKING

Answer _____

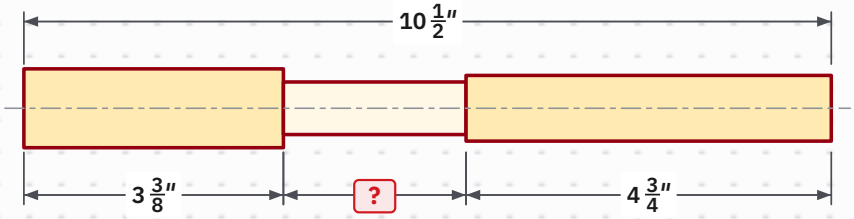
4 You cut $2\frac{1}{4}$ in pieces from a 30 in bar. You make one cut for each piece and the kerf is $\frac{1}{8}$ in. How many pieces do you get, and how long is the offcut?

WORKING

Answer _____

5 The shaft is $10\frac{1}{2}$ in long overall. Find the length of the middle step.

SHAPE CANVAS — PRACTICE: FIND THE ?

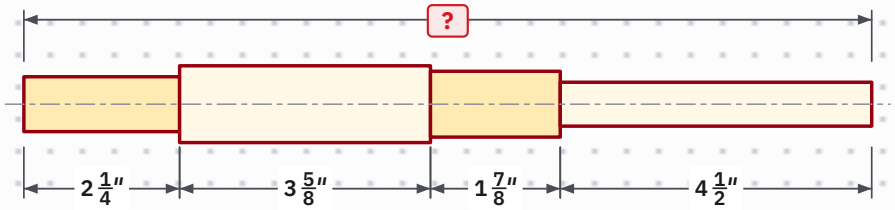


WORKING

Answer _____

6 Find the overall length of the shaft.

SHAPE CANVAS — PRACTICE: FIND THE ?

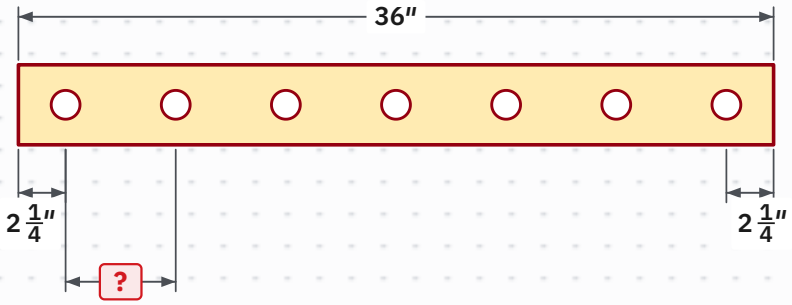


WORKING

Answer _____

7 Seven holes are evenly spaced along a 36 in plate. The end holes are $2\frac{1}{4}$ in from the ends. Find the spacing.

SHAPE CANVAS — PRACTICE: FIND THE ?



WORKING

Answer _____

CHAPTER

5

Decimals

A metric drawing might give a size of 76.2 mm, flat bar might cost $3.99 a metre, and a calculator answer might come out as 2.83 in. Each of these is a decimal.

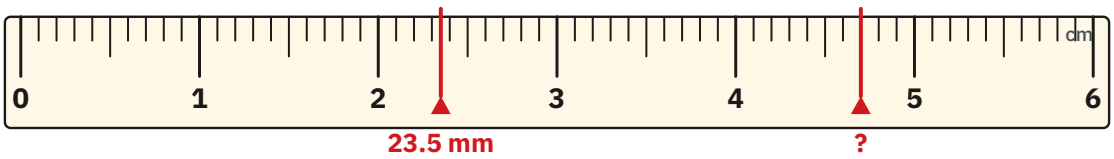
This chapter covers place value after the decimal point, reading a metric rule, rounding, and adding, subtracting, multiplying and dividing decimals. It then links decimals to the fractions in Chapter 4, so you can change a decimal inch to the nearest sixteenth and mark it on a tape.

In this chapter you will

- Read and write decimals using place value
- Read a metric rule to the nearest 0.5 mm
- Compare and round decimals
- Add, subtract, multiply and divide decimals
- Change fractions to decimals and decimals to fractions
- Change a decimal inch to the nearest 1/16 or 1/32 in
- Work with money to the nearest cent

Key words

- decimal
- decimal point
- decimal place
- decimal inch



5.1 Decimals and place value

A **decimal** is a number written with a **decimal point**. The point separates the whole number part from the part that is less than one. Each place after the point is worth one tenth of the place to its left: tenths, then hundredths, then thousandths.

Number	Tens	Ones	.	Tenths	Hundredths	Thousandths
25.4	2	5	.	4		
76.2	7	6	.	2		
3.175		3	.	1	7	5
0.375		0	.	3	7	5

Table 5.1 Place value on both sides of the decimal point.

Each digit after the point fills one **decimal place**. 2.375 has three decimal places. A decimal is a fraction with 10, 100 or 1,000 on the bottom: $0.7 = \frac{7}{10}$, $0.25 = \frac{25}{100}$ and $0.375 = \frac{375}{1{,}000}$. Write a 0 in front of the point when there is no whole number, as in 0.375. It makes the point easy to see.

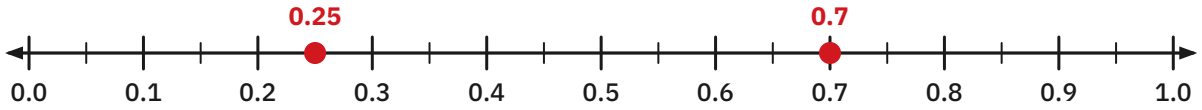


Figure 5.1 Tenths from 0 to 1. 0.25 sits halfway between 0.2 and 0.3, and 0.7 is seven tenths.

Example 5.1 Reading a decimal size

A drawing gives a hole position as 12.07 mm. Read the number in words and say what the 7 is worth.

SOLUTION

1. The whole number part is 12.
2. The 0 is in the tenths place and the 7 is in the hundredths place.
3. Read it as twelve and seven hundredths. The 7 is worth $\frac{7}{100}$, or 0.07.

Answer **Twelve and seven hundredths millimetres. The 7 is worth 0.07 mm.**

Reading a metric rule

On the metric rules in this book, each small line is 1 mm, the longer lines mark every 5 mm, and the numbered lines mark every 10 mm (1 cm). This book reads metric rules to the nearest 0.5 mm. A mark halfway between two millimetre lines reads as .5.

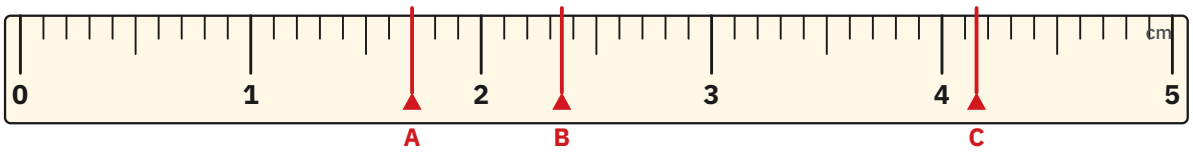


Figure 5.2 A metric rule. Each small space is 1 mm.

Example 5.2 Reading the metric rule

Read marks A, B and C on the rule above.

SOLUTION

1. A: past the 1 cm line (10 mm) by 7 small spaces: 10 + 7 = 17 mm.
2. B: past the 20 mm line by 3 spaces and then half a space: 23.5 mm.
3. C: past the 40 mm line by 1 space and half a space: 41.5 mm.

Answer **A = 17 mm, B = 23.5 mm, C = 41.5 mm**

TRY IT 5.1

1. Write 3.06 in words.
2. What is the 5 worth in 0.375?
3. How many decimal places does 9.525 have?
4. Write $\frac{9}{10}$ as a decimal.
5. Write $\frac{45}{100}$ as a decimal.
6. Write 0.125 as a fraction over 1,000.

Answers are in the Answer Key.

5.2 Comparing and rounding decimals

To compare decimals, line up the decimal points and compare the digits from the left. Add zeros on the right so both numbers have the same number of decimal places. Zeros on the right of a decimal do not change its value: 0.38 = 0.380. To compare a 0.375 in plate with a 0.38 in plate, write 0.375 and 0.380. The tenths are the same, and 8 hundredths is more than 7 hundredths, so the 0.38 in plate is thicker, by 0.005 in.

MORE DIGITS IS NOT BIGGER

0.375 has more digits than 0.38, but it is the smaller number. Compare place by place from the left, not by counting digits.

Decimals round the same way as whole numbers. Find the place you are rounding to, look at the digit to its right, and round up if that digit is 5 or more. Then drop every digit after the rounding place.

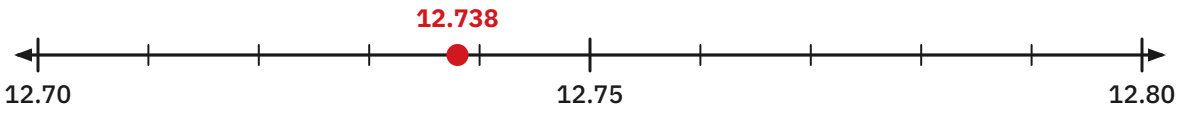


Figure 5.3 12.738 is closer to 12.74 than to 12.73, and closer to 12.7 than to 12.8.

Example 5.3 Rounding a size

Round 12.738 mm to one decimal place and to two decimal places.

SOLUTION

1. One place: the tenths digit is 7. The digit to its right is 3, so the 7 stays: 12.7 mm.
2. Two places: the hundredths digit is 3. The digit to its right is 8, so the 3 goes up to 4: 12.74 mm.

Answer **12.7 mm to one place, 12.74 mm to two places.**

Example 5.4 Rounding to the nearest 0.5 mm

A calculation gives 23.38 mm. Round it to the nearest 0.5 mm so you can mark it with a metric rule.

SOLUTION

1. Double the number: 23.38 × 2 = 46.76.
2. Round to the nearest whole number: 47.
3. Halve it: 47 ÷ 2 = 23.5.

Answer **23.5 mm**

TRY IT 5.2

1. Which is bigger, 0.6 or 0.58?
2. Put in order from smallest: 0.5, 0.125, 0.25.
3. Round 6.35 to one decimal place.
4. Round 9.525 to two decimal places.
5. Round 3.175 mm to the nearest 0.5 mm.
6. Round 18.74 mm to the nearest whole millimetre.

Answers are in the Answer Key.

5.3 Adding and subtracting decimals

To add or subtract decimals, line up the decimal points, one under the other. Fill empty places with zeros, then add or subtract as with whole numbers. The point in the answer goes straight below the points above it.

Example 5.5 Total of a metric cut list

Find the total length of four pieces: 152.4 mm, 76.2 mm, 38.1 mm, 12.7 mm.

SOLUTION

1. Line up the points: each number has one decimal place.
2. Tenths: 4 + 2 + 1 + 7 = 14. Write 4, carry 1.
3. Whole numbers: 152 + 76 + 38 + 12 + 1 (carried) = 279.

CALCULATOR

TI-30Xa

152.4 + 76.2 + 38.1 + 12.7 =

Display 279.4

Sharp EL-531XT

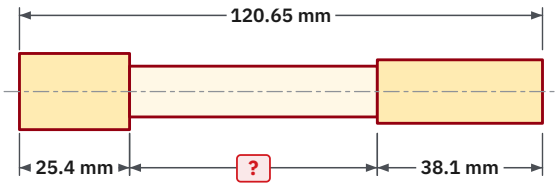
152.4 + 76.2 + 38.1 + 12.7 =

Display 279.4

Answer **279.4 mm**

Example 5.6 Finding a hidden length

A stepped pin is 120.65 mm long overall. Two of its steps are 25.4 mm and 38.1 mm long. How long is the third step?



SOLUTION

1. Add the known steps: 25.4 + 38.1 = 63.5.
2. Subtract from the overall length: write 63.5 as 63.50 and line up the points.
3. 120.65 − 63.50 = 57.15.

CALCULATOR

TI-30Xa

120.65 − 25.4 − 38.1 =

Display 57.15

Sharp EL-531XT

120.65 − 25.4 − 38.1 =

Display 57.15

Answer **57.15 mm**

Decimal inches add the same way. Pieces of 2.375 in, 1.25 in and 0.625 in come to 2.375 + 1.250 + 0.625 = 4.25 in.

TRY IT 5.3

1. 4.5 + 0.75
2. 12.7 + 6.35 + 3.175
3. 10 − 3.25
4. 76.2 − 19.05
5. A 600 mm bar has pieces of 152.4 mm and 228.6 mm cut off. Ignore the saw cuts. How much is left?
6. Shims of 0.5 mm, 0.25 mm and 0.1 mm are stacked. How thick is the stack?

Answers are in the Answer Key.

5.4 Multiplying decimals

To multiply decimals, multiply as if there were no points. Then count the decimal places in the numbers you multiplied and give the answer that many decimal places. For 1.5 × 0.25: 15 × 25 = 375. 1.5 has one place and 0.25 has two, so the answer has three: 0.375.

Multiplying by 10, 100 or 1,000 moves the point 1, 2 or 3 places to the right: 2.375 × 1,000 = 2,375.

Example 5.7 Twelve equal pieces

You need 12 pieces, each 152.4 mm long. What total length do they use? Ignore the saw cuts.

SOLUTION

1. Estimate first: 12 × 150 = 1,800.
2. Multiply without the point: 12 × 1,524 = 18,288.
3. 152.4 has one decimal place, so the answer has one: 1,828.8. It is close to the estimate.

CALCULATOR

TI-30Xa

12 × 152.4 =

Display 1828.8

Sharp EL-531XT

12 × 152.4 =

Display 1’828.8

The TI-30Xa shows 1828.8. The Sharp shows 1’828.8, with a small mark between the thousands and the hundreds.

Answer 1,828.8 mm

Example 5.8 Inches to millimetres

A drawing calls for a $\frac{3}{8}$ in plate. How thick is that in millimetres? Round to the nearest 0.5 mm.

SOLUTION

1. There are exactly 25.4 mm in 1 in. Chapter 7 covers unit conversion.
2. $\frac{3}{8}$ in = 0.375 in. 0.375 × 25.4 = 9.525 mm.
3. To the nearest 0.5 mm: 9.525 × 2 = 19.05, which rounds to 19, and 19 ÷ 2 = 9.5.

CALCULATOR

TI-30Xa

3 ÷ 8 × 25.4 =

Display 9.525

Sharp EL-531XT

3 ÷ 8 × 25.4 =

Display 9.525

Answer **9.525 mm, or 9.5 mm to the nearest 0.5 mm.**

TRY IT 5.4

1. 0.5 × 0.5
2. 3 × 2.375
3. 1.2 × 0.15
4. 4.75 × 100
5. 8 pieces at 76.2 mm. Total length?
6. Change $\frac{5}{8}$ in (0.625 in) to millimetres.

Answers are in the Answer Key.

5.5 Dividing decimals

When you divide by a whole number, divide as usual and keep the point in the same place in the answer. When you divide by a decimal, move the point to the right in both numbers until the number you divide by is a whole number. Dividing by 10, 100 or 1,000 moves the point 1, 2 or 3 places to the left: 254 ÷ 100 = 2.54.

Example 5.9 Dividing by a decimal

How many 0.25 m lengths can be cut from a 7.5 m bar? Ignore the saw cuts.

SOLUTION

1. 0.25 has two decimal places, so move the point two places to the right in both numbers: 7.5 ÷ 0.25 becomes 750 ÷ 25 = 30.

Answer **30 lengths**

Example 5.10 Pieces with saw cuts

How many 152.4 mm pieces can you cut from a 2,438.4 mm bar? You make one cut for each piece and the kerf is 3.2 mm. How long is the offcut?

SOLUTION

1. Each piece uses its length plus one saw cut: 152.4 + 3.2 = 155.6 mm.
2. 2,438.4 ÷ 155.6 = 15.67. Only whole pieces count: 15 pieces.
3. 15 pieces use 15 × 155.6 = 2,334 mm. The offcut is 2,438.4 − 2,334 = 104.4 mm.

CALCULATOR

TI-30Xa

Pieces

2438.4 ÷ 155.6 =

Display 15.67095116

Offcut

2438.4 − 15 × 155.6 =

Display 104.4

Sharp EL-531XT

Pieces

2438.4 ÷ 155.6 =

Display 15.67095116

Offcut

2438.4 − 15 × 155.6 =

Display 104.4

Answer **15 pieces, with an offcut of 104.4 mm.**

TRY IT 5.5

1. 9.6 ÷ 4
2. 1.5 ÷ 0.3
3. 25.4 ÷ 0.5
4. 635 ÷ 100
5. A 914.4 mm bar is cut into 6 equal pieces. Ignore the saw cuts. How long is each piece?
6. 100 mm split into 3 equal parts. How long is each, to two decimal places?

Answers are in the Answer Key.

5.6 Fractions and decimals

The fraction bar means divide, so to change a fraction to a decimal, divide the numerator by the denominator: $\frac{3}{8}$ = 3 ÷ 8 = 0.375. An inch size written this way is a **decimal inch**: 0.375 in is $\frac{3}{8}$ in. For a mixed number, change only the fraction part and keep the whole number: $2\frac{5}{8}$ in = 2.625 in.

CALCULATOR Fraction to decimal

TI-30Xa

$\frac{3}{8}$ by dividing

3 ÷ 8 =

Display 0.375

$2\frac{5}{8}$ with the fraction key

2 a b/c 5 a b/c 8 2nd F⇄D

Display 2.625

Sharp EL-531XT

$\frac{3}{8}$ by dividing

3 ÷ 8 =

Display 0.375

$2\frac{5}{8}$ with the fraction key

2 a b/c 5 a b/c 8 = a b/c

Display 2.625

Fraction	Decimal inch	Millimetres
$\frac{1}{8}$ in	0.125 in	3.175 mm
$\frac{1}{4}$ in	0.25 in	6.35 mm
$\frac{3}{8}$ in	0.375 in	9.525 mm
$\frac{1}{2}$ in	0.5 in	12.7 mm
$\frac{5}{8}$ in	0.625 in	15.875 mm
$\frac{3}{4}$ in	0.75 in	19.05 mm
$\frac{7}{8}$ in	0.875 in	22.225 mm
1 in	1 in	25.4 mm

Table 5.2 Eighths of an inch as decimal inches and millimetres. All of these are exact.

To change a decimal with three places or fewer to a fraction, write it over 10, 100 or 1,000 and put it in **lowest terms**: $0.375 = \frac{375}{1{,}000} = \frac{3}{8}$ (divide both by 125). The fraction key does this for you (2nd F⇄D on the TI-30Xa, a b/c on the Sharp), as Chapter 4 showed.

Decimal inches to the nearest sixteenth

A tape is marked in sixteenths, but a calculator gives decimal inches such as 2.83 in. To mark the length, change the decimal part to the nearest $\frac{1}{16}$ in. For finer work on a steel rule, use the nearest $\frac{1}{32}$ in. Always say which one you used.

1. Keep the whole number of inches.
2. Multiply the decimal part by 16 (or by 32).
3. Round to the nearest whole number. That is the number of sixteenths (or 32nds).
4. Write it over 16 (or 32) and put it in lowest terms.

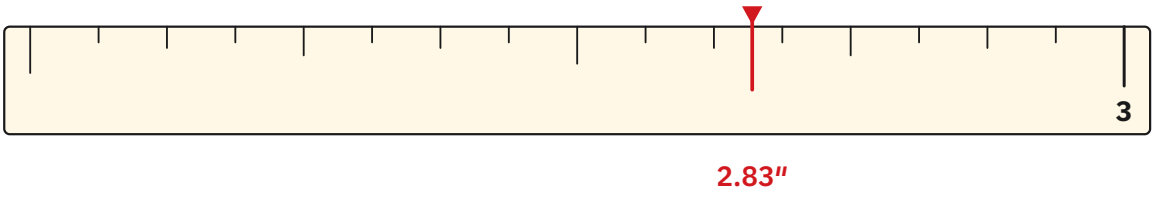


Figure 5.4 The half inch from $2\frac{1}{2}$ in to 3 in, marked in 32nds. 2.83 in lies between the $2\frac{26}{32}$ line and the $2\frac{27}{32}$ line, a little closer to $2\frac{27}{32}$.

Example 5.11 Marking 2.83 in on a tape

A calculation gives 2.83 in. Change it to the nearest $\frac{1}{16}$ in, and to the nearest $\frac{1}{32}$ in.

SOLUTION

1. Keep the 2 inches. The decimal part is 0.83.
2. Nearest $\frac{1}{16}$: 0.83 × 16 = 13.28, which rounds to 13. That gives $\frac{13}{16}$, so $2\frac{13}{16}$ in.
3. Nearest $\frac{1}{32}$: 0.83 × 32 = 26.56, which rounds to 27. That gives $\frac{27}{32}$, so $2\frac{27}{32}$ in.

CALCULATOR

TI-30Xa

Sixteenths

0.83 × 16 =

Display 13.28

32nds

0.83 × 32 =

Display 26.56

Sharp EL-531XT

Sixteenths

0.83 × 16 =

Display 13.28

32nds

0.83 × 32 =

Display 26.56

Answer $2\frac{13}{16}$ **in to the nearest** $\frac{1}{16}$ **in, or** $2\frac{27}{32}$ **in to the nearest** $\frac{1}{32}$ **in.**

Example 5.12 Millimetres to the nearest sixteenth

A metric part is 50 mm long. What is that to the nearest $\frac{1}{16}$ in?

SOLUTION

1. Change to inches by dividing by 25.4, and multiply by 16 in the same line to count sixteenths: 50 ÷ 25.4 × 16 = 31.496 (to three places).
2. Round to the nearest whole number: 31.496 is less than 31.5, so 31 sixteenths.
3. $\frac{31}{16} = 1\frac{15}{16}$.

CALCULATOR

TI-30Xa

50 ÷ 25.4 × 16 =

Display: 31.49606299

Sharp EL-531XT

50 ÷ 25.4 × 16 =

Display: 31.49606299

The display shows 31.49606299. That is just under 31.5, so it rounds down to 31 sixteenths.

Answer $1\frac{15}{16}$ **in to the nearest** $\frac{1}{16}$ **in.**

THE FRACTION KEY DOES NOT ROUND TO THE TAPE

The fraction key gives the exact fraction, not the nearest sixteenth. On the TI-30Xa, 0.83 2nd F⇄D shows 83⌟100. On the Sharp, 0.83 = a b/c shows 83 ⌜100. Both mean $\frac{83}{100}$, not a tape reading. Use the multiply-by-16 method instead.

CALCULATOR The fraction key on 0.83

TI-30Xa

0.83 2nd F⇄D

Display: 83⌟100

Sharp EL-531XT

0.83 = a b/c

Display: 83⌜100

TRY IT 5.6

1. Change $\frac{3}{4}$ to a decimal.
2. Change $1\frac{7}{8}$ in to a decimal inch.
3. Change 0.625 to a fraction in lowest terms.
4. Change 3.4 in to the nearest $\frac{1}{16}$ in.
5. Change 0.59 in to the nearest $\frac{1}{32}$ in.
6. Change 100 mm to the nearest $\frac{1}{16}$ in.

Answers are in the Answer Key.

5.7 Money

Money is written with two decimal places: dollars, then cents. $3.99 is 3 dollars and 99 cents. Round money answers to the nearest cent, which is two decimal places. The calculator does not show zeros at the end of a decimal: 4 × 12.85 shows 51.4, which means $51.40. Always write money with two decimal places.

Flat bar at $3.99 per metre: 7.5 m cost 7.5 × 3.99 = $29.925. The third decimal digit is 5, so round up to the nearest cent: $29.93.

Example 5.13 Adding up an order

An order has 4 boxes of bolts at $12.85 a box and 2 cans of cutting oil at $8.49 a can. What is the total?

SOLUTION

1. Bolts: 4 × 12.85 = 51.40.
2. Oil: 2 × 8.49 = 16.98.
3. Total: 51.40 + 16.98 = 68.38.

CALCULATOR

TI-30Xa

4 × 12.85 + 2 × 8.49 =

Display 68.38

Sharp EL-531XT

4 × 12.85 + 2 × 8.49 =

Display 68.38

The calculator multiplies before it adds, so no brackets are needed. Chapter 3 explains why. The display shows 68.38, which is $68.38.

Answer **$68.38**

TRY IT 5.7

1. Round $14.736 to the nearest cent.
2. 3 m of pipe at $6.25 per metre. Cost?
3. 12 washers at $0.18 each. Cost?
4. 2.4 m of angle at $5.49 per metre. Cost, to the nearest cent?
5. You pay for a $37.65 order with $50. How much change?
6. Five workers split a $68.40 lunch bill equally. How much each?

Answers are in the Answer Key.

WORKSHEET 5A

Decimal parts on the canvas

Metric parts with decimal sizes. Find every red ?

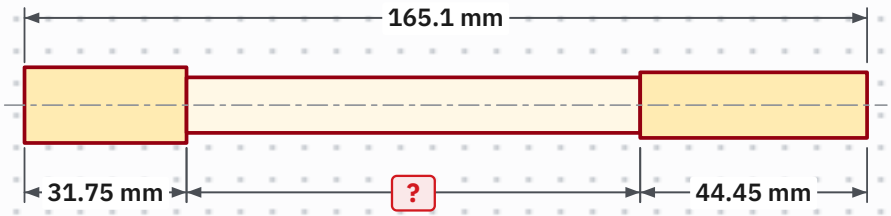
Name _____

Date _____

Line up the decimal points when you add or subtract. Estimate first, then check with your calculator.

1 The pin is 165.1 mm long overall. Find the length of the middle step.

SHAPE CANVAS — PRACTICE: FIND THE ?

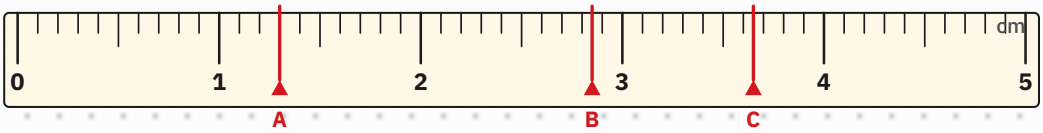


WORKING

Answer _____

2 Read marks A, B and C to the nearest 0.5 mm.

SHAPE CANVAS — READ THE DRAWING

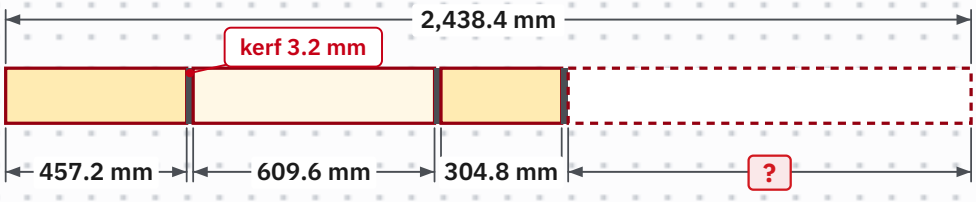


WORKING

Answer _____

3 Three pieces are cut from a 2,438.4 mm bar. You make 3 cuts and the kerf is 3.2 mm. How long is the offcut?

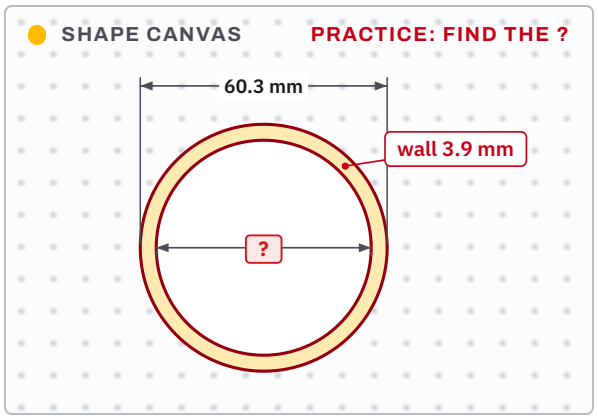
SHAPE CANVAS — PRACTICE: FIND THE ?



WORKING

Answer _____

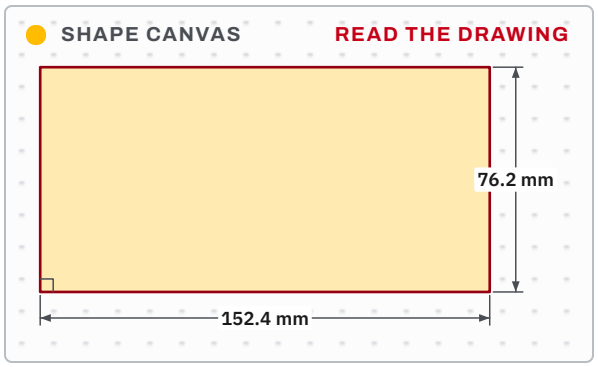
4 A pipe has an outside diameter of 60.3 mm and a wall 3.9 mm thick. Find the inside diameter.



WORKING

Answer _____

5 Find the distance around the plate by adding its four sides.



WORKING

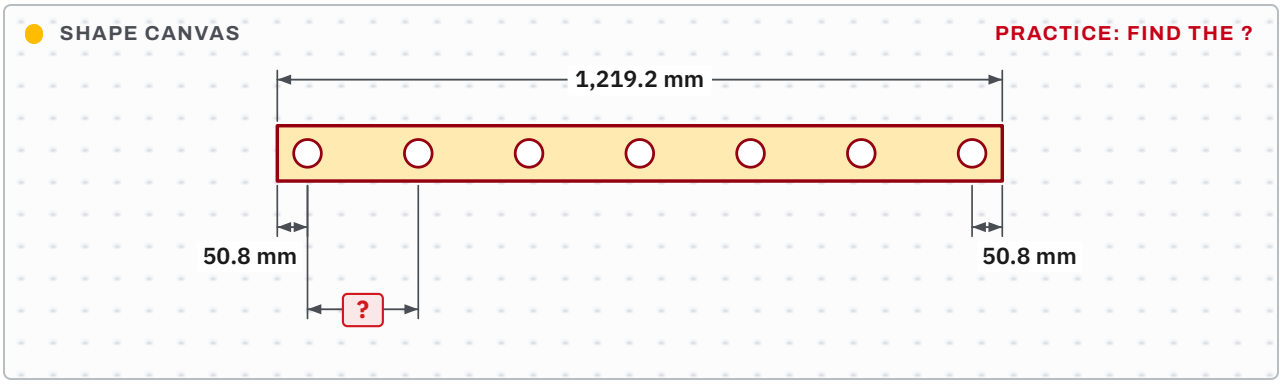
Answer _____

6 Shims of 0.8 mm, 0.5 mm, 0.25 mm, 1.5 mm are stacked. How thick is the stack? Is it more or less than 3 mm?

WORKING

Answer _____

7 Seven holes are evenly spaced along a 1,219.2 mm plate. The end holes are 50.8 mm from the ends. Find the spacing to two decimal places.



WORKING

Answer _____

WORKSHEET 5B

Decimal inches and money

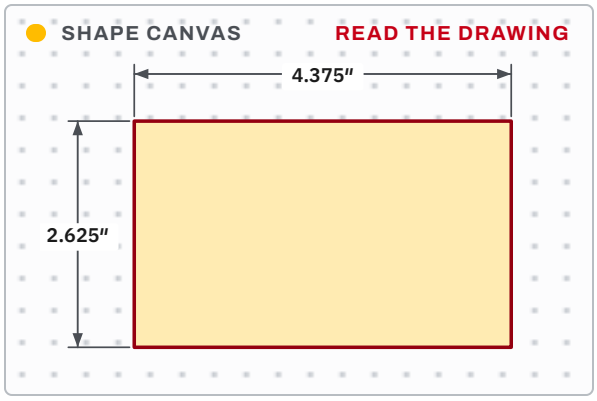
Change decimal inches to tape fractions and work out costs

Name _____

Date _____

For each tape reading, say whether you rounded to the nearest 1/16 in or the nearest 1/32 in. Round money to the nearest cent.

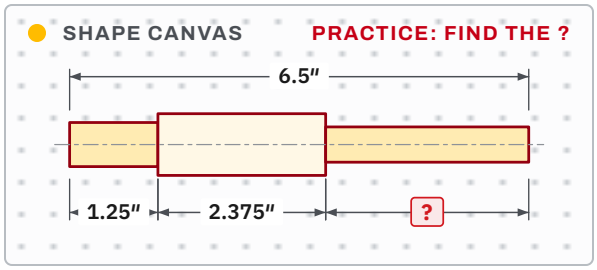
1 Write each size on the plate as a fraction in lowest terms.



WORKING

Answer _____

2 The pin is 6.5 in long overall. Find the missing step as a decimal inch, then as a fraction.



WORKING

Answer _____

3 Change 1.71 in to the nearest $\frac{1}{16}$ in and to the nearest $\frac{1}{32}$ in.

WORKING

Answer _____

4 A metric part is 75 mm long. What is that to the nearest $\frac{1}{16}$ in?

WORKING

Answer _____

5 Change $\frac{5}{16}$ in to millimetres, to two decimal places.

WORKING

Answer _____

6 Three 6 m lengths of flat bar at $4.15 per metre and 12 bolts at $0.38 each. What is the total cost?

WORKING

Answer _____

7 Change one value: a cut list has 6 pieces at 2.375 in. Find the total length as a decimal inch and as a fraction. If each piece is cut 0.125 in shorter, what is the new total?

WORKING

Answer _____

CHAPTER

6

Percent, Ratio and Proportion

Shop math is full of comparisons. A waste allowance adds 10% to an order. A coating is mixed 4 parts to 1. A pump moves so many litres per minute. A drawing shows a 2,400 mm wall as a 120 mm line.

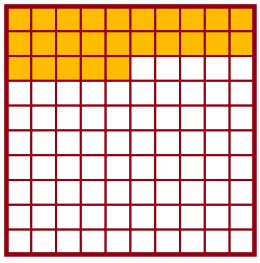
This chapter shows how percents, ratios, rates and proportions work, how they connect to the fractions and decimals you already know, and how to read a scale drawing. Percent is shown on both calculators, because the Sharp EL-531XT and the TI-30Xa use the % key in different ways.

In this chapter you will

- Change between percents, decimals and fractions
- Find a percent of a number, and find what percent one number is of another
- Add a waste allowance and work out increases and decreases
- Write and simplify ratios, and share an amount in a ratio
- Work with rates and solve proportions
- Read and make scale drawings in metric and imperial

Key words

- percent
- ratio
- rate
- proportion
- scale



25 of 100 squares
= 25%
= 0.25
= 1/4

6.1 What percent means

Percent means “out of 100”. The sign % stands for percent. 25% means 25 out of every 100. If a square is split into 100 equal parts and 25 of them are shaded, 25% of the square is shaded.

A percent is one more way to write a part of a whole. You already know two others: a **fraction** and a **decimal**. 25% = 25/100 = 0.25. All three describe the same amount.

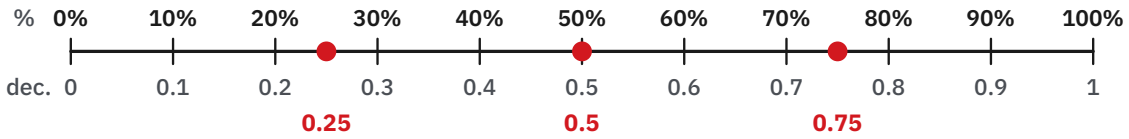


Figure 6.1 Percent and decimal on the same line. 100% is the whole, which is 1.

MOVING BETWEEN PERCENT AND DECIMAL

Percent to decimal: divide by 100, which moves the decimal point two places left. 35% = 0.35.

Decimal to percent: multiply by 100, which moves the decimal point two places right. 0.08 = 8%.

Example 6.1 Percent to decimal and fraction

Write 35%, 6.5% and 150% as decimals. Write 35% as a fraction in lowest terms.

SOLUTION

1. 35% = 35 ÷ 100 = 0.35.
2. 6.5% = 6.5 ÷ 100 = 0.065. The point moves two places left, so a zero fills the empty place.
3. 150% = 150 ÷ 100 = 1.5. A percent over 100 is more than one whole.
4. 35% = 35/100. Divide the top and bottom by 5: 35/100 = 7/20.

Answer **0.35, 0.065 and 1.5. 35% = $\frac{7}{20}$.**

Example 6.2 Fraction to percent

A crew has finished 3/8 of a pipe run. What percent is finished?

SOLUTION

1. Change the fraction to a decimal: 3 ÷ 8 = 0.375.
2. Multiply by 100 to change the decimal to a percent: 0.375 × 100 = 37.5.

CALCULATOR

TI-30Xa

3 ÷ 8 × 100 =

Display 37.5

Sharp EL-531XT

3 ÷ 8 × 100 =

Display 37.5

Answer **37.5% is finished.**

Percent	1%	5%	10%	12.5%	20%	25%	50%	75%	100%	150%
Decimal	0.01	0.05	0.1	0.125	0.2	0.25	0.5	0.75	1	1.5
Fraction	$\frac{1}{100}$	$\frac{1}{20}$	$\frac{1}{10}$	$\frac{1}{8}$	$\frac{1}{5}$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	1	$1\frac{1}{2}$

Table 6.1 Percents you will meet often, with their decimals and fractions in lowest terms.

TRY IT 6.1

1. Write 42% as a decimal.
2. Write 7.5% as a decimal.
3. Write 0.06 as a percent.
4. Write 1.25 as a percent.
5. Write 40% as a fraction in lowest terms.
6. Write $\frac{5}{8}$ as a percent.

Answers are in the Answer Key.

6.2 Finding a percent of a number

In a percent question, the word “of” means multiply. To find 15% of 240, change 15% to a decimal and multiply: 0.15 × 240.

Both calculators also have a % key. They use it in different ways. On the TI-30Xa the % key is a second function, and you finish with = (TI-30Xa guidebook, p. 4). On the Sharp EL-531XT, 2ndF % works the calculation out by itself, so there is no = at the end (Sharp EL-531XT Calculation Examples [4]).

Example 6.3 Brackets for a second job

A pallet holds 240 brackets. 15% of them are set aside for a second job. How many brackets is that?

SOLUTION

1. Change the percent to a decimal: 15% = 0.15.
2. Multiply: 0.15 × 240 = 36.

CALCULATOR

TI-30Xa

240 × 15 2nd % =

Display 36.

Sharp EL-531XT

240 × 15 2ndF %

Display 36.

With the % key, the TI-30Xa keys end with =, and the Sharp keys end with 2ndF %. The decimal method, 0.15 × 240 =, gives 36 on both calculators too.

Answer **36 brackets.**

THE SHARP % KEY ENDS THE CALCULATION

On the Sharp EL-531XT, do not press = after 2ndF %. The answer is already on the display. On the TI-30Xa, press = after % to finish.

A QUICK 10% CHECK

10% of a number is the number divided by 10: move the decimal point one place left. 10% of 240 is 24, so 15% must be about one and a half times that, 36. Use this to check your calculator answer.

TRY IT 6.2

1. 20% of 450
2. 8% of 1,250
3. 75% of 64
4. 2.5% of 3,000
5. A box of 500 bolts. 4% are rejected. How many bolts are rejected?
6. 10% of 6,096 mm

Answers are in the Answer Key.

6.3 Finding the percent and the whole

Sometimes you know the part and the whole and you want the percent. Divide the part by the whole, then multiply by 100.

PERCENT OF A WHOLE

Percent = Part ÷ Whole × 100

The part and the whole must be in the same unit.

Example 6.4 Welds that failed inspection

An inspector checks 240 welds and 18 of them fail. What percent fail?

SOLUTION

1. The part is 18 and the whole is 240.
2. 18 ÷ 240 = 0.075.
3. 0.075 × 100 = 7.5.

CALCULATOR

TI-30Xa

18 ÷ 240 2nd % =

Display 7.5

Sharp EL-531XT

18 ÷ 240 2ndF %

Display 7.5

Here the % key does the × 100. Keying 18 ÷ 240 × 100 = gives the same 7.5.

Answer **7.5% of the welds fail.**

When you know the part and the percent, you can find the whole. Change the percent to a decimal and divide the part by it. In Example 6.3, 36 brackets were 15% of the pallet. If you did not know the pallet size, you could find it: 36 ÷ 0.15 = 240 brackets.

THREE QUESTIONS, ONE RELATIONSHIP

Part = decimal × Whole. Find the part by multiplying. Find the decimal by dividing the part by the whole. Find the whole by dividing the part by the decimal.

TRY IT 6.3

1. 45 is what percent of 180?
2. 12 is what percent of 400?
3. 9 of 60 parts are scrap. What percent is scrap?
4. 30 is 25% of what number?
5. 18 is 6% of what number?
6. A 750 mm piece is cut from a 3,000 mm bar. What percent of the bar is the piece?

Answers are in the Answer Key.

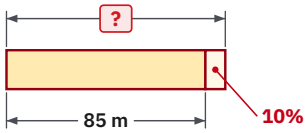
6.4 Waste allowance, increase and decrease

Some material is always lost to cutting, fitting and mistakes. A waste allowance is extra material added to an order to cover that loss. A 10% waste allowance means you order the amount you need plus 10% of it.

Adding a percent is a percent increase. Taking a percent away is a percent decrease. You can do either in two steps (find the percent, then add or subtract it) or in one step by multiplying. To add 10%, multiply by 1.10, which is 110%. To take off 30%, multiply by 0.70, which is 70%.

Example 6.5 Ordering pipe with a waste allowance

A job needs 85 m of pipe. Add a 10% waste allowance. The pipe is sold in 6 m lengths. How many lengths do you order?



SOLUTION

1. 10% of 85 = 0.1 × 85 = 8.5 m.
2. Add it on: 85 + 8.5 = 93.5 m.
3. Lengths: 93.5 ÷ 6 = 15.5833…
4. A part length still has to be bought, so round up to 16.

CALCULATOR

TI-30Xa

85 + 10 2nd % =

Display 93.5

Sharp EL-531XT

85 + 10 2ndF %

Display 93.5

Pressing + and then % adds 10% of 85 to 85 on both calculators (TI-30Xa guidebook, p. 4, and Sharp EL-531XT Calculation Examples [4]). In one step without the % key: 85 × 1.1 = 93.5.

Answer **93.5 m of pipe, which is 16 lengths.**

Example 6.6 A sale price

A tool is priced at $400.00. It is on sale at 30% off. What is the sale price?

SOLUTION

1. 30% of $400.00 = 0.3 × 400 = $120.00.
2. Subtract: $400.00 − $120.00 = $280.00. In one step: 0.7 × 400 = 280.

CALCULATOR

TI-30Xa

400 | − | 30 | 2nd | % | =

Display: 280.

Sharp EL-531XT

400 | − | 30 | 2ndF | %

Display: 280.

Pressing − and then % takes 30% of 400 away from 400 on both calculators (TI-30Xa guidebook, p. 4, and Sharp EL-531XT Calculation Examples [4]).

Answer **The sale price is $280.00.**

To find a percent change, find the amount of the change, divide it by the original amount (the amount before the change), and multiply by 100.

PERCENT CHANGE

$$\text{Percent change} = (\text{New} - \text{Original}) \div \text{Original} \times 100$$

A positive answer is an increase. A negative answer is a decrease.

Example 6.7 A pay raise

An apprentice's wage goes from $24.00 to $25.20 an hour. What is the percent increase?

SOLUTION

1. Change: $25.20 − $24.00 = $1.20.
2. Divide by the original wage: 1.20 ÷ 24.00 = 0.05.
3. Multiply by 100: 5%.

CALCULATOR

TI-30Xa

(25.2 − 24) ÷ 24 × 100 =

Display 5.

Sharp EL-531XT

(25.2 − 24) ÷ 24 × 100 =

Display 5.

The brackets make both calculators find the change before dividing.

Answer **A 5% increase.**

UP 10% AND DOWN 10% IS NOT A ROUND TRIP

Start with 200. Up 10% gives 200 × 1.1 = 220. Down 10% from there gives 220 × 0.9 = 198, not 200. Each percent is taken of a different starting amount.

TRY IT 6.4

1. Add a 15% waste allowance to 60 m of cable.
2. You need 24 sheets. Add a 10% allowance and round up to whole sheets.
3. A $250.00 price is cut by 20%. New price?
4. A 1,200 mm piece grows by 5%. New length?
5. A price rises from $80.00 to $92.00. Percent increase?
6. A tank level drops from 1,500 L to 1,200 L. Percent decrease?

Answers are in the Answer Key.

6.5 Ratio

A **ratio** compares two amounts. A coating mixed 4 parts base to 1 part hardener has a ratio of base to hardener of 4 to 1. You can write it three ways: 4 : 1, 4 to 1 or 4/1.

Order matters. The ratio of hardener to base is 1 : 4, not 4 : 1. A ratio is simplified the same way as a fraction: divide both numbers by the same number until they share no common factor. Before you compare two amounts, put them in the same unit.

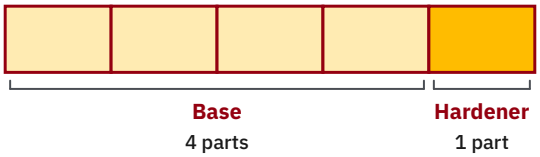
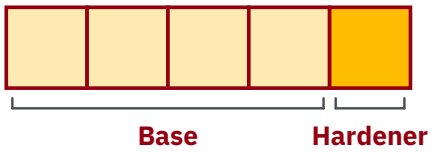


Figure 6.2 A mix of 4 : 1. Every part is the same size.

For example, to compare a 750 mm piece with a 2.5 m piece, first write 2.5 m as 2,500 mm. The ratio 750 : 2,500 simplifies to 3 : 10 when you divide both numbers by 250.

Example 6.8 Sharing an amount in a ratio

You need 750 mL of mixed coating. The label gives a mix of 4 parts base to 1 part hardener. How much of each do you measure?



SOLUTION

1. Add the parts: 4 + 1 = 5 parts.
2. One part: 750 ÷ 5 = 150 mL.
3. Base: 4 × 150 = 600 mL. Hardener: 1 × 150 = 150 mL.
4. Check: 600 + 150 = 750 mL.

CALCULATOR

TI-30Xa

750 ÷ 5 × 4 =

Display 600.

Sharp EL-531XT

750 ÷ 5 × 4 =

Display 600.

Answer **600 mL of base and 150 mL of hardener.**

TRY IT 6.5

1. Write 12 : 18 in lowest terms.
2. Write 48 teeth to 16 teeth in lowest terms.
3. Write 400 mm to 1.2 m in lowest terms.
4. Share 900 mL in the ratio 2 : 1.
5. Share 2 L in the ratio 3 : 1. Give the answers in mL.
6. A crew has 3 journeypersons and 5 apprentices. Ratio of apprentices to journeypersons?

Answers are in the Answer Key.

6.6 Rate

A **rate** compares two amounts in different units: litres per minute, dollars per metre, millimetres per minute. A unit rate is the amount for one unit, such as 150 L in one minute. Find it by dividing.

Example 6.9 Pump rate

A pump moves 1,200 L in 8 minutes. What is its rate? At that rate, how long does it take to fill a 2,700 L tank?

SOLUTION

1. Rate: 1,200 ÷ 8 = 150 L per minute.
2. Time: 2,700 ÷ 150 = 18 minutes.

CALCULATOR

TI-30Xa

Rate

1200 ÷ 8 =

Display 150.

Time

2700 ÷ 150 =

Display 18.

Sharp EL-531XT

Rate

1200 ÷ 8 =

Display 150.

Time

2700 ÷ 150 =

Display 18.

Answer **150 L per minute. The tank takes 18 minutes.**

TRY IT 6.6

1. 450 mm of weld in 3 minutes. Rate in mm per minute?
2. 96 fittings in 8 hours. Fittings per hour?
3. A pump moves 75 L per minute. How long for 1,800 L?
4. Supplier A: 12 m of cable for $86.40. Supplier B: 20 m for $138.00. Which costs less per metre?
5. At $7.25 per metre, what do 18 m cost?
6. A drill bit cuts 120 holes in 40 minutes. Holes per minute?

Answers are in the Answer Key.

6.7 Proportion

A **proportion** says that two ratios are equal. 1/4 = 5/20 is a proportion, because both ratios simplify to 1 : 4. In a proportion the cross products are equal: 1 × 20 = 4 × 5 = 20.

When one of the four numbers is missing, set up the two ratios with the same units in the same places, then solve for the missing number.

SOLVING A PROPORTION

$$a \div b = c \div x$$

$$x = b \times c \div a$$

Multiply the two numbers that sit across from each other, then divide by the third.

Example 6.10 Welding time for a larger order

A welder finishes 8 brackets in 2.5 hours. At the same rate, how long do 20 brackets take?

SOLUTION

1. Write the proportion with brackets on top and hours below: 8/2.5 = 20/x.
2. Cross multiply: 8 × x = 2.5 × 20 = 50.
3. Divide: x = 50 ÷ 8 = 6.25 hours.
4. 6.25 hours is 6 hours and 0.25 × 60 = 15 minutes.

CALCULATOR

TI-30Xa

2.5 × 20 ÷ 8 =

Display 6.25

Sharp EL-531XT

2.5 × 20 ÷ 8 =

Display 6.25

Answer **6.25 hours, which is 6 hours 15 minutes.**

KEEP THE UNITS LINED UP

If brackets are on top in the first ratio, brackets go on top in the second ratio too. Mixing them up gives an answer that is far too big or far too small. Check it with common sense: more brackets must take more time.

TRY IT 6.7

1. Is 3/4 = 15/20 a proportion?
2. Solve 2/5 = x/45.
3. Solve 6/x = 9/24.
4. 5 brackets use 3 m of flat bar. How much bar do 35 brackets use?
5. A crew lays 18 m of pipe in 4 hours. How long for 45 m?
6. A mix is 3 : 1. You have 1.8 L of the first part. How much of the second part?

Answers are in the Answer Key.

6.8 Scale drawings

A drawing of a large part is smaller than the part. The **scale** is the ratio of a length on the drawing to the real length. A scale of 1 : 20 means 1 mm on the drawing stands for 20 mm on the real part.

Drawing to real: multiply by the second number. Real to drawing: divide by it. A larger second number gives a smaller drawing. A scale such as 2 : 1 makes the drawing larger than the part, which is useful for small details.

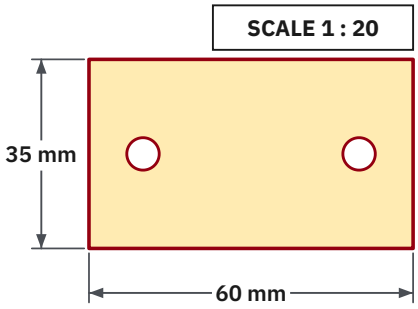
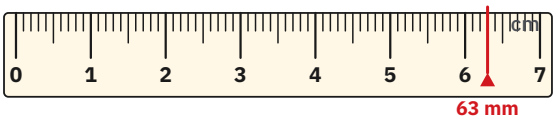


Figure 6.3 A plate drawn at 1 : 20. The sizes shown are the lengths measured on the paper.

The plate above measures 60 mm by 35 mm on the paper, so the real plate is 60 × 20 = 1,200 mm long and 35 × 20 = 700 mm wide.

Example 6.11 Measuring a drawing

A line on a drawing at 1 : 50 measures 63 mm with a rule, as shown. How long is the real part?



SOLUTION

1. Drawing to real, so multiply by 50.
2. 63 × 50 = 3,150 mm.

CALCULATOR

TI-30Xa

63 × 50 =

Display 3150.

Sharp EL-531XT

63 × 50 =

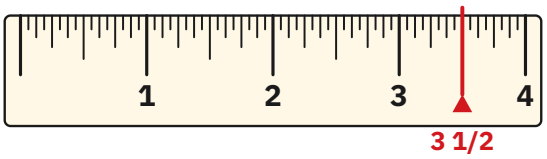
Display 3′150.

Answer **3,150 mm, which is 3.15 m.**

A scale can also be written as two lengths. A drawing that states its scale as 1/4″ = 1′-0″ uses 1/4 inch on the drawing for 1 foot on the real job. Since 1 ft = 12 in, that ratio is 1/4 : 12, which is 1 : 48.

Example 6.12 An imperial scale

A drawing is marked 1/4″ = 1′-0″. A line measures 3 1/2 in, as shown. How long is the real length?



SOLUTION

1. Count how many quarter inches are in the line: 3.5 ÷ 0.25 = 14.
2. Each quarter inch is 1 ft, so the real length is 14 ft.

CALCULATOR

TI-30Xa

3.5 ÷ 0.25 =

Display 14.

Sharp EL-531XT

3.5 ÷ 0.25 =

Display 14.

Answer **14 ft**

TRY IT 6.8

1. At 1 : 10, a line measures 48 mm. Real length?
2. At 1 : 25, a line measures 72 mm. Real length?
3. A 3,000 mm beam is drawn at 1 : 50. Drawing length?
4. A 12 mm part is drawn at 5 : 1. Drawing length?
5. A 1,500 mm part is drawn 75 mm long. What is the scale?
6. At 1/4″ = 1′-0″, a line measures 2 3/4 in. Real length?

Answers are in the Answer Key.

WORKSHEET 6A

Percent on the canvas

Find each red ?

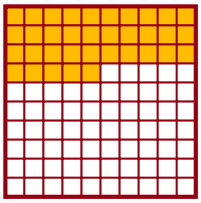
Name _____

Date _____

Show the decimal you multiply or divide by. Check each answer with a 10% estimate.

1 Write the shaded part as a percent, a decimal and a fraction in lowest terms.

SHAPE CANVAS — READ THE DRAWING

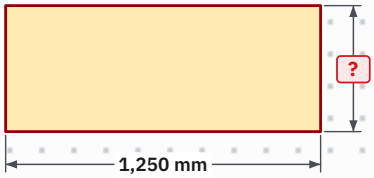


WORKING

Answer _____

2 The width is 40% of the length. Find the width.

SHAPE CANVAS — PRACTICE: FIND THE ?

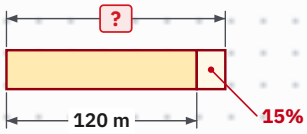


WORKING

Answer _____

3 A run needs 120 m of conduit. Add a 15% waste allowance. How much do you order?

SHAPE CANVAS — PRACTICE: FIND THE ?

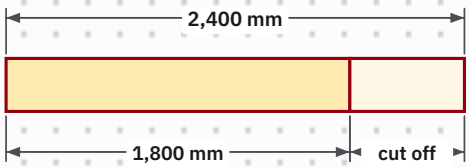


WORKING

Answer _____

4 A 2,400 mm bar is cut down to 1,800 mm. What is the percent decrease?

SHAPE CANVAS — READ THE DRAWING



WORKING

Answer _____

5 Change one value: a 750 mm piece is made 20% longer. Then the new piece is made 20% shorter. Find both lengths. Is it back to 750 mm?

WORKING

Answer _____

WORKSHEET 6B

Ratio, proportion and scale

Mixes, similar shapes and scale drawings

Name _____

Date _____

Write each ratio or proportion before you calculate. Keep the same units in the same places.

1 A coating is mixed as shown. How many mL of base and of hardener make 2.4 L of mix?

SHAPE CANVAS — READ THE DRAWING

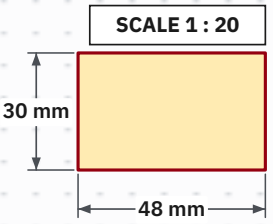


WORKING

Answer _____

2 The sizes shown are measured on a 1 : 20 drawing. Find the real length and width.

SHAPE CANVAS — READ THE DRAWING

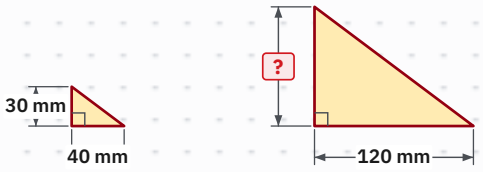


WORKING

Answer _____

3 Triangle B is triangle A made larger, so their sides are in proportion. Find the height of B.

SHAPE CANVAS — PRACTICE: FIND THE ?



WORKING

Answer _____

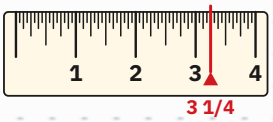
4 Change one value: the plate in Problem 2 is redrawn at 1 : 10. How long is its 960 mm side on the new drawing? What happens to every line?

WORKING

Answer _____

5 At 1/4″ = 1′-0″, find the real length of this line.

SHAPE CANVAS — READ THE DRAWING



WORKING

Answer _____

6 A pump fills 1,500 L in 12 minutes. Find its rate, and the time to fill 2,250 L.

WORKING

Answer _____

CHAPTER

7

Measurement and Unit Conversion

A metric drawing, an inch tape, a pump rated in US gallons and a heater set in °F can all turn up on the same job. Before you can add, compare or cut, every number has to be in the same unit.

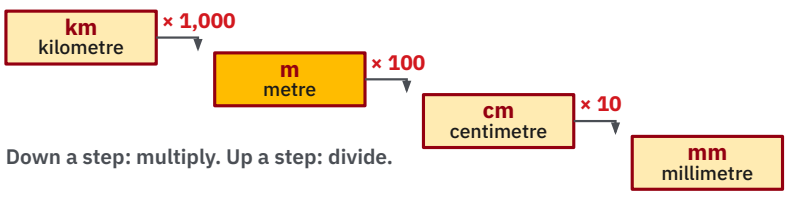
This chapter covers the metric system and its prefixes, imperial units and feet-and-inches, and how to convert between them with exact conversion factors. It finishes with units of area and volume, litres and gallons, and temperature. Every factor in this chapter is exact or is taken from a published source, and the source is given with each table.

In this chapter you will

- Use SI units and the prefixes kilo, centi and milli
- Work with inches, feet and yards, and with feet and inches
- Convert between metric and imperial with exact factors
- Convert units of area and volume
- Convert between litres, US gallons and imperial gallons
- Convert temperatures between °C and °F

Key words

- unit
- metric system
- SI
- prefix
- imperial system
- conversion factor



7.1 Units and the metric system

A **unit** is a fixed amount used to measure something: a metre, an inch, a litre. A measurement is always a number and a unit. “2,400” on its own means nothing on the job. 2,400 mm and 2,400 in are very different lengths.

The **metric system** is built on powers of ten. Its modern form is the **SI**, the International System of Units. The SI base units include the metre (m) for length, the kilogram (kg) for mass and the second (s) for time (NIST SP 1038, section 4.1.1). The litre (L) is accepted for use with the SI for volume (NIST SP 1038, section 4.1.2 and the note to the volume table in section 5.1.5).

A **prefix** in front of a unit multiplies it by a power of ten. Kilo means 1,000, so 1 km = 1,000 m. Milli means 1/1,000, so 1 mm = 1/1,000 m.

Prefix	Symbol	Multiplies the unit by	Example
kilo	k	1,000	1 km = 1,000 m
hecto	h	100	1 hL = 100 L
deca	da	10	1 dam = 10 m
deci	d	0.1	1 dm = 0.1 m
centi	c	0.01	1 cm = 0.01 m
milli	m	0.001	1 mm = 0.001 m
micro	μ	0.000 001	1 μm = 0.000 001 m

Table 7.1 SI prefixes. Source: Weights and Measures Act (Canada), Schedule I, Part V, and NIST SP 1038, Table 2. NIST notes that deci, deka (deca) and hecto are rarely used.

Each step on the ladder in the chapter opener is a power of ten, so a metric conversion only moves the decimal point. Going to a smaller unit, the number gets bigger, so multiply. Going to a bigger unit, the number gets smaller, so divide.

Example 7.1 Metric conversions

Convert: (a) 2.45 m to mm, (b) 375 mm to m, (c) 1,250 mL to L, (d) 0.85 t to kg.

SOLUTION

1. (a) 1 m = 1,000 mm. Smaller unit, so multiply: 2.45 × 1,000 = 2,450 mm. The point moves 3 places right.
2. (b) Bigger unit, so divide: 375 ÷ 1,000 = 0.375 m. The point moves 3 places left.
3. (c) 1 L = 1,000 mL. 1,250 ÷ 1,000 = 1.25 L.
4. (d) 1 t (tonne) = 1,000 kg (Weights and Measures Act, Schedule I, Part IV). 0.85 × 1,000 = 850 kg.

Answer **2,450 mm, 0.375 m, 1.25 L and 850 kg.**

TRY IT 7.1

1. 3.6 m to mm
2. 850 mm to m
3. 42 cm to mm
4. 6,096 mm to m
5. 2.5 L to mL
6. 1,800 kg to t

Answers are in the Answer Key.

7.2 Imperial units, feet and inches

The **imperial system** uses the inch, foot, yard, pound and gallon. Canadian law defines the yard as 0.9144 m, the foot as 1/3 yard and the inch as 1/36 yard (Weights and Measures Act, Schedule II). The US defines the same inch, foot and yard (NIST SP 1038), so a US tape and a Canadian tape agree. Table 7.2 lists the length units.

Lengths on the job are often written in feet and inches, such as 7′ 9 1/2″ (7 feet 9 1/2 inches). To work with them on a calculator, change them to inches first. Multiply the feet by 12 and add the inches.

For example, 7′ 9 1/2″ is 7 × 12 + 9.5 = 93.5 in.

Example 7.2 Inches to feet and inches

A cut list gives a length of 142 3/4 in. Write it in feet and inches.

SOLUTION

1. Divide by 12: 142.75 ÷ 12 = 11.8958... The whole number part is 11 ft.
2. Inches left over: 142.75 − 11 × 12 = 10.75 in.
3. 10.75 in is 10 3/4 in.

CALCULATOR

TI-30Xa

142.75 ÷ 12 =

Display 11.89583333

Sharp EL-531XT

142.75 ÷ 12 =

Display 11.89583333

Only the whole number part, 11, is feet. Find the inches left with 142.75 − 11 × 12 =, which shows 10.75.

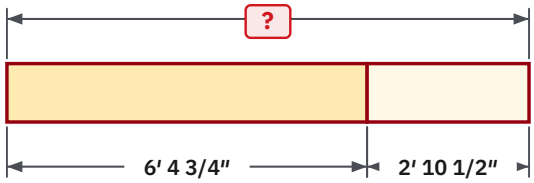
Answer $11' \; 10\frac{3}{4}''$

Unit	Symbol	Equals
inch	in or ″	1/36 yard
foot	ft or ′	12 in = 1/3 yard
yard	yd	3 ft = 36 in
mile	mi	1,760 yd = 5,280 ft

Table 7.2 Imperial length units. Source: Weights and Measures Act (Canada), Schedule II, Measurement of Length.

Example 7.3 Adding feet and inches

Two pieces of 6′ 4 3/4″ and 2′ 10 1/2″ are laid end to end. Find the total length.



SOLUTION

1. Add the feet: 6 + 2 = 8 ft. Add the inches: 4 3/4 + 10 1/2 = 15 1/4 in.
2. 15 1/4 in is more than a foot: 15 1/4 − 12 = 3 1/4 in, and carry 1 ft.
3. Total: 8 + 1 = 9 ft and 3 1/4 in.
4. Check in inches: 76.75 + 34.5 = 111.25 in = 9′ 3 1/4″.

CALCULATOR

TI-30Xa

76.75 + 34.5 =

Display 111.25

Sharp EL-531XT

76.75 + 34.5 =

Display 111.25

Answer $9' \, 3\frac{1}{4}''$

TRY IT 7.2

1. Write 5′ 7″ in inches.
2. Write 10′ 2 1/4″ in inches.
3. Write 86 in in feet and inches.
4. Write 125 1/2 in in feet and inches.
5. How many feet are in 4 yd?
6. Add 3′ 8 1/2″ and 4′ 7 3/4″.

Answers are in the Answer Key.

7.3 Converting with exact factors

A **conversion factor** is the number you multiply or divide by to change units. The inch is defined as exactly 25.4 mm, so 25.4 is the factor between inches and millimetres. It is exact, not rounded, and it gives the same answer on every job.

INCHES AND MILLIMETRES

$$\text{mm} = \text{in} \times 25.4$$

$$\text{in} = \text{mm} \div 25.4$$

1 in = 25.4 mm exactly (NIST SP 1038, section 5.1.3). It also follows from the Weights and Measures Act: 0.9144 m ÷ 36 = 0.0254 m.

From	To	Multiply by	Exact?
inch (in)	millimetre (mm)	25.4	exact
inch (in)	centimetre (cm)	2.54	exact
foot (ft)	metre (m)	0.3048	exact
yard (yd)	metre (m)	0.9144	exact
mile (mi)	kilometre (km)	1.609 344	exact
pound (lb)	kilogram (kg)	0.453 592 37	exact

Table 7.3 Length and mass factors. Source: NIST SP 1038, sections 5.1.3 and 5.2, which prints exact factors in bold. The yard and the pound are also defined this way in the Weights and Measures Act (Canada), Schedule II. To go the other way, divide. Long decimals are printed in groups of three digits, as in the source. Key them without the spaces.

Example 7.4 A fraction of an inch in millimetres

A drawing calls for 3/8 in plate. What is that in millimetres?

SOLUTION

1. Inches to millimetres: multiply by 25.4.
2. 3/8 × 25.4 = 0.375 × 25.4 = 9.525 mm.

CALCULATOR

TI-30Xa

3 | a b/c | 8 | × | 25.4 | =

Display: 9.525

Sharp EL-531XT

3 | a b/c | 8 | × | 25.4 | =

Display: 9.525

The fraction key keys 3/8 directly. Keying 3 ÷ 8 × 25.4 = gives the same result.

Answer 9.525 mm

Example 7.5 Millimetres to the nearest sixteenth

A metric drawing shows 1,480 mm. You have an inch tape. What do you mark, to the nearest 1/16 in?

SOLUTION

1. Divide by 25.4: 1,480 ÷ 25.4 = 58.2677… in.
2. Find the sixteenths: multiply by 16, 58.2677 × 16 = 932.28…, which rounds to 932 sixteenths.
3. 932 ÷ 16 = 58.25 in, which is 58 4/16 = 58 1/4 in.

CALCULATOR

TI-30Xa

Inches

1480 ÷ 25.4 =

Display 58.26771654

Sixteenths

1480 ÷ 25.4 × 16 =

Display 932.2834646

Sharp EL-531XT

Inches

1480 ÷ 25.4 =

Display 58.26771654

Sixteenths

1480 ÷ 25.4 × 16 =

Display 932.2834646

Answer **Mark $58\frac{1}{4}$″ (that is 4′ $10\frac{1}{4}$″).**

Example 7.6 Feet and inches to millimetres

A pipe run is 7′ 9 1/2″ long. How long is it in millimetres?

SOLUTION

1. Change to inches first, as in Section 7.2: 7 × 12 + 9.5 = 93.5 in.
2. Multiply by 25.4: 93.5 × 25.4 = 2,374.9 mm.

CALCULATOR

TI-30Xa

(7 × 12 + 9.5) × 25.4 =

Display 2374.9

Sharp EL-531XT

(7 × 12 + 9.5) × 25.4 =

Display 2′374.9

The brackets make both calculators finish the inches before multiplying by 25.4.

Answer **2,374.9 mm, which is about 2.37 m.**

Example 7.7 Pounds to kilograms

A bag of grout is marked 50 lb. What is its mass in kilograms, to two decimal places?

SOLUTION

1. Pounds to kilograms: multiply by 0.453 592 37.
2. 50 × 0.453 592 37 = 22.6796185 kg.
3. Round to two decimal places: 22.68 kg.

CALCULATOR

TI-30Xa

50 × 0.45359237 =

Display 22.6796185

Sharp EL-531XT

50 × 0.45359237 =

Display 22.6796185

Answer **22.68 kg**

MULTIPLY OR DIVIDE?

A millimetre is much smaller than an inch, so a length in millimetres is always the bigger number. If you convert 3/8 in and get 0.015 mm, you divided when you should have multiplied. Check every answer this way.

TRY IT 7.3

1. 12 in to mm
2. 5/16 in to mm
3. 20 ft to m
4. 762 mm to inches
5. 2,000 mm to inches, to the nearest 1/16 in
6. 110 lb to kg, to two decimal places

Answers are in the Answer Key.

7.4 Units of area and volume

Area is measured in square units and volume in cubic units. A **square unit** is a square 1 unit on each side. A **cubic unit** is a cube 1 unit on each side. Because a square has two sides that are measured, the length factor is used twice for area. For volume it is used three times.

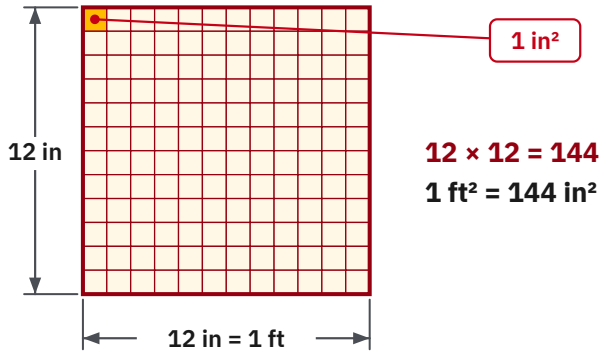


Figure 7.1 One square foot is a square 12 in on each side. It holds 12 rows of 12 square inches.

Area	Volume
1 m² = 1,000 × 1,000 = 1,000,000 mm²	1 m³ = 1,000 L
1 ft² = 12 × 12 = 144 in²	1 L = 1,000,000 mm³ = 1,000 cm³
1 in² = 645.16 mm² (exact)	1 ft³ = 12 × 12 × 12 = 1,728 in³
1 ft² = 0.092 903 04 m² (exact)	1 in³ = 16.387 064 cm³ (exact)

Table 7.4 Area and volume factors. Sources: NIST SP 1038, sections 5.1.4 and 5.1.5, for the square inch, square foot and cubic inch. The Weights and Measures Act (Canada), Schedule I, Part IV, for the litre (1/1,000 m³), and Schedule II for 1 in² = 1/144 ft² and 1 in³ = 1/1,728 ft³.

Example 7.8 A plate in square metres

A plate is 2,400 mm by 1,200 mm. Find its area in mm² and in m².

SOLUTION

1. Area = length × width = 2,400 × 1,200 = 2,880,000 mm².
2. 1 m² = 1,000,000 mm², so divide: 2,880,000 ÷ 1,000,000 = 2.88 m².
3. Check with metres first: 2.4 m × 1.2 m = 2.88 m².

CALCULATOR

TI-30Xa

2400 × 1200 ÷ 1000000 =

Display 2.88

Sharp EL-531XT

2400 × 1200 ÷ 1000000 =

Display 2.88

Answer 2,880,000 mm² = 2.88 m²

Example 7.9 A sheet in square feet and square metres

A sheet is 4 ft by 8 ft. Find its area in ft², then in m² to two decimal places.

SOLUTION

1. Area = 4 × 8 = 32 ft².
2. Multiply by 0.092 903 04: 32 × 0.092 903 04 = 2.97289728 m².
3. Round: 2.97 m².

CALCULATOR

TI-30Xa

32 × 0.09290304 =

Display 2.97289728

Sharp EL-531XT

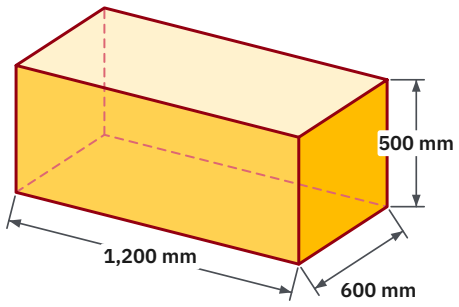
32 × 0.09290304 =

Display 2.97289728

Answer **32 ft² = 2.97 m²**

Example 7.10 Litres in a tank

A rectangular tank measures 1,200 mm × 600 mm × 500 mm inside. How many litres does it hold?



SOLUTION

1. Volume = 1,200 × 600 × 500 = 360,000,000 mm³.
2. 1 L = 1,000,000 mm³, so divide:
   360,000,000 ÷ 1,000,000 = 360 L.

CALCULATOR

TI-30Xa

1200 × 600 × 500 ÷ 1000000 =

Display 360.

Sharp EL-531XT

1200 × 600 × 500 ÷ 1000000 =

Display 360.

Answer **360 L**

DO NOT USE 1,000 FOR SQUARE MILLIMETRES

1 m = 1,000 mm, but 1 m² = 1,000,000 mm². A square metre is 1,000 mm long and 1,000 mm wide, so it holds 1,000 × 1,000 square millimetres.

TRY IT 7.4

1. 3 m² to mm²
2. 450,000 mm² to m²
3. 5 ft² to in²
4. 10 in² to mm²
5. A tank 1,000 mm × 800 mm × 600 mm. Litres?
6. 2 ft³ to in³

Answers are in the Answer Key.

7.5 Litres and gallons

There are two gallons, and they are not the same size. The US gallon is 231 in³ (NIST SP 1038, section 3.2). Since 1 in³ = 16.387 064 cm³ exactly, the US gallon is 231 × 16.387 064 = 3,785.411 784 cm³, which is exactly 3.785 411 784 L. The imperial gallon is defined in the Weights and Measures Act (Canada), Schedule II, as 454 609/100 000 000 m³, which is exactly 4.546 09 L.

Gallon	Litres (exact)	Source
US gallon (US gal)	3.785 411 784	NIST SP 1038: 231 in³, and 1 in³ = 16.387 064 cm³
Imperial gallon (imp gal)	4.546 09	Weights and Measures Act (Canada), Schedule II

Table 7.5 The two gallons. The volume table in NIST SP 1038 (section 5.1.5) rounds the US gallon to 3.785 412 L.

Example 7.11 US gallons to litres

A drum holds 55 US gal. How many litres is that, to two decimal places?

SOLUTION

1. Multiply by 3.785 411 784: 55 × 3.785 411 784 = 208.197648… L.
2. Round: 208.20 L.

CALCULATOR

TI-30Xa

55 × 3.785411784 =

Display 208.1976481

Sharp EL-531XT

55 × 3.785411784 =

Display 208.1976481

Answer 208.20 L

Example 7.12 Imperial gallons, litres and US gallons

An older tank is marked 40 imp gal. How many litres does it hold? How many US gallons? Round to two decimal places.

SOLUTION

1. Imperial gallons to litres: 40 × 4.546 09 = 181.8436 L.
2. Litres to US gallons: divide by 3.785 411 784. 181.8436 ÷ 3.785 411 784 = 48.038…
3. Round: 181.84 L and 48.04 US gal.

CALCULATOR

TI-30Xa

Litres

40 × 4.54609 =

Display 181.8436

US gallons

40 × 4.54609 ÷ 3.785411784 =

Display 48.03799702

Sharp EL-531XT

Litres

40 × 4.54609 =

Display 181.8436

US gallons

40 × 4.54609 ÷ 3.785411784 =

Display 48.03799702

Answer **181.84 L, which is 48.04 US gal. The imperial gallon is the larger gallon.**

TRY IT 7.5

1. 10 US gal to L, to two decimal places
2. 10 imp gal to L, to two decimal places
3. The 360 L tank in Example 7.10, in US gal to two decimal places
4. 1 imp gal in US gal, to three decimal places
5. 1 ft³ in US gal, to two decimal places (1 ft³ = 1,728 in³)
6. 500 L to imp gal, to two decimal places

Answers are in the Answer Key.

7.6 Temperature

Temperature is measured in degrees Celsius (°C) in the metric system and degrees Fahrenheit (°F) in the imperial system. The two scales are linked by an exact formula. NIST SP 1038 (section 5.3, note 33) gives °C = (°F − 32) ÷ 1.8, and turned around it gives °F = 1.8 × °C + 32. Both are exact definitions, so round only the final answer.

CELSIUS AND FAHRENHEIT

$$°F = 1.8 \times °C + 32$$

$$°C = (°F - 32) \div 1.8$$

Exact. Follow BEDMAS: in the first formula multiply before you add, and in the second subtract inside the brackets before you divide.

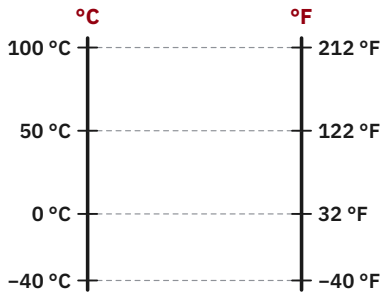


Figure 7.2 The two scales side by side. Each °F mark comes from °F = 1.8 × °C + 32. At −40 the two scales read the same number.

Example 7.13 Celsius to Fahrenheit

A water heater is set to 60 °C. What is that in °F?

SOLUTION

1. °F = 1.8 × 60 + 32. Multiply first: 1.8 × 60 = 108.
2. Then add: 108 + 32 = 140.

CALCULATOR

TI-30Xa

1.8 × 60 + 32 =

Display 140.

Sharp EL-531XT

1.8 × 60 + 32 =

Display 140.

Answer 140 °F

Example 7.14 Fahrenheit to Celsius

An oven for drying electrodes is set to 350 °F. What is that in °C, to one decimal place?

SOLUTION

1. °C = (350 − 32) ÷ 1.8. Brackets first: 350 − 32 = 318.
2. Divide: 318 ÷ 1.8 = 176.6667..., which rounds to 176.7.

CALCULATOR

TI-30Xa

(350 − 32) ÷ 1.8 =

Display 176.6666667

Sharp EL-531XT

(350 − 32) ÷ 1.8 =

Display 176.6666667

Without the brackets, both calculators divide 32 by 1.8 first and give a wrong answer.

Answer **176.7 °C**

A CHANGE IN TEMPERATURE HAS NO 32

A rise or drop is converted with the 1.8 alone: a change of 1 °C is a change of 1.8 °F (NIST SP 1038, note 33). A part that warms up by 25 °C warms up by 25 × 1.8 = 45 °F.

TRY IT 7.6

1. 20 °C to °F
2. −10 °C to °F
3. 212 °F to °C
4. 50 °F to °C
5. 400 °F to °C, to one decimal place
6. A temperature drops by 15 °C. How many °F is that drop?

Answers are in the Answer Key.

WORKSHEET 7A

Convert on the canvas

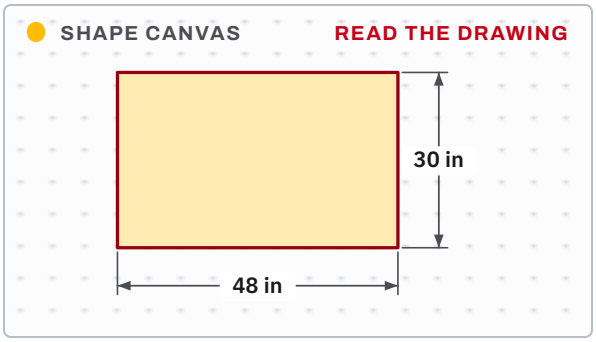
Lengths and masses in both systems

Name _____

Date _____

Write the factor you use on every line. Use 1 in = 25.4 mm and 1 lb = 0.453 592 37 kg.

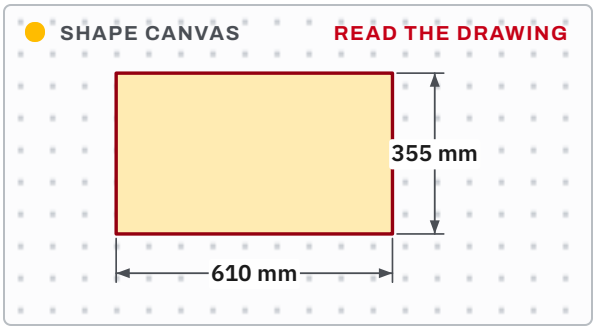
1 The sheet is sized in inches. Write both sizes in millimetres.



WORKING

Answer _____

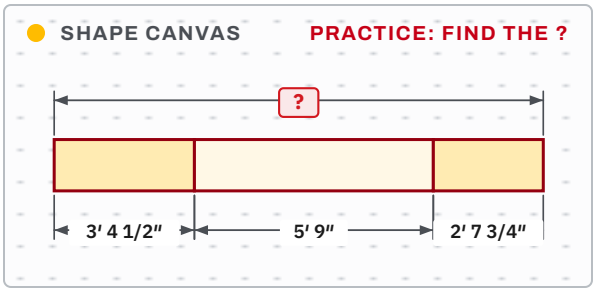
2 The plate is sized in millimetres. Write both sizes in inches, to the nearest 1/16 in.



WORKING

Answer _____

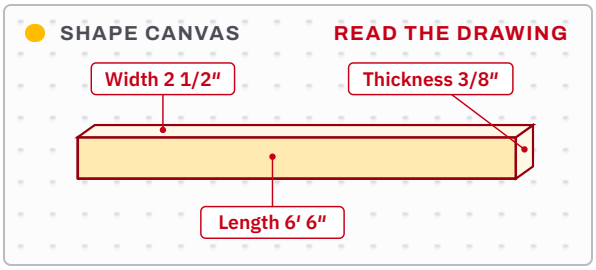
3 Find the total length in feet and inches, then in millimetres.



WORKING

Answer _____

4 Convert each size on the flat bar to millimetres.



WORKING

Answer _____

5 Three bags of 50 lb are loaded on a cart. What is their total mass in kilograms, to two decimal places?

WORKING

Answer _____

6 A drawing gives 2,400 mm. Write it in metres, in centimetres, and in feet and inches to the nearest 1/16 in.

WORKING

Answer _____

WORKSHEET 7B

Area, volume, capacity and temperature

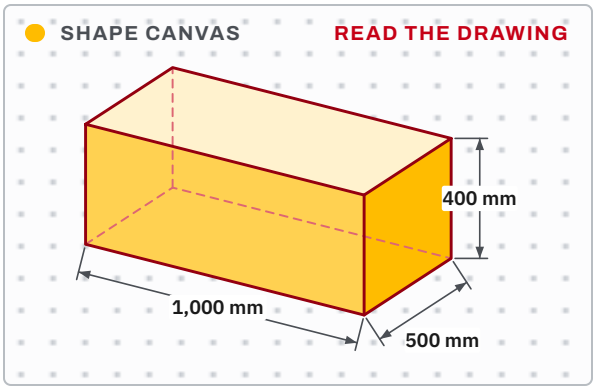
Square and cubic units, gallons and degrees

Name _____

Date _____

Use 1 L = 1,000,000 mm³, 1 US gal = 231 in³ and 1 ft² = 0.092 903 04 m².

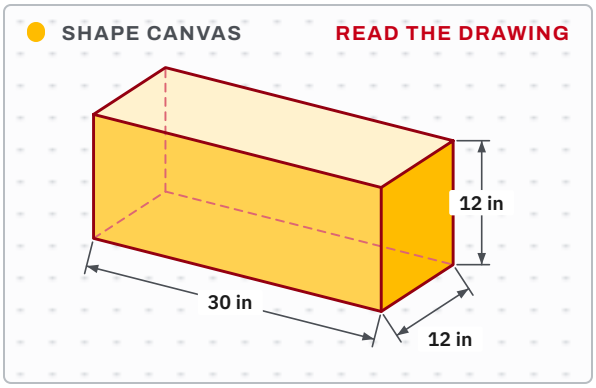
1 How many litres does the tank hold?



WORKING

Answer _____

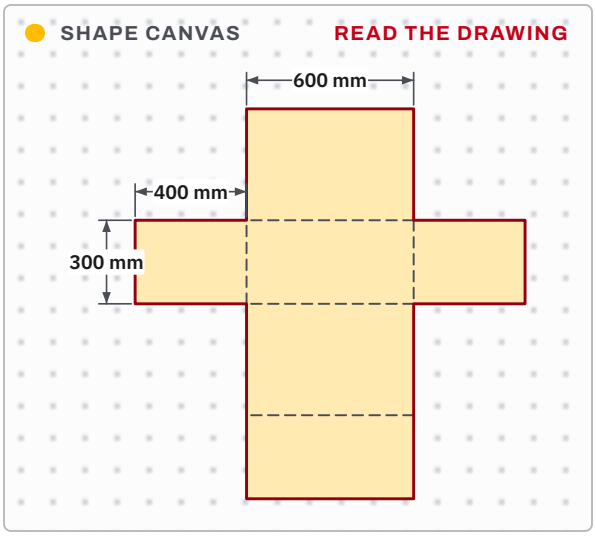
2 The tank is measured in inches. How many US gallons does it hold, to two decimal places?



WORKING

Answer _____

3 Unfold it: the net of a closed box. Add the six rectangles. Give the area in mm² and m².



WORKING

Answer _____

4 A procedure gives a preheat of 150 °C. What is that in °F? A part cools by 40 °C. How many °F is that drop?

WORKING

Answer _____

5 Change one value: a cube is 100 mm on each side. How many litres does it hold? How many if each side is 200 mm?

WORKING

Answer _____

CHAPTER

8

Formulas and Transposing

A formula is a short rule that tells you how to work something out. Shop drawings, code books and manuals are full of them: the area of a plate, the distance around a pipe, the volume of a tank.

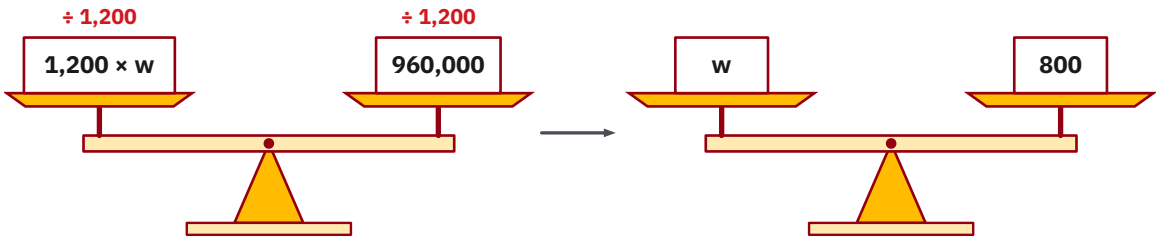
This chapter shows you how to read a formula, put your own numbers into it, and turn it around to find a different value. Turning a formula around is called transposing. It lets you start from the answer you need, such as an area, and find the size that gives it. Every result is checked by putting it back into the formula.

In this chapter you will

- Read a formula and name its subject, variables and constants
- Substitute values into a formula and work it out with BEDMAS
- Use inverse operations to keep an equation balanced
- Transpose one-step and two-step formulas
- Transpose formulas with squares and square roots, such as $A = \pi \times r^2$
- Check every answer by substituting it back

Key words

- formula
- variable
- constant
- subject of a formula
- substitute
- inverse operation
- transpose



8.1 Reading a formula

A **formula** is a rule written with letters and symbols. The formula for the area of a rectangle is $A = l \times w$. It says: the area A equals the length l times the width w.

A letter that stands for a quantity that can change is a **variable**. A value that never changes is a **constant**. In $C = \pi \times D$, C and D are variables and π is a constant.

The letter on its own on the left of the equals sign is the **subject** of the formula. The formula tells you how to work out the subject from the other values.

Formula	What it works out	What the letters mean
$A = l \times w$	Area of a rectangle	A area, l length, w width
$P = 2 \times (l + w)$	Perimeter of a rectangle	P perimeter (distance around)
$C = \pi \times D$	Distance around a circle	C circumference, D diameter
$A = \pi \times r^2$	Area of a circle	r radius (half the diameter)
$V = l \times w \times h$	Volume of a box shape	V volume, h height
$t_F = 1.8 \times t_C + 32$	Temperature in °F from °C	t_F degrees Fahrenheit, t_C degrees Celsius

Table 8.1 Formulas used in this chapter. Chapters 9, 10 and 11 explain where the shape formulas come from.

Some books leave out the × sign: lw means $l \times w$ and 2l means $2 \times l$. A small raised 2 means squared: r^2 means $r \times r$. This book writes the × sign every time.

TRY IT 8.1

1. What is the subject of $V = l \times w \times h$?
2. Name the constant in $A = \pi \times r^2$.
3. Name the variables in $C = \pi \times D$.
4. Rewrite 2lw with × signs.
5. What does s^2 mean?
6. Name the two constants in $t_F = 1.8 \times t_C + 32$.

Answers are in the Answer Key.

8.2 Substituting values

To **substitute** means to put a number in place of a letter. Write the formula, write it again with the numbers in place of the letters, then work it out using **BEDMAS** from Chapter 3.

1. Write the formula.
2. Write it again with each letter replaced by its value. Keep every bracket.
3. Work it out in BEDMAS order: brackets, exponents, division and multiplication, then addition and subtraction.
4. Write the answer with its unit.

Example 8.1 Area and perimeter of a plate

A plate is 1,200 mm long and 800 mm wide. Use A = l × w to find its area and P = 2 × (l + w) to find the distance around it.

SOLUTION

1. Area: A = 1,200 × 800 = 960,000.
2. Perimeter: P = 2 × (1,200 + 800). Brackets first, 1,200 + 800 = 2,000. Then 2 × 2,000 = 4,000.

CALCULATOR Perimeter

TI-30Xa

2 × (1200 + 800) =

Display 4000.

Sharp EL-531XT

2 × (1200 + 800) =

Display 4'000.

Key the brackets. Without them the calculator works 2 × 1,200 first and then adds 800.

Answer **A = 960,000 mm² (square millimetres) and P = 4,000 mm.**

Example 8.2 Formulas with π and a square

A disc has a diameter of 150 mm, so its radius is 75 mm. Use C = π × D to find the distance around it and A = π × r² to find its area. Round to two decimal places.

SOLUTION

1. Distance around: C = π × 150 = 471.24. Use the π key. It holds many more digits than 3.14.
2. Area: A = π × 75². The exponent comes first, so square only the 75: 75² = 5,625. Then π × 5,625 = 17,671.46.

CALCULATOR

TI-30Xa

Distance around

π × 150 =

Display 471.238898

Area

π × 75 x² =

Display 17671.45868

Sharp EL-531XT

Distance around

π × 150 =

Display 471.238898

Area

π × 75 x² =

Display 17'671.45868

The x² key squares only the number just before it, so the calculator follows BEDMAS for you.

Answer **C = 471.24 mm and A = 17,671.46 mm².**

Example 8.3 Converting a temperature

The formula $t_F = 1.8 \times t_C + 32$ changes a Celsius temperature to Fahrenheit. Change 20 °C to °F.

SOLUTION

1. $t_F = 1.8 \times 20 + 32$
2. Multiply first: $1.8 \times 20 = 36$. Then add: $36 + 32 = 68$.

Answer **68 °F**

USE THE SAME UNITS

Every length in a formula must be in the same unit. A plate 1.2 m long and 800 mm wide must be changed to 1,200 mm by 800 mm, or 1.2 m by 0.8 m, before you multiply. Chapter 7 shows how.

TRY IT 8.2

1. A = l × w with l = 2.4 m and w = 1.2 m.
2. P = 2 × (l + w) with l = 600 mm and w = 450 mm.
3. C = π × D with D = 300 mm. Round to two decimal places.
4. A = π × r² with r = 20 cm. Round to two decimal places.
5. V = l × w × h with l = 2 m, w = 1.5 m and h = 0.5 m.
6. $t_F = 1.8 \times t_C + 32$ with $t_C = 35$ °C.

Answers are in the Answer Key.

8.3 Keeping an equation balanced

An equals sign says that both sides have the same value. Think of a balance scale that is level. If you take something off one pan, you must take the same off the other pan, or the scale tips. The same rule keeps an equation true: whatever you do to one side, do to the other side.

To find an unknown value you undo what has been done to it. Each operation has an **inverse operation** that undoes it.

If the letter has been	Undo it by	Example
added to (+ 250)	subtracting (− 250)	x + 250 − 250 = x
subtracted from (− 75)	adding (+ 75)	x − 75 + 75 = x
multiplied (× 4)	dividing (÷ 4)	4 × x ÷ 4 = x
divided (÷ 3)	multiplying (× 3)	x ÷ 3 × 3 = x
squared (x^2)	taking the square root (√)	$\sqrt{(x^2)} = x$

Table 8.2 Operations and their inverses.

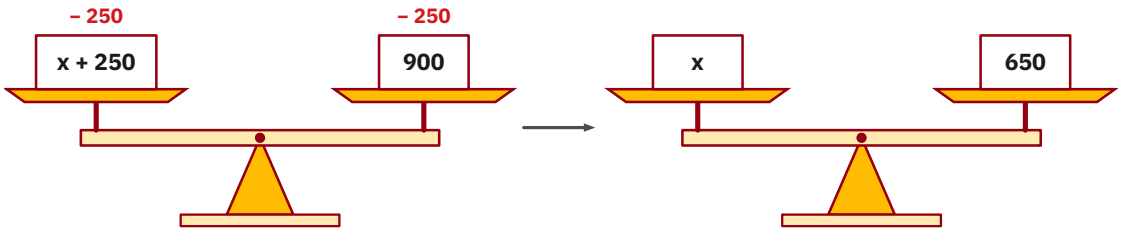
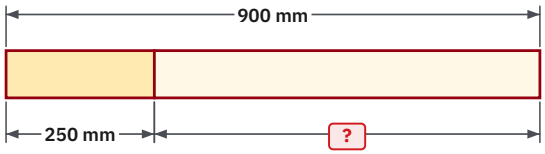


Figure 8.1 Take 250 off both pans and the scale stays level. The unknown x is left on its own.

Example 8.4 Undoing an operation

A 900 mm bar is cut into a 250 mm piece and a second piece, so x + 250 = 900. Solve it. Then solve 4 × x = 1,000 and x ÷ 3 = 125.



SOLUTION

1. 250 was added to x. Subtract 250 from both sides: x = 900 − 250 = 650. Check: 650 + 250 = 900.
2. x was multiplied by 4. Divide both sides by 4: x = 1,000 ÷ 4 = 250. Check: 4 × 250 = 1,000.
3. x was divided by 3. Multiply both sides by 3: x = 125 × 3 = 375. Check: 375 ÷ 3 = 125.

Answer **The second piece is 650 mm. Then x = 250 and x = 375.**

KEY IDEA

Do the same thing to both sides. Pick the inverse operation that leaves the letter on its own.

TRY IT 8.3

1. Solve x + 38 = 120.
2. Solve x − 45 = 155.
3. Solve 6 × x = 1,500.
4. Solve x ÷ 4 = 35.
5. Solve x + 1,275 = 6,000.
6. Solve 12 × x = 3,000.

Answers are in the Answer Key.

8.4 Transposing one-step formulas

To **transpose** a formula means to rearrange it so that a different letter becomes the subject. You use the balance rule from the last section. The letters stay letters until the end, then you substitute the numbers.

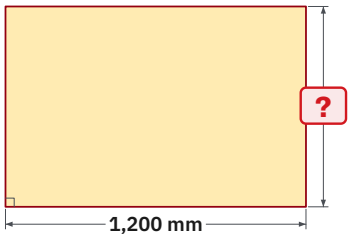
1. Find the letter you want on its own.
2. Ask what has been done to it.
3. Undo that with the inverse operation, on both sides.

4. Write the new formula with the letter you want on the left.
5. Substitute the numbers, work it out, then check.

In the Shape Canvas, change one size and the numbers that depend on it update. Transposing lets you work the other way, from a value you need, such as an area, back to the size that gives it.

Example 8.5 Given the area, find the width

A rectangular plate must have an area of 960,000 mm². It is 1,200 mm long. Transpose A = l × w to find the width.



SOLUTION

1. w has been multiplied by l. Divide both sides by l.
2. A ÷ l = l × w ÷ l, so w = A ÷ l.
3. w = 960,000 ÷ 1,200 = 800

CALCULATOR

TI-30Xa

960000 ÷ 1200 =

Display 800.

Sharp EL-531XT

960000 ÷ 1200 =

Display 800.

Answer **w = 800 mm. Check: 1,200 × 800 = 960,000.**

Example 8.6 Given the distance around, find the diameter

A tape wrapped once around a round tank reads 1,000 mm. Transpose C = π × D and find the diameter to two decimal places.

SOLUTION

1. D has been multiplied by π. Divide both sides by π, so D = C ÷ π.
2. D = 1,000 ÷ π = 318.31

CALCULATOR

TI-30Xa

1000 ÷ π =

Display 318.3098862

Sharp EL-531XT

1000 ÷ π =

Display 318.3098862

Answer **D = 318.31 mm. Check: π × 318.31 = 1,000.00 to two decimal places, so the answer is right.**

Example 8.7 Find the height of a tank

A tank with a rectangular base 1.2 m long and 0.8 m wide must hold 0.576 m³. Transpose V = l × w × h to find the height.

SOLUTION

1. h has been multiplied by l and by w. Divide both sides by l × w, so h = V ÷ (l × w).
2. h = 0.576 ÷ (1.2 × 0.8) = 0.576 ÷ 0.96 = 0.6

CALCULATOR

TI-30Xa

0.576 ÷ (1.2 × 0.8) =

Display 0.6

Sharp EL-531XT

0.576 ÷ (1.2 × 0.8) =

Display 0.6

The brackets matter. Keyed as 0.576 ÷ 1.2 × 0.8, both calculators divide by 1.2 and then multiply by 0.8, which gives 0.384.

Answer **0.6 m. Check: 1.2 × 0.8 × 0.6 = 0.576.**

TRY IT 8.4

1. Make l the subject of A = l × w.
2. A = 2.88 m² and w = 1.2 m. Find l.
3. A sheet has an area of 1,152 in² and is 24 in wide. Find its length.
4. C = 942.48 mm. Find D to the nearest millimetre.
5. D = 2 × r. D = 168 mm. Find r.
6. V = 1.5 m³, l = 2 m and w = 1.5 m. Find h.
7. A square has P = 4 × s. P = 2,400 mm. Find s.

Answers are in the Answer Key.

8.5 Transposing two-step formulas

In a two-step formula, two things have been done to the letter you want. Undo them in the reverse order. The last thing done is the first thing you undo, just as you take off your boots before your socks.

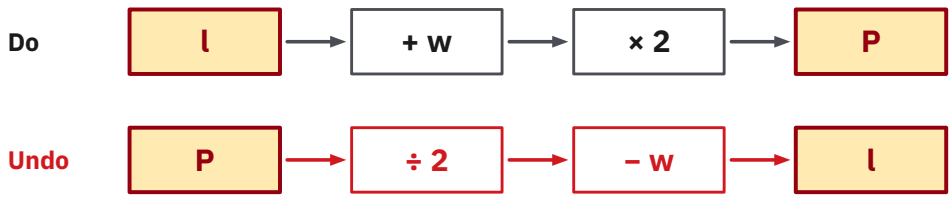


Figure 8.2 In P = 2 × (l + w), l has w added and then is multiplied by 2. To get l back, divide by 2 and then subtract w.

Example 8.8 Find the length from the perimeter

A rectangular frame uses 5,000 mm of flat bar all the way around. It is 900 mm wide. Transpose $P = 2 \times (l + w)$ to find the length.

SOLUTION

1. Undo the × 2: divide both sides by 2. $P \div 2 = l + w$
2. Undo the + w: subtract w from both sides. $l = P \div 2 - w$
3. $l = 5{,}000 \div 2 - 900 = 2{,}500 - 900 = 1{,}600$

CALCULATOR

TI-30Xa

5000 ÷ 2 − 900 =

Display 1600.

Sharp EL-531XT

5000 ÷ 2 − 900 =

Display 1’600.

Answer **1,600 mm. Check: 2 × (1,600 + 900) = 5,000.**

Example 8.9 Fahrenheit to Celsius

Transpose $t_F = 1.8 \times t_C + 32$ to make t_C the subject. Then change 350 °F to °C, to two decimal places.

SOLUTION

1. t_C was multiplied by 1.8 and then 32 was added. Undo the + 32 first: $t_F - 32 = 1.8 \times t_C$
2. Then undo the × 1.8: $t_C = (t_F - 32) \div 1.8$
3. $t_C = (350 - 32) \div 1.8 = 318 \div 1.8 = 176.67$

CALCULATOR

TI-30Xa

(350 − 32) ÷ 1.8 =

Display 176.6666667

Sharp EL-531XT

(350 − 32) ÷ 1.8 =

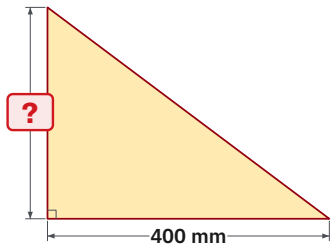
Display 176.6666667

Keyed without brackets, 350 − 32 ÷ 1.8 gives 332.22, because the calculator divides first.

Answer **176.67 °C. NIST publishes the same conversion: $t_C = (t_F - 32) / 1.8$.**

Example 8.10 Height of a gusset

A triangular gusset has an area given by A = b × h ÷ 2. Its base is 400 mm and its area must be 60,000 mm². Find the height.



SOLUTION

1. Undo the ÷ 2: 2 × A = b × h
2. Undo the × b: h = 2 × A ÷ b
3. h = 2 × 60,000 ÷ 400 = 300

CALCULATOR

TI-30Xa

2 × 60000 ÷ 400 =

Display 300.

Sharp EL-531XT

2 × 60000 ÷ 400 =

Display 300.

Answer **300 mm. Check: 400 × 300 ÷ 2 = 60,000.**

TRY IT 8.5

1. P = 2 × (l + w). P = 3,600 mm and l = 1,100 mm. Find w.
2. Make w the subject of P = 2 × (l + w).
3. Change 212 °F to °C.
4. Change −40 °F to °C.
5. An invoice uses T = 60 + 85 × h, with T in dollars and h in hours. T = $740. Find h.
6. A = b × h ÷ 2. A = 6,000 mm² and b = 120 mm. Find h.

Answers are in the Answer Key.

8.6 Formulas with squares and square roots

Squaring and taking the **square root** undo each other. If a letter has been squared, take the square root of both sides to get it back.

Example 8.11 Side of a square plate

A square plate must have an area of 62,500 mm². Transpose A = s² and find the side.

SOLUTION

1. s has been squared. Take the square root of both sides, so s = √A.
2. s = √62,500 = 250

CALCULATOR

TI-30Xa

62500 √x

Display 250.

Sharp EL-531XT

√ 62500 =

Display 250.

Answer **250 mm. Check: 250² = 250 × 250 = 62,500.**

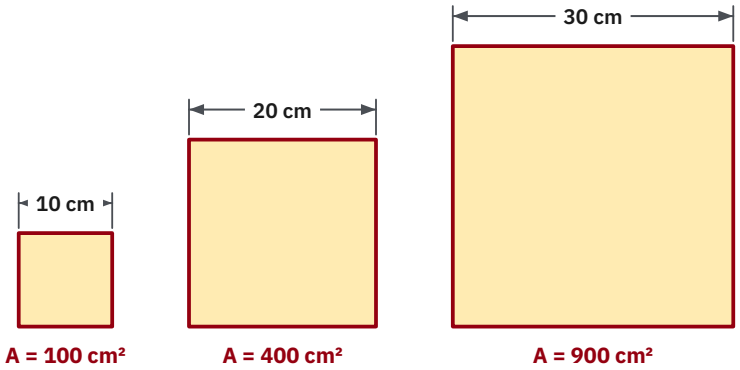


Figure 8.3 Change the side of a square and its area changes with it. Four times the area goes with twice the side. Nine times the area goes with three times the side.

In A = π × r², two things are done to r: it is squared, then multiplied by π. Undo them in reverse order. Divide by π first, then take the square root.

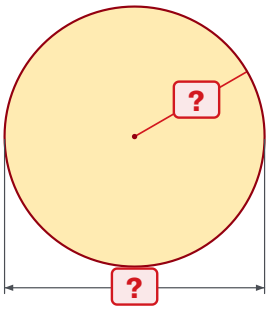
RADIUS FROM THE AREA OF A CIRCLE

$$r = \sqrt{(A \div \pi)} \qquad D = 2 \times r$$

Divide by π first, then take the square root. Double r to get the diameter.

Example 8.12 Given the area, find the radius

A round plate must have an area of 50,000 mm². Find its radius and its diameter, to two decimal places.



SOLUTION

1. Undo the × π: A ÷ π = r²
2. Undo the square: r = √(A ÷ π)
3. r = √(50,000 ÷ π) = 126.16
4. D = 2 × r = 2 × 126.1566... = 252.31 (use the unrounded r)

CALCULATOR Diameter

TI-30Xa

50000 ÷ π = √x × 2 =

Display 252.3132522

Sharp EL-531XT

√ (50000 ÷ π) × 2 =

Display 252.3132522

The calculator keeps every digit of r, so the diameter comes from the unrounded r.

Answer **r = 126.16 mm and D = 252.31 mm. Check: π × 126.16² = 50,002.7, very close to 50,000.**

Example 8.13 An imperial cover plate

A round cover plate must have an area of 1,000 in². Find its diameter to the nearest 1/16 in.

SOLUTION

1. r = √(1,000 ÷ π) = 17.841 in, so D = 2 × r = 35.682 in.
2. Change the decimal part to sixteenths: 0.682 × 16 = 10.91, which rounds to 11.

CALCULATOR

TI-30Xa

1000 ÷ π = √x × 2 =

Display 35.68248232

Sharp EL-531XT

√ (1000 ÷ π) × 2 =

Display 35.68248232

Answer **D = $35\frac{11}{16}$″**

DIVIDE BY Π BEFORE THE SQUARE ROOT

In r = √(A ÷ π) the division is inside the root. Taking √50,000 first and then dividing by π gives 71.18, which is wrong. On the TI-30Xa press = before √x. On the Sharp put the division in brackets after √.

TRY IT 8.6

1. A square has an area of 1,600 cm². Find its side.
2. A square has an area of 0.36 m². Find its side.
3. A circle has an area of 31,416 mm². Find r to the nearest millimetre.
4. A circle has an area of 2,000 mm². Find D to two decimal places.
5. A round tank has V = π × r² × h, r = 0.5 m and V = 1.5 m³. Find h to two decimal places.
6. The area of a square is made 9 times as big. What happens to its side?

Answers are in the Answer Key.

8.7 Checking by substituting back

A transposed formula can look right and still be wrong. The check takes a few seconds: put your answer back into the formula you started with, together with the values you were given. If the formula works out to the value you were given, your answer is right. If you rounded the answer, the check comes out close rather than exact.

Example 8.14 Catching a mistake

A learner transposes $V = l \times w \times h$ and writes $h = V \times l \div w$. With $V = 0.45\ m^3$, $l = 1.5$ m and $w = 0.6$ m this gives $h = 1.125$ m. Check the answer and correct it.

SOLUTION

1. Substitute back: $1.5 \times 0.6 \times 1.125 = 1.0125$. That is not 0.45, so the answer is wrong.
2. The mistake: l multiplies h, so it must divide. $h = V \div (l \times w) = 0.45 \div (1.5 \times 0.6) = 0.5$
3. Substitute back: $1.5 \times 0.6 \times 0.5 = 0.45$. It checks.

CALCULATOR

TI-30Xa

0.45 ÷ (1.5 × 0.6) =

Display 0.5

Sharp EL-531XT

0.45 ÷ (1.5 × 0.6) =

Display 0.5

Answer **h = 0.5 m**

Also ask whether the size is reasonable before you cut. A tank 1.5 m long and 0.6 m wide that holds less than half a cubic metre cannot be more than a metre tall.

KEY IDEA

Transpose with letters, substitute the numbers, then check by putting the answer back into the original formula.

WORKSHEET 8A

Given the area, find the side

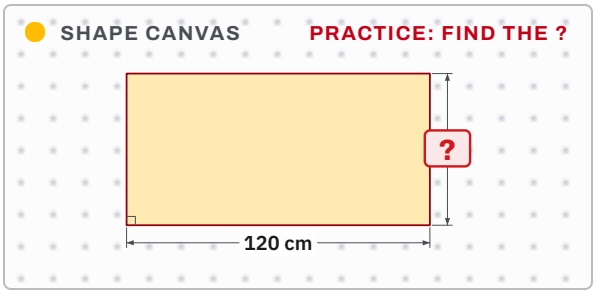
Use the value you know to find the ?

Name _____

Date _____

Transpose the formula first, then substitute. Check each answer by putting it back into the formula you started with. Round to two decimal places where the answer does not come out exactly.

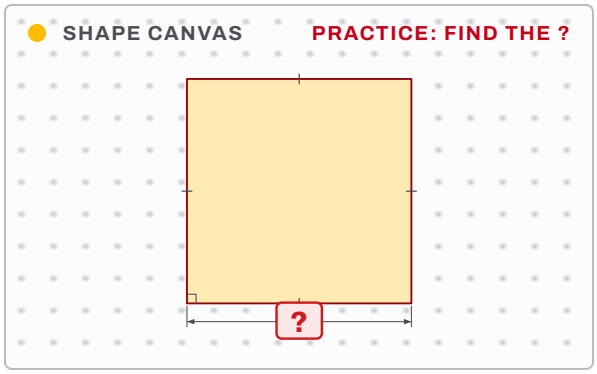
1 The rectangle has an area of 7,200 cm² and a length of 120 cm. Use A = l × w to find the width.



WORKING

Answer _____

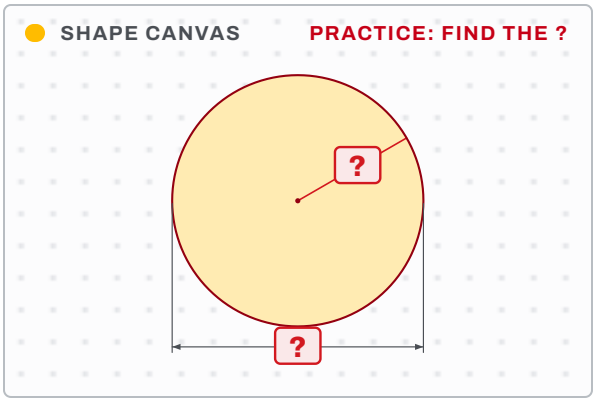
2 The square has an area of 2,025 cm². Use A = s² to find the side.



WORKING

Answer _____

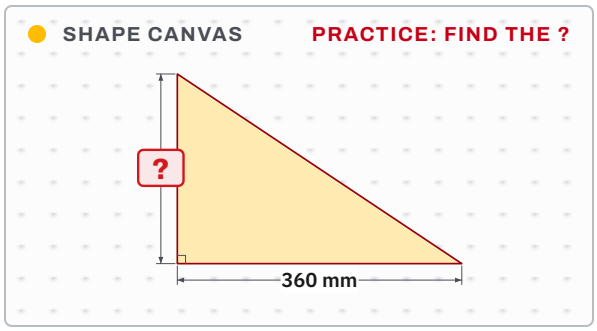
3 The circle has an area of 5,000 mm². Find the radius and the diameter.



WORKING

Answer _____

4 The gusset has an area of 43,200 mm² and a base of 360 mm. Use A = b × h ÷ 2 to find the height.

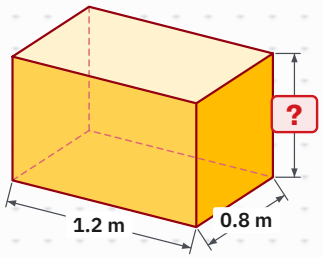


WORKING

Answer _____

5 The box holds 0.72 m³. Use V = l × w × h to find the height.

SHAPE CANVAS PRACTICE: FIND THE ?

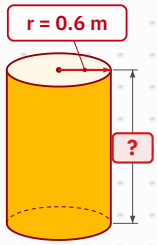


WORKING

Answer _____

6 The round tank holds 2 m³. Use V = π × r² × h to find the height.

SHAPE CANVAS PRACTICE: FIND THE ?



WORKING

Answer _____

7 Change one value: the square in problem 2 is given an area 4 times as big, 8,100 cm². Find the new side. What happened to the side?

WORKING

Answer _____

8 Change one value: the rectangle in problem 1 keeps its area of 7,200 cm², but its length is doubled to 240 cm. Find the new width. What happened to the width?

WORKING

Answer _____

9 Find the mistake. A learner keys 0.72 ÷ 1.2 × 0.8 = to find the height of a box 1.2 m by 0.8 m that holds 0.72 m³, and gets 0.48 m. What went wrong? Find the correct height and check it.

WORKING

Answer _____

WORKSHEET 8B

Rearrange, solve and check

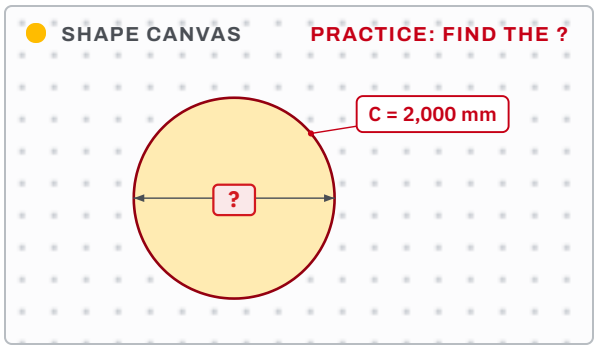
Transpose with letters first, then put in the numbers

Name _____

Date _____

Show the transposed formula, the substitution and the check. Round to two decimal places unless the problem says otherwise.

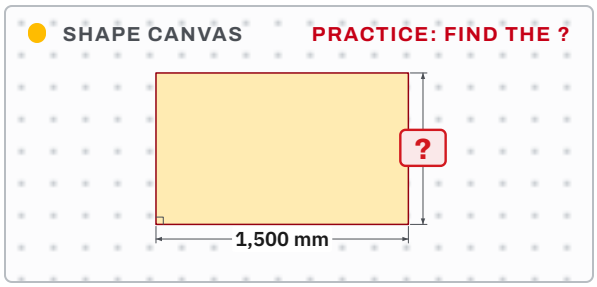
1 Make D the subject of C = π × D. Find the diameter of a pipe whose circumference is 2,000 mm.



WORKING

Answer _____

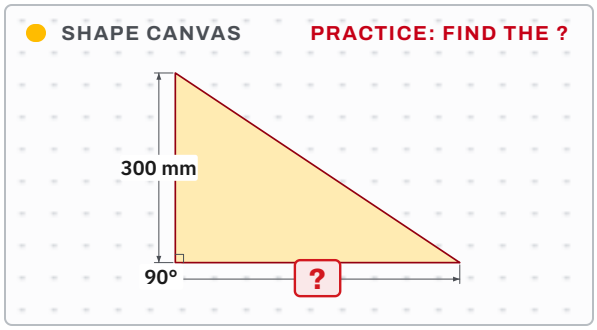
2 A frame uses 4,800 mm of bar all the way around and is 1,500 mm long. Use P = 2 × (l + w) to find the width.



WORKING

Answer _____

3 The gusset has an area of 67,500 mm² and a height of 300 mm. Make b the subject of A = b × h ÷ 2 and find the base.



WORKING

Answer _____

4 Make t_C the subject of $t_F = 1.8 \times t_C + 32$. Then change 450 °F to °C.

WORKING

Answer _____

5 An invoice uses T = 60 + 85 × h, with T in dollars and h in hours. The total is $527.50. How many hours were worked?

WORKING

Answer _____

CHAPTER

9

Perimeter and Circumference

How much flat bar goes around a guard? How long is the band that wraps a tank? How big a sheet rolls into a round duct? Each question asks for a distance around the outside of a shape.

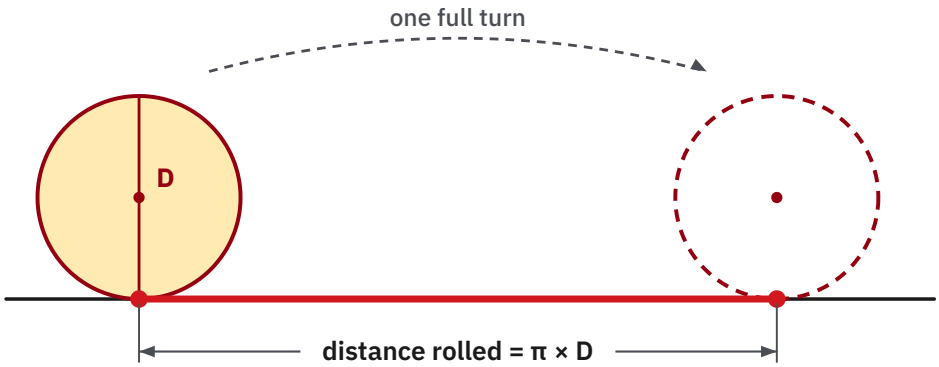
This chapter works out the distance around rectangles, triangles and other straight-sided shapes, and then around circles. The distance around a circle is always π × D, the number π times the diameter. The same formula gives the stretch-out, the length of flat sheet that rolls into a pipe or a round duct.

In this chapter you will

- Find the perimeter of straight-sided shapes
- Find a missing side from a perimeter
- Name the parts of a circle
- Use C = π × D, and find D from C
- Find the stretch-out of a pipe or duct
- Find arcs and composite perimeters

Key words

- perimeter
- radius
- diameter
- circumference
- pi (π)
- stretch-out
- arc



9.1 Perimeter of straight-sided shapes

The **perimeter** is the distance all the way around the outside of a flat shape. Picture walking along every edge until you are back where you started. Add up every edge you walked.

PERIMETER

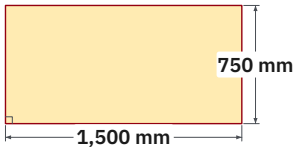
Rectangle: P = 2 × (l + w) **Square: P = 4 × s**

Triangle: P = a + b + c **Regular polygon: P = n × s**

l length, w width, s side, a, b and c the three sides, n the number of equal sides

Example 9.1 A rectangular guard frame

A guard frame is 1,500 mm long and 750 mm wide. How much flat bar goes around it?



SOLUTION

1. Two lengths and two widths: P = 2 × (l + w).
2. P = 2 × (1,500 + 750) = 2 × 2,250 = 4,500

Answer **4,500 mm, which is 4.5 m.**

Example 9.2 A frame in fractions of an inch

A picture-style frame is $24\frac{3}{8}$ in by $18\frac{1}{2}$ in. Find the perimeter.

SOLUTION

1. Add a length and a width: $24\frac{3}{8} + 18\frac{1}{2} = 24\frac{3}{8} + 18\frac{4}{8} = 42\frac{7}{8}$ in.
2. Double it: $2 \times 42\frac{7}{8} = 84\frac{14}{8} = 85\frac{6}{8} = 85\frac{3}{4}$ in. Both calculators below show this as a mixed number, 85 and 3/4.

CALCULATOR

TI-30Xa

[2] [×] [(] [24] [a b/c] [3] [a b/c] [8] [+]
[18] [a b/c] [1] [a b/c] [2] [)] [=]

Display 85⌟3⌟4

Sharp EL-531XT

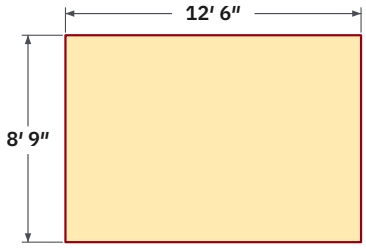
[2] [×] [(] [24] [a b/c] [3] [a b/c] [8] [+]
[18] [a b/c] [1] [a b/c] [2] [)] [=]

Display 85⌜3⌜4

Answer **$85\frac{3}{4}$ in, which is 7′ $1\frac{3}{4}$″.**

Example 9.3 Feet and inches

A concrete pad is 12 ft 6 in long and 8 ft 9 in wide. How much edge form goes around it? Give the answer in feet and inches.



SOLUTION

1. Change to inches. 12 ft 6 in = 12 × 12 + 6 = 150 in. 8 ft 9 in = 8 × 12 + 9 = 105 in.
2. P = 2 × (150 + 105) = 510 in.
3. 510 ÷ 12 = 42.5 ft, which is 42 ft and 0.5 × 12 = 6 in.

CALCULATOR

TI-30Xa

2 × (150 + 105) =

Display 510.

Sharp EL-531XT

2 × (150 + 105) =

Display 510.

Answer **42 ft 6 in**

When you know the perimeter and one side, transpose the formula as in Chapter 8. For a rectangle, w = P ÷ 2 − l. A rectangle with a perimeter of 3,000 mm and a length of 900 mm has a width of 3,000 ÷ 2 − 900 = 600 mm.

TRY IT 9.1

1. A square plate has sides of 850 mm. Find the perimeter.
2. A triangle has sides of 1.2 m, 0.9 m and 1.5 m. Find the perimeter.
3. A regular hexagon has sides of 125 mm. Find the perimeter.
4. A regular octagon has sides of 80 mm. Find the perimeter.
5. A rectangle is 3 ft 4 in by 2 ft 8 in. Find the perimeter in feet.
6. A rectangle has a perimeter of 2,600 mm and a length of 800 mm. Find the width.
7. A frame is $36\frac{1}{4}$ in by $24\frac{1}{2}$ in. Find the perimeter.

Answers are in the Answer Key.

9.2 The parts of a circle and π

The **radius** (r) is the distance from the centre of a circle to its edge. The **diameter** (D) is the distance across the circle through the centre, so it is two radii: D = 2 × r and r = D ÷ 2. The **circumference** (C) is the distance around the circle. It is the perimeter of a circle.

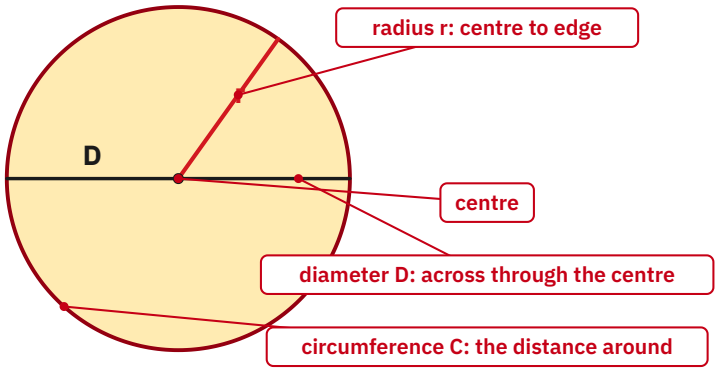


Figure 9.1 The parts of a circle.

Wrap a string once around any circle and lay it along the diameter. It fits a little more than 3 times, every time, for a washer or a tank. That number is **pi**, written π. It is about 3.14159, and its digits never end.

CALCULATOR The π key

TI-30Xa

π

Display 3.141592654

Sharp EL-531XT

π =

Display 3.141592654

Both calculators show π to 10 digits. Use the key every time instead of typing 3.14.

3.14 IS NOT Π

π × 600 = 1,884.96, but 3.14 × 600 = 1,884. The short value is almost 1 mm short on a 600 mm circle, and the error grows with the size of the job.

TRY IT 9.2

1. A circle has a radius of 45 mm. Find the diameter.
2. A pipe has a diameter of 168.3 mm. Find the radius.
3. A tank has a diameter of 2 ft. Find the radius in inches.
4. Which is longer, the diameter of a circle or its circumference? About how many times longer?

Answers are in the Answer Key.

9.3 Circumference

CIRCUMFERENCE

$$C = \pi \times D$$

If you know the radius, double it first to get D. Double D and C doubles: C and D always keep the same ratio, π.

Example 9.4 From a diameter and from a radius

Find the circumference of a disc 450 mm in diameter, and of a ring with a radius of 125 mm. Round to two decimal places.

SOLUTION

1. Disc: $C = \pi \times 450 = 1{,}413.72$.
2. Ring: double the radius first, $D = 2 \times 125 = 250$. Then $C = \pi \times 250 = 785.40$.

CALCULATOR

TI-30Xa

Disc

π × 450 =

Display 1413.716694

Ring

π × 250 =

Display 785.3981634

Sharp EL-531XT

Disc

π × 450 =

Display 1′413.716694

Ring

π × 250 =

Display 785.3981634

Answer **1,413.72 mm and 785.40 mm.**

Example 9.5 A band around a tank

A round tank is 48 in in diameter. How long is a band that goes once around it? Give the answer to the nearest 1/16 in.

SOLUTION

1. $C = \pi \times 48 = 150.796$ in.
2. Change the decimal part to sixteenths: $0.796 \times 16 = 12.74$, which rounds to 13.
3. $150\frac{13}{16}$ in is 12 ft $6\frac{13}{16}$ in, because $150 = 12 \times 12 + 6$.

CALCULATOR

TI-30Xa

π × 48 =

Display 150.7964474

Sharp EL-531XT

π × 48 =

Display 150.7964474

Answer **$150\frac{13}{16}$″, which is 12′ $6\frac{13}{16}$″.**

Example 9.6 Turns of a wheel

A wheel 600 mm in diameter rolls along a floor. In one turn it rolls one circumference. How far does it roll in 25 turns? How many turns does it make in 50 m?

SOLUTION

1. One turn: π × 600 mm. 25 turns: 25 × π × 600 = 47,123.89 mm, about 47.12 m.
2. Turns in 50 m = 50,000 mm: 50,000 ÷ (π × 600) = 26.53.

CALCULATOR

TI-30Xa

Distance

25 × π × 600 =

Display 47123.8898

Turns

50000 ÷ (π × 600) =

Display 26.52582385

Sharp EL-531XT

Distance

25 × π × 600 =

Display 47’123.8898

Turns

50000 ÷ (π × 600) =

Display 26.52582385

Answer **47,123.89 mm (47.12 m) and 26.53 turns.**

TRY IT 9.3

1. D = 200 mm. Find C to two decimal places.
2. r = 60 mm. Find C to two decimal places.
3. D = 10 in. Find C to the nearest 1/16 in.
4. A tape wrapped once around a pipe reads 2,500 mm. Find D to two decimal places.
5. C = 1,256.64 mm. Find D to the nearest millimetre.
6. A 500 mm wheel turns 10 times. How far does it roll, to two decimal places?
7. A circle with C = 1,000 mm has its diameter doubled. What is the new C?

Answers are in the Answer Key.

9.4 Stretch-out of a pipe or duct

Open out the side of a cylinder and it lies flat as a rectangle. One side of the rectangle is the length of the cylinder. The other side wrapped once around the cylinder, so it is the circumference. This length is the **stretch-out**. To make a round duct, you cut a sheet as long as the stretch-out and roll it.

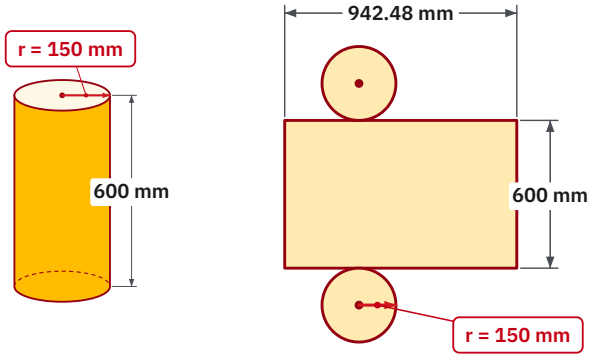


Figure 9.2 Unfold it. The cylinder on the left opens out to the net on the right. The stretch-out is π × D = π × 300 = 942.48 mm.

MEASURE AROUND WHEN YOU CAN

A tape wrapped once around a part reads the stretch-out directly. You need π only when you start from the diameter.

Example 9.7 Sheet for a round duct

A round duct is 300 mm in diameter and 1,200 mm long. The drawing adds 25 mm to the stretch-out for a lap seam. What size sheet is cut? Give the sheet length to the nearest millimetre.

SOLUTION

1. Stretch-out: π × 300 = 942.48 mm. Add the 25 mm seam allowance from the drawing: 942.48 + 25 = 967.48 mm.

CALCULATOR

TI-30Xa	Sharp EL-531XT
π × 300 + 25 =	π × 300 + 25 =
Display 967.4777961	Display 967.4777961

Answer **967 mm by 1,200 mm.**

Example 9.8 A jacket around a pipe

A sheet-metal jacket wraps around the outside of a pipe. The outside diameter is $6\frac{5}{8}$ in. Find the stretch-out to the nearest 1/16 in.

SOLUTION

1. The jacket wraps the outside surface, so use the outside diameter: π × 6.625 = 20.813 in.
2. Sixteenths: 0.813 × 16 = 13.01, which rounds to 13.

CALCULATOR

TI-30Xa

π × 6 a b/c 5 a b/c 8 =

Display 20.81305133

Sharp EL-531XT

π × 6 a b/c 5 a b/c 8 =

Display 20.81305133

Answer $20\frac{13}{16}$″

TRY IT 9.4

1. A duct is 250 mm in diameter. Find the stretch-out to two decimal places.
2. A duct is 400 mm in diameter and the drawing adds 30 mm for the seam. Find the cut length to two decimal places.
3. A pipe has an outside diameter of $4\frac{1}{2}$ in. Find the stretch-out of a jacket to the nearest 1/16 in.
4. A stretch-out is 942.48 mm. What diameter does it roll into, to the nearest millimetre?

Answers are in the Answer Key.

9.5 Arcs

An **arc** is a part of the circumference. A 90° arc is a quarter of the circle, because a full turn is 360°. Find the fraction of the circle, then take that fraction of π × D.

ARC LENGTH

$$\text{arc} = \text{angle} \div 360 \times \pi \times D$$

Half circle: π × D ÷ 2. Quarter circle: π × D ÷ 4.

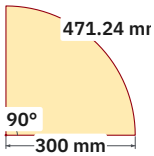


Figure 9.3 A 90° arc of a circle with a 300 mm radius. D = 600 mm, so the arc is 90 ÷ 360 × π × 600 = 471.24 mm.

Example 9.9 A curved handrail

A curved handrail follows a 120° arc of a circle 2,400 mm in diameter. How long is the rail? Round to two decimal places.

SOLUTION

1. 120° is 120 ÷ 360 = one third of the circle.
2. arc = 120 ÷ 360 × π × 2,400 = 2,513.27

CALCULATOR

TI-30Xa

120 ÷ 360 × π × 2400 =

Display 2513.274123

Sharp EL-531XT

120 ÷ 360 × π × 2400 =

Display 2'513.274123

Answer 2,513.27 mm

TRY IT 9.5

1. A half circle has D = 500 mm. Find the curved edge.
2. A quarter circle has r = 200 mm. Find the curved edge.
3. A 60° arc has D = 900 mm. Find its length.
4. A 270° arc has D = 100 mm. Find its length.

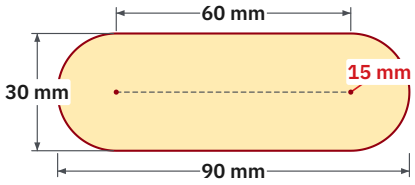
Answers are in the Answer Key.

9.6 Composite perimeters

A composite shape is made of simple parts. Break it down: split it into straight edges and arcs, find any missing lengths, then add only the edges on the outside. The dashed lines used to split a shape are not edges.

Example 9.10 A slot

A slot has straight sides 60 mm long and round ends 30 mm across. Find the distance around the slot.



SOLUTION

1. Two straight sides: 2 × 60 = 120 mm.
2. Two half circles make one whole circle with D = 30 mm: π × 30 = 94.25 mm.
3. Add: 120 + 94.25 = 214.25.

CALCULATOR

TI-30Xa

2 × 60 + π × 30 =

Display 214.2477796

Sharp EL-531XT

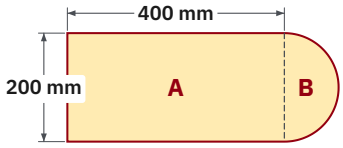
2 × 60 + π × 30 =

Display 214.2477796

Answer 214.25 mm

Example 9.11 A plate with a round end

A plate is a 400 mm by 200 mm rectangle with a half circle on one end. Find its perimeter.



SOLUTION

1. Part A gives three outside edges: 400 + 400 + 200 = 1,000 mm. The dashed line is inside the plate. Part B is a half circle with D = 200 mm: π × 200 ÷ 2 = 314.16 mm.
2. Add: 1,000 + 314.16 = 1,314.16.

Answer **1,314.16 mm**

Example 9.12 A belt around two pulleys

A belt runs around two pulleys, each 200 mm in diameter. The centres are 800 mm apart. How long is the belt?



SOLUTION

1. Each straight run is as long as the centre distance: 2 × 800 = 1,600 mm.
2. The belt wraps half of each pulley, and two halves make one circle: π × 200 = 628.32 mm. Add: 1,600 + 628.32 = 2,228.32.

Answer **2,228.32 mm**

TRY IT 9.6

1. A slot: straight sides 50 mm, ends 20 mm across. Find P.
2. A 600 mm by 400 mm plate, 50 mm corner radius. Find P.
3. Two 150 mm pulleys, centres 600 mm apart. Belt length?
4. A quarter circle, r = 300 mm. Find P, edges and arc.

Answers are in the Answer Key.

WORKSHEET 9A

Around the outside

Find the perimeter, or find the ? from the perimeter

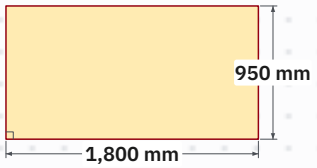
Name _____

Date _____

Add every outside edge. If a perimeter is given, transpose to find the ?

1 Find the perimeter of the plate.

SHAPE CANVAS READ THE DRAWING

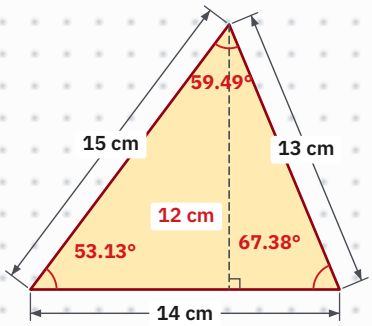


WORKING

Answer _____

2 Find the perimeter. The dashed height is not an edge.

SHAPE CANVAS READ THE DRAWING

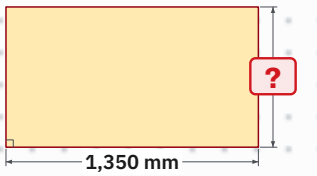


WORKING

Answer _____

3 The perimeter is 4,200 mm. Find the width.

SHAPE CANVAS PRACTICE: FIND THE ?

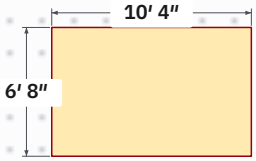


WORKING

Answer _____

4 Find the perimeter in feet and inches.

SHAPE CANVAS READ THE DRAWING



WORKING

Answer _____

5 A square has a perimeter of 2,000 mm. Find the side.

WORKING

Answer _____

6 Change one value: both sides of the plate in problem 1 are doubled. Find the new perimeter. What happened to it?

WORKING

Answer _____

WORKSHEET 9B

Circles, arcs and stretch-outs

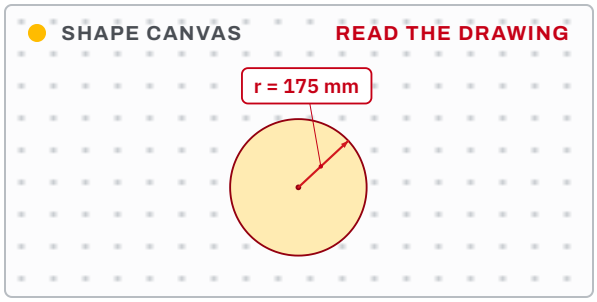
Distance around a circle is always π × D

Name _____

Date _____

Round to two decimal places unless a problem says otherwise. If a drawing gives r, double it to get D.

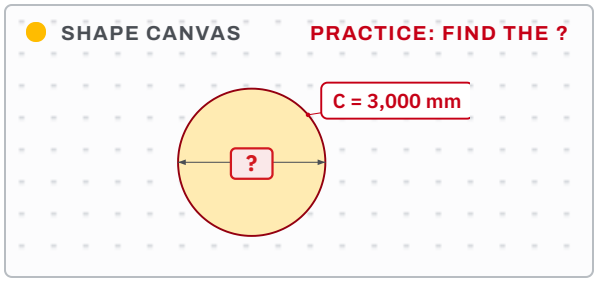
1 Find the circumference.



WORKING

Answer _____

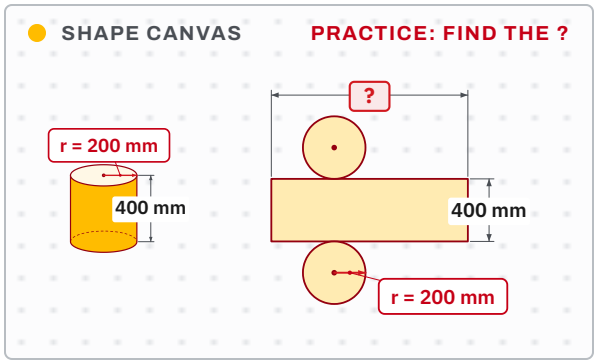
2 A tape around the tank reads 3,000 mm. Find the diameter.



WORKING

Answer _____

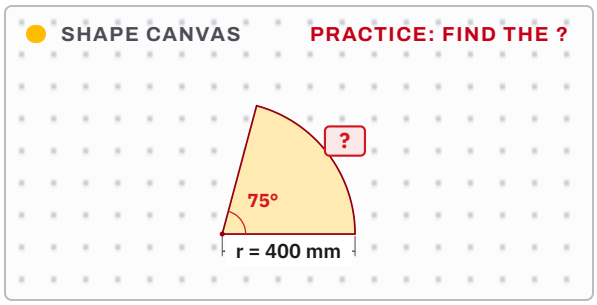
3 Unfold it. Find the stretch-out marked ? above the net.



WORKING

Answer _____

4 Find the length of the arc.



WORKING

Answer _____

5 A wheel is 700 mm in diameter. How many turns does it make in 10 m?

WORKING

Answer _____

6 A jacket wraps a pipe with an outside diameter of $8\frac{5}{8}$ in. Find the stretch-out to the nearest 1/16 in.

WORKING

Answer _____

WORKSHEET 9C

Break it down: composite shapes

Split each shape into straight edges and arcs

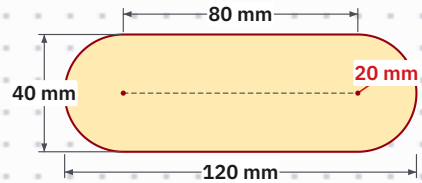
Name _____

Date _____

Add only the outside edges. Dashed lines are not edges. Round to two decimal places.

1 Find the perimeter of the slot.

SHAPE CANVAS READ THE DRAWING

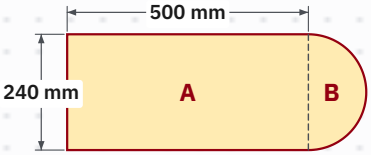


WORKING

Answer _____

2 Find the perimeter. Part B is a half circle.

SHAPE CANVAS READ THE DRAWING

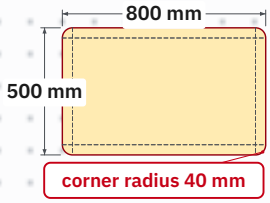


WORKING

Answer _____

3 Each corner is a quarter circle. Find the perimeter.

SHAPE CANVAS READ THE DRAWING

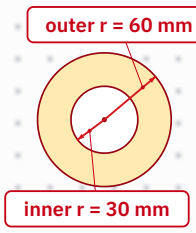


WORKING

Answer _____

4 Find the total length of both edges of the washer.

SHAPE CANVAS READ THE DRAWING



WORKING

Answer _____

5 Find the length of the belt.

SHAPE CANVAS READ THE DRAWING



WORKING

Answer _____

6 A quarter circle has a radius of 250 mm. Find its perimeter, both straight edges and the arc.

WORKING

Answer _____

CHAPTER

10

Area

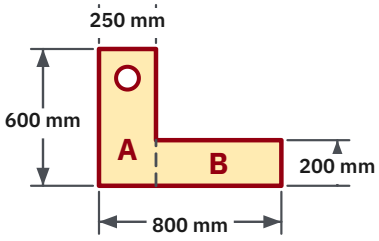
Area tells you how much flat surface a part covers. You use it to order plate and sheet, to work out how much coating a floor needs, and to lay out the sheet metal for a duct or a tank. This chapter builds the area formulas one shape at a time, then breaks composite parts into pieces and unfolds solids into flat nets.

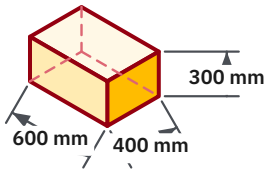
In this chapter you will

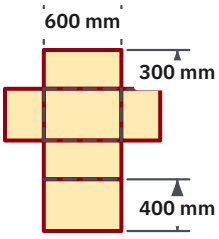
- Explain what area is and use square units
- Find the area of rectangles, triangles, parallelograms and trapezoids
- Find the area of circles and rings with $\pi \times r^2$
- Change mm^2 to m^2 and in^2 to ft^2
- Break a composite shape into basic shapes
- Unfold a solid into a net and find its surface area

Key words

- area
- square unit
- base and height
- composite shape
- surface area
- net







10.1 Area and square units

Area is the amount of flat surface inside a shape. A length is measured in one direction. Area covers two directions at once, so it is measured with squares.

A **square unit** is a square that is 1 unit long on each side. A square 1 cm on each side has an area of 1 square centimetre, written 1 cm². The small 2 shows that two lengths were multiplied: centimetres times centimetres.

The square units used in this book are mm², cm² and m² (square millimetres, square centimetres and square metres), and in² and ft² (square inches and square feet). A length in mm and an area in mm² measure different things, so always write the small 2 on an area.

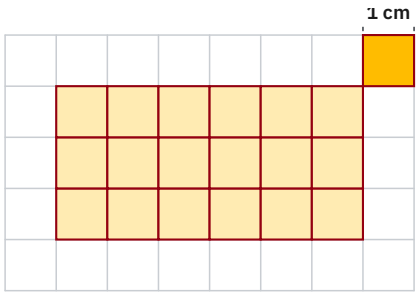


Figure 10.1 A plate drawn on a 1 cm grid covers 3 rows of 6 squares, so its area is 18 cm². The gold square is 1 cm².

TRY IT 10.1

1. A plate on a 1 cm grid covers 4 rows of 7 squares. What is its area?
2. How do you say 35 m² out loud?
3. Which is an area: 12 ft, 12 ft² or 12 ft³?
4. A floor is covered by 30 tiles. Each tile is exactly 1 ft². What is the area of the floor?

Answers are in the Answer Key.

10.2 Rectangles and squares

A rectangle that is 6 squares long and 3 squares wide holds 3 rows of 6 squares. Multiplying the length by the width counts them all at once, so you do not need a grid.

AREA OF A RECTANGLE AND A SQUARE

$$A = l \times w$$

$$A = s \times s = s^2$$

A = area, l = length, w = width, s = side of a square. Both lengths must be in the same unit.

Square millimetres make big numbers, and a large area is easier to read in square metres. One metre is 1,000 mm, so a square metre is 1,000 mm by 1,000 mm. In the same way, a square foot is 12 in by 12 in.

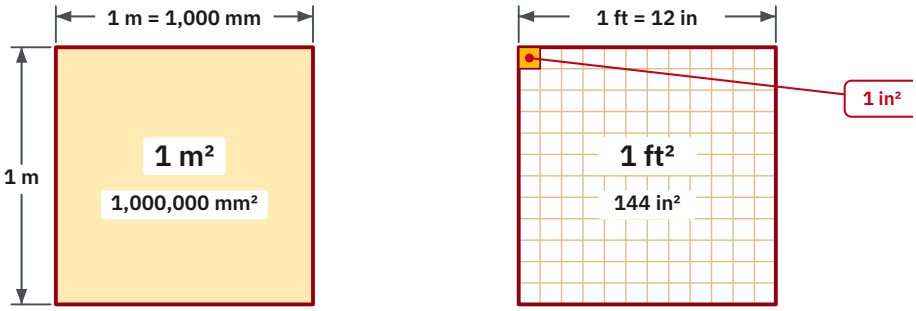
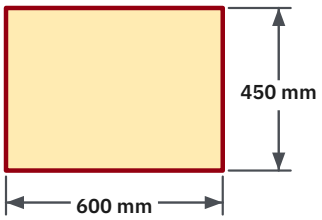


Figure 10.2 1 m² = 1,000 mm × 1,000 mm = 1,000,000 mm². 1 ft² = 12 in × 12 in = 144 in².

Example 10.1 Area of a base plate

A machine base plate is 600 mm long and 450 mm wide. Find its area in mm² and in m².



SOLUTION

1. A = l × w = 600 × 450 = 270,000 mm².
2. To change mm² to m², divide by 1,000,000: 270,000 ÷ 1,000,000 = 0.27 m².
3. Check: in metres the plate is 0.6 m by 0.45 m, and 0.6 × 0.45 = 0.27 m².

CALCULATOR

TI-30Xa

Area in mm²

600 × 450 =

Display 270000.

mm² to m²

270000 ÷ 1000000 =

Display 0.27

Sharp EL-531XT

Area in mm²

600 × 450 =

Display 270'000.

mm² to m²

270000 ÷ 1000000 =

Display 0.27

Answer **270,000 mm² = 0.27 m²**

DIVIDE BY 1,000,000, NOT BY 1,000

Changing mm to m divides a length by 1,000. An area has two lengths multiplied, so the factor is used twice: 1,000 × 1,000 = 1,000,000. Dividing 270,000 mm² by 1,000 gives 270, far too big for this plate in m².

The same idea works in imperial units. A sheet 4 ft by 8 ft has an area of 4 × 8 = 32 ft². In inches it is 48 in by 96 in, and 48 × 96 = 4,608 in². The check 32 × 144 = 4,608 shows the two answers agree.

Example 10.2 Coating a shop floor

A shop floor is 6 m by 4.5 m. The coating label says one litre covers 5 m². The coating is sold in 1 L tins. How many tins do you buy?

SOLUTION

1. Area: 6 × 4.5 = 27 m².
2. Litres: 27 ÷ 5 = 5.4 L.
3. A part tin still has to be bought, so round up.

CALCULATOR

TI-30Xa	Sharp EL-531XT
6 × 4.5 ÷ 5 =	6 × 4.5 ÷ 5 =
Display 5.4	Display 5.4

Answer **6 tins**

TRY IT 10.2

1. A plate is 750 mm by 400 mm. Find its area in mm².
2. A square cover plate has 250 mm sides. Find its area.
3. A table top is 3 ft by 5 ft. Find its area in ft² and in in².
4. A floor is 8 m by 5 m. One litre of coating covers 6 m². How many 1 L tins?

Answers are in the Answer Key.

10.3 Triangles

Every triangle is half of a rectangle or of a parallelogram. To use that idea you need two measurements, the base and the height.

The **base** is the side you measure from, usually the bottom side. The **height** is the distance from the base to the opposite corner, measured at right angles to the base. The height is not a sloping side.

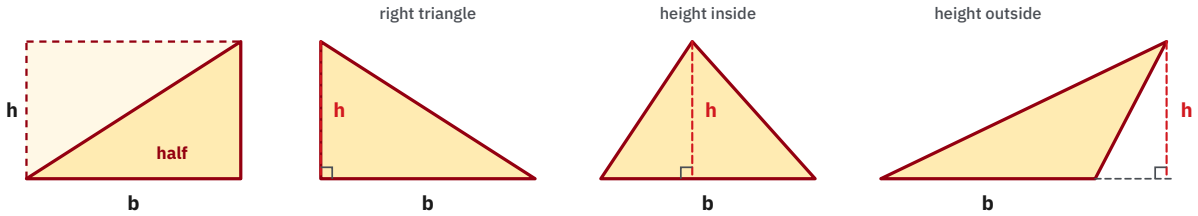


Figure 10.3 Left: a rectangle cut on its diagonal makes two equal triangles. Right: three triangles with the same base b and the same height h have the same area.

AREA OF A TRIANGLE

$$A = b \times h \div 2$$

b = base, h = height at right angles to the base.

Example 10.3 A triangular end panel

The end panel of a trough is a triangle with a base of 600 mm and a height of 400 mm. Its two sloping sides are each 500 mm long. Find the area in mm² and in m².



SOLUTION

1. Use the base and the height. The 500 mm sloping sides are not needed.
2. A = 600 × 400 ÷ 2 = 120,000 mm².
3. In m²: 120,000 ÷ 1,000,000 = 0.12 m².

CALCULATOR

TI-30Xa

600 × 400 ÷ 2 =

Display 120000.

Sharp EL-531XT

600 × 400 ÷ 2 =

Display 120’000.

Answer **120,000 mm² = 0.12 m²**

THE SLOPING SIDE IS NOT THE HEIGHT

Using the 500 mm side in the example above would give 600 × 500 ÷ 2 = 150,000 mm², which is too big. The height always meets the base at a right angle. A drawing often shows it as a dashed line with a small square at its foot.

TRY IT 10.3

1. A right-angled gusset: base 300 mm, height 200 mm. Area?
2. Base 18 in, height 10.5 in. Area?
3. Base 1.2 m, height 0.9 m. Area?
4. Base 300 mm, sloping sides 250 mm, height 200 mm. Area?

Answers are in the Answer Key.

10.4 Parallelograms and trapezoids

A parallelogram has two pairs of parallel sides. Cut the triangle off one end and move it to the other end: the parallelogram becomes a rectangle with the same base and the same height. So its area is base times height: **A = b × h**, where h is the height at right angles to the base, not the slanted side.

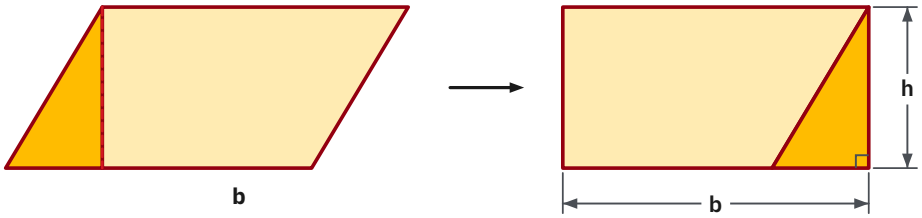


Figure 10.4 The gold triangle moves from the left end to the right end. The parallelogram and the rectangle have the same area.

A trapezoid has one pair of parallel sides, called a and b. Two copies of the same trapezoid, one turned around, fit together to make a parallelogram with base a + b. One trapezoid is half of it.

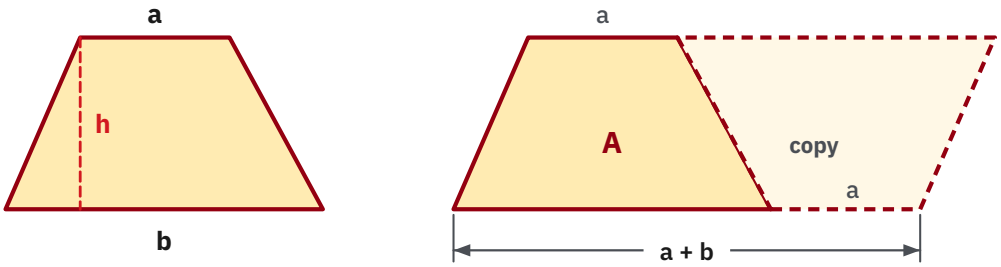


Figure 10.5 A trapezoid and a turned copy make a parallelogram with base a + b and the same height.

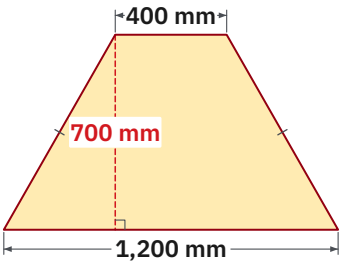
AREA OF A TRAPEZOID

$$A = (a + b) \div 2 \times h$$

a and b = the two parallel sides, h = the height between them. (a + b) ÷ 2 is the average width.

Example 10.4 A hopper side panel

A hopper side panel is a trapezoid. The parallel sides are 400 mm and 1,200 mm, and the height between them is 700 mm. Find the area in mm² and m².



SOLUTION

1. A = (a + b) ÷ 2 × h.
2. Brackets first: 400 + 1,200 = 1,600.
3. Average width: 1,600 ÷ 2 = 800.
4. Times the height: 800 × 700 = 560,000 mm² = 0.56 m².

CALCULATOR

TI-30Xa

(400 + 1200) ÷ 2 × 700 =

Display 560000.

Sharp EL-531XT

(400 + 1200) ÷ 2 × 700 =

Display 560’000.

The brackets make both calculators add the parallel sides before dividing.

Answer **560,000 mm² = 0.56 m²**

TRY IT 10.4

1. Parallelogram: base 500 mm, slanted side 340 mm, height 300 mm. Area?
2. Parallelogram: base 30 in, slanted side 15 in, height 12 in. Area?
3. Trapezoid: parallel sides 300 mm and 500 mm, height 200 mm. Area?
4. Trapezoid: parallel sides 14 in and 22 in, height 9 in. Area?

Answers are in the Answer Key.

10.5 Circles and rings

The area of a circle uses the **radius**, the distance from the centre to the edge. The radius is half the **diameter**.

Picture the circle inside a square. The square is 2 r on each side, so it holds 4 small squares that are r by r. The circle covers a bit more than 3 of those small squares: exactly **π** of them, and π is about 3.14.

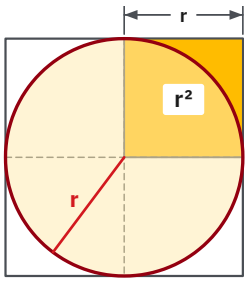


Figure 10.6 The gold square is r × r = r². The circle covers π of these squares.

AREA OF A CIRCLE

$$A = \pi \times r^2$$

r = radius. Square the radius first (exponents come before multiplying in BEDMAS), then multiply by π. Use the π key for the full value.

Example 10.5 A round blank

A round blank is cut with a radius of 150 mm. Find its area to two decimal places.

SOLUTION

1. A = π × r².
2. Square the radius: 150² = 150 × 150 = 22,500.
3. Multiply by π: π × 22,500 = 70,685.83.

CALCULATOR

TI-30Xa	Sharp EL-531XT
π × 150 x² =	π × 150 x² =
Display 70685.83471	Display 70'685.83471

Both calculators square 150 before multiplying by π, because they follow BEDMAS.

Answer 70,685.83 mm²

HALVE THE DIAMETER FIRST

A round cover for an opening 18 in across has r = 9 in, so its area is π × 9² = 254.47 in². Putting the diameter into the formula gives π × 18² = 1,017.88 in², four times too big. When a drawing gives Ø (diameter), divide by 2 before you square.

A half circle has half the area of the full circle, and a quarter circle has a quarter. A ring, such as a washer, a flange or the end of a pipe, is a big circle with a small circle taken out. Find the area of the outside circle (radius R) and subtract the area of the hole (radius r).

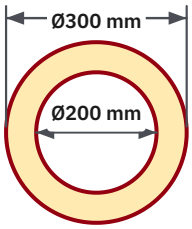
AREA OF A RING

$$A = \pi \times R^2 - \pi \times r^2 = \pi \times (R^2 - r^2)$$

R = outside radius, r = inside radius. Square each radius before subtracting.

Example 10.6 A flange ring

A flat flange ring has an outside diameter of 300 mm and an inside diameter of 200 mm. Find its area to two decimal places.



SOLUTION

1. Halve the diameters: R = 150 mm and r = 100 mm.
2. Square each one: $150^2 = 22{,}500$ and $100^2 = 10{,}000$.
3. Subtract: 22,500 − 10,000 = 12,500.
4. Multiply by π: π × 12,500 = 39,269.91.

CALCULATOR

TI-30Xa

π × (150 x² − 100 x²) =

Display 39269.90817

Sharp EL-531XT

π × (150 x² − 100 x²) =

Display 39'269.90817

Answer **39,269.91 mm²**

SQUARE EACH RADIUS, THEN SUBTRACT

Subtracting the radii first, 150 − 100 = 50, and then squaring gives π × 50^2 = 7,853.98 mm². That is the area of a small circle, not the ring.

TRY IT 10.5

1. Circle, radius 80 mm. Area to two decimal places?
2. Half circle, radius 200 mm. Area to two decimal places?
3. Washer: outside diameter 40 mm, inside diameter 22 mm. Area to two decimal places?
4. Tube end: outside diameter 60 mm, wall 5 mm thick. Area of metal to two decimal places?

Answers are in the Answer Key.

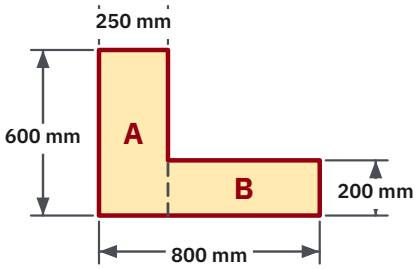
10.6 Composite shapes

A **composite shape** is made of two or more basic shapes: an L-shaped bracket, a plate with a hole, a lug with a round end. The Shape Canvas calls the method **break it down**.

1. Split the shape with dashed lines into basic shapes. Label them A, B, C.
2. Find any missing dimensions from the overall dimensions.
3. Find the area of each piece.
4. Add the pieces. Subtract any hole or cut-out.

Example 10.7 An L-shaped plate

Find the area of the L-shaped plate.



SOLUTION

1. Split on the dashed line into A (the upright) and B (the foot).
2. A: 250 × 600 = 150,000 mm².
3. B is 800 − 250 = 550 mm long: 550 × 200 = 110,000 mm².
4. Add: 150,000 + 110,000 = 260,000 mm².

CALCULATOR

TI-30Xa

250 × 600 + 550 × 200 =

Display 260000.

Sharp EL-531XT

250 × 600 + 550 × 200 =

Display 260'000.

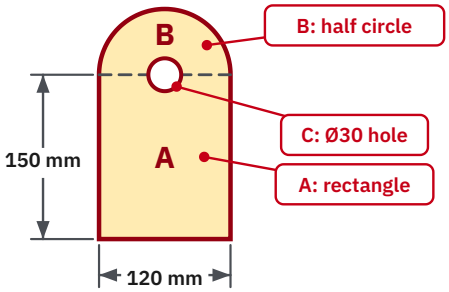
Answer **260,000 mm²**

TWO WAYS, ONE ANSWER

The L is also an 800 × 600 rectangle with a 550 × 400 corner missing.
800 × 600 − 550 × 400 = 260,000 mm², the same answer. When two methods agree, you can trust the answer.

Example 10.8 A lifting lug

A lifting lug is a 120 mm by 150 mm rectangle with a half circle on top. A 30 mm diameter hole is centred where the half circle meets the rectangle. Find the area of the lug to two decimal places.



SOLUTION

1. A: 120 × 150 = 18,000 mm².
2. B: the half circle is 120 mm across, so r = 60 mm. π × 60² ÷ 2 = 5,654.87 mm².
3. C: r = 15 mm. π × 15² = 706.86 mm².
4. A + B − C = 22,948.01 mm².

CALCULATOR

TI-30Xa

120 × 150 + π × 60 x^2 ÷ 2 − π × 15 x^2 =

Display 22948.00843

Sharp EL-531XT

120 × 150 + π × 60 x^2 ÷ 2 − π × 15 x^2 =

Display 22’948.00843

Answer **22,948.01 mm²**

TRY IT 10.6

1. A 500 mm by 300 mm rectangle with a 200 mm by 100 mm corner cut away. Area?
2. A 400 mm by 300 mm plate with a 160 mm diameter hole in the middle. Area to two decimal places?
3. A 300 mm by 200 mm plate with four 20 mm diameter holes. Area to two decimal places?
4. A 700 mm by 400 mm plate with a 200 mm by 150 mm notch cut out of one edge. Area?

Answers are in the Answer Key.

10.7 Surface area and nets

A solid, such as a box or a tank, is covered by faces. Its **surface area** is the total area of all its outside faces. Surface area tells you how much sheet a tank needs, or how much area there is to paint.

A **net** is a flat pattern that folds or rolls up to make the solid. The Shape Canvas calls this **unfold it**. Unfold the solid, find the area of each piece of the net, and add them.

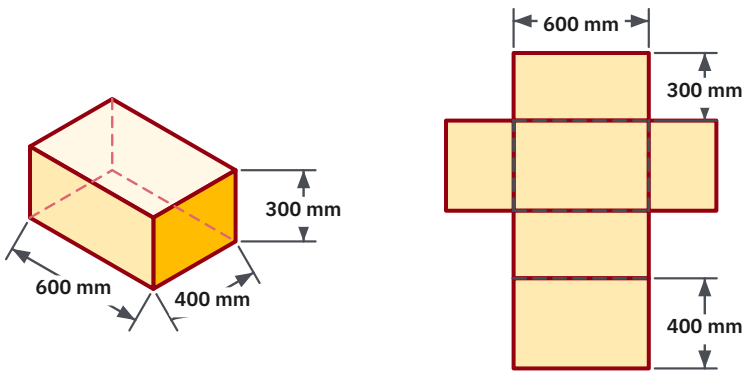


Figure 10.7 A closed box and its net. Opposite faces are the same size, so the net has three pairs of rectangles. Dashed lines are folds.

Example 10.9 Surface area of a closed box

A closed box is 600 mm long, 400 mm wide and 300 mm high. Find its surface area in mm² and m².

SOLUTION

1. Top and bottom: 2 × 600 × 400 = 480,000 mm².
2. Front and back: 2 × 600 × 300 = 360,000 mm².
3. Two ends: 2 × 400 × 300 = 240,000 mm².
4. Add the three pairs: 480,000 + 360,000 + 240,000 = 1,080,000 mm².
5. In m²: 1,080,000 ÷ 1,000,000 = 1.08 m².

CALCULATOR

TI-30Xa

2 × (600 × 400 + 600 × 300 + 400 × 300) =

Display 1080000.

Sharp EL-531XT

2 × (600 × 400 + 600 × 300 + 400 × 300) =

Display 1'080'000.

Answer **1,080,000 mm² = 1.08 m²**

A cylinder unrolls into two circles and one rectangle. The rectangle wraps around the circle, so its length is the distance around the circle, π × D. This length is the **stretch-out**. The width of the rectangle is the height of the cylinder.

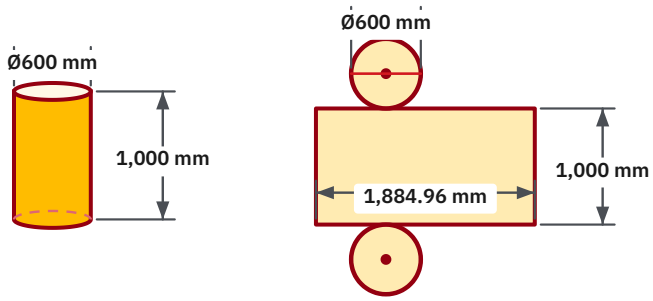


Figure 10.8 A closed cylinder and its net. The circles are 600 mm across, and the stretch-out is π × 600 = 1,884.96 mm.

SURFACE AREA OF A CLOSED CYLINDER

$$\text{Two ends: } 2 \times (\pi \times r^2)$$

$$\text{Side: } \pi \times D \times h$$

$$\text{SA} = \text{two ends} + \text{side}$$

r = radius, D = diameter, h = height. The side is the stretch-out π × D times the height.

Example 10.10 Sheet for a cylinder tank

A closed cylindrical tank has a radius of 300 mm and a height of 1,000 mm. Find its surface area to two decimal places in mm², then in m².

SOLUTION

1. Two ends: π × 300² = 282,743.34 mm² each, so 565,486.68 mm² for both.
2. Side: D = 2 × 300 = 600 mm, and π × 600 × 1,000 = 1,884,955.59 mm².
3. Total: 565,486.68 + 1,884,955.59 = 2,450,442.27 mm².
4. In m²: 2,450,442.27 ÷ 1,000,000 = 2.45 m² to two decimal places.

CALCULATOR

TI-30Xa

Two ends

π × 300 x² × 2 =

Display 565486.6776

Side

π × 600 × 1000 =

Display 1884955.592

Sharp EL-531XT

Two ends

π × 300 x² × 2 =

Display 565'486.6776

Side

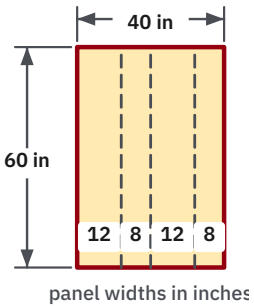
π × 600 × 1000 =

Display 1'884'955.592

Answer **2,450,442.27 mm², about 2.45 m²**

Example 10.11 Sheet for a duct

A rectangular duct is 12 in by 8 in and 60 in long, open at both ends. Its net is one long rectangle: the four sides side by side. Ignore seams. Find the area of sheet in in² and in ft² to two decimal places.



panel widths in inches

SOLUTION

1. The width of the net is the **perimeter** of the duct: 12 + 8 + 12 + 8 = 40 in.
2. Area: 40 × 60 = 2,400 in².
3. In ft²: 2,400 ÷ 144 = 16.67 ft².

CALCULATOR

TI-30Xa

2 × (12 + 8) × 60 ÷ 144 =

Display 16.66666667

Sharp EL-531XT

2 × (12 + 8) × 60 ÷ 144 =

Display 16.66666667

Answer **2,400 in² = 16.67 ft²**

TRY IT 10.7

1. Closed box 500 mm by 400 mm by 200 mm. Surface area in mm²?
2. Open-top box 400 mm by 300 mm by 250 mm (no lid). Surface area in mm²?
3. Duct 10 in by 6 in, 72 in long, open ends. Sheet area in ft² to two decimal places?
4. Closed cylinder, radius 100 mm, height 400 mm. Surface area to two decimal places?

Answers are in the Answer Key.

WORKSHEET 10A

Find the area

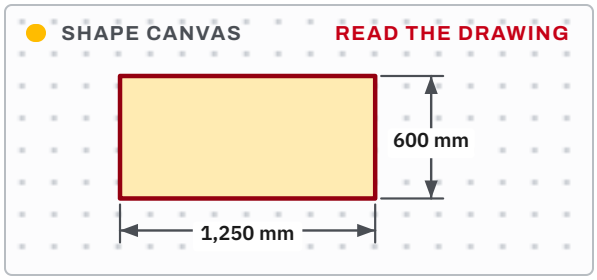
Practice mode: one drawing hides a dimension behind a red ?

Name _____

Date _____

Show the formula, the numbers you put into it and the answer with its square unit.

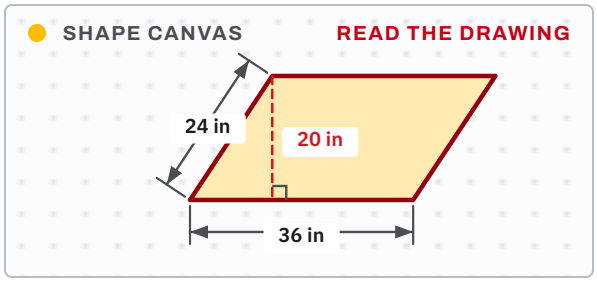
1 Find the area in mm² and in m².



WORKING

Answer _____

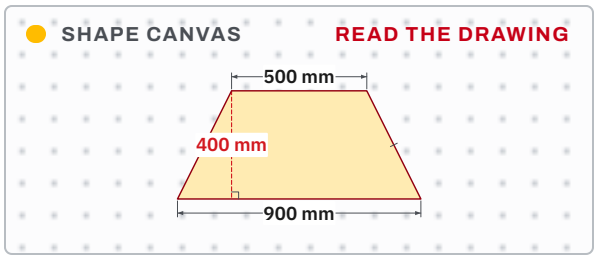
2 Find the area in in² and in ft².



WORKING

Answer _____

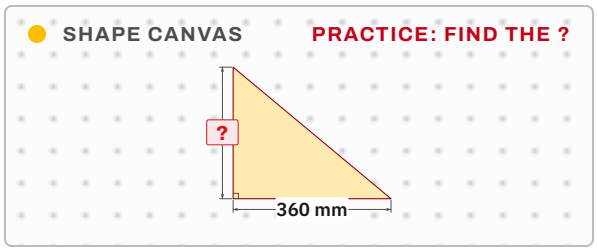
3 Find the area of the trapezoid.



WORKING

Answer _____

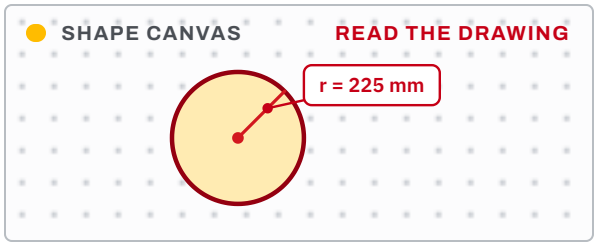
4 The triangle has an area of 54,000 mm². Find the hidden height.



WORKING

Answer _____

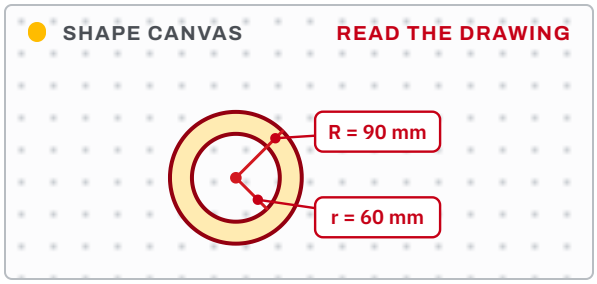
5 Find the area of the disc to the nearest mm².



WORKING

Answer _____

6 Find the area of the washer to two decimal places.



WORKING

Answer _____

WORKSHEET 10B

Break it down

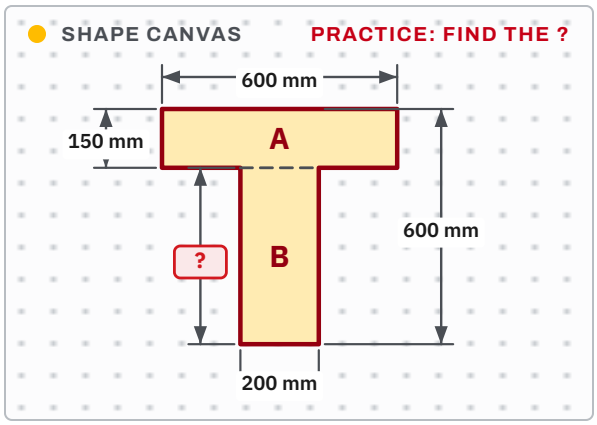
Split each part into basic shapes A, B, C

Name _____

Date _____

Find any ? first. Then find the area of each piece and add, or subtract a hole. Give answers to two decimal places where π is used.

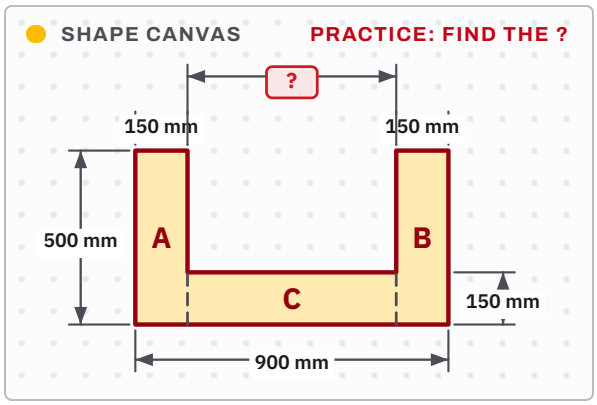
1 Find the hidden height of B, then the area of the T-plate.



WORKING

Answer _____

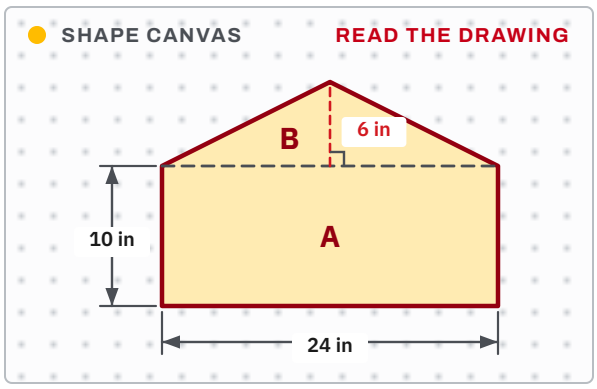
2 Find the hidden width, then the area of the U-shaped plate.



WORKING

Answer _____

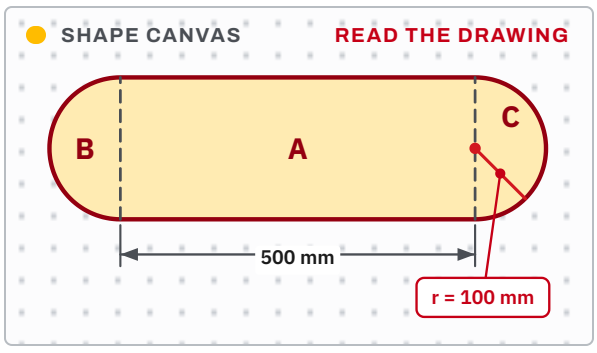
3 Find the area of the gusset in in², then in ft² to two decimal places.



WORKING

Answer _____

4 Find the area of the slotted plate. B and C are half circles.



WORKING

Answer _____

WORKSHEET 10C

Unfold it

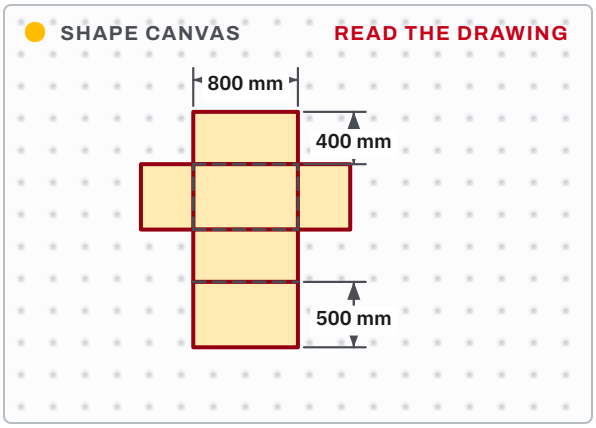
Solids and their nets

Name _____

Date _____

Find the area of each piece of the net, then add. Give answers to two decimal places where π is used.

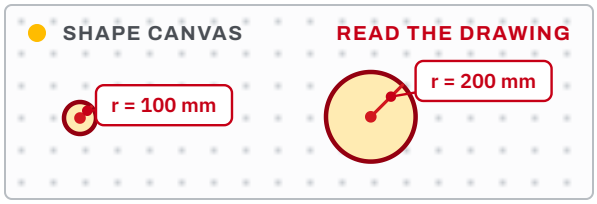
1 This is the net of a closed box. Find the surface area of the box in mm² and in m².



WORKING

Answer _____

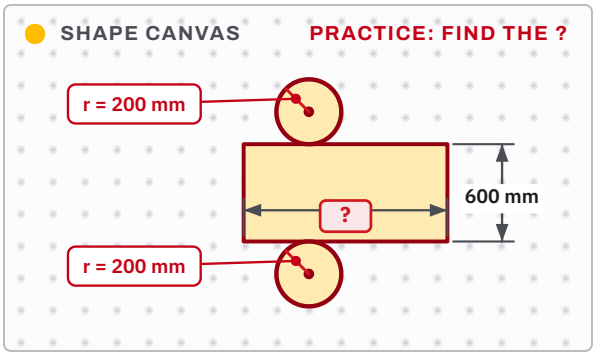
2 Change one value: a disc has a radius of 100 mm. The radius is doubled to 200 mm. Find both areas to two decimal places. How many times bigger is the new area?



WORKING

Answer _____

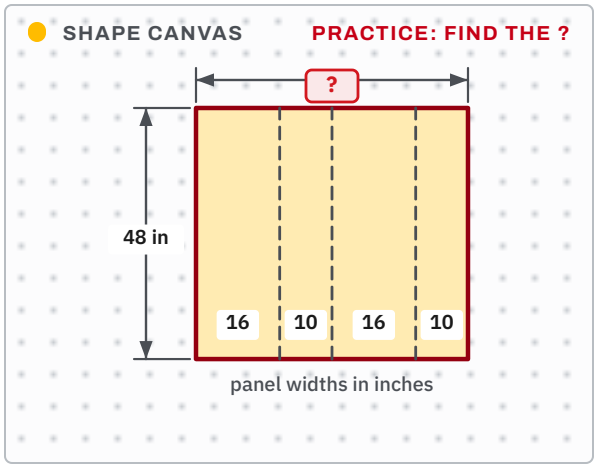
3 This is the net of a closed cylinder with radius 200 mm and height 600 mm. Find the hidden stretch-out to two decimal places, then the surface area.



WORKING

Answer _____

4 The duct is 16 in by 10 in and open at both ends. Find the hidden width of its net, then the sheet area in ft² to two decimal places.



WORKING

Answer _____

CHAPTER

11

Volume and Capacity

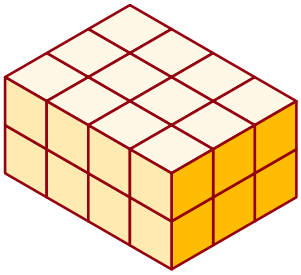
Volume is the amount of space a solid takes up or holds. You need it to size a tank, to know how much water fills a pipe run before a pressure test, and to work out what a hopper holds. This chapter counts cubic units, builds the volume formulas for prisms, cylinders, cones, pyramids and spheres, and changes volumes into litres and US gallons.

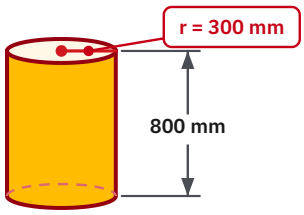
In this chapter you will

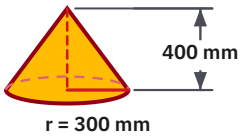
- Explain volume and use cubic units
- Find the volume of prisms with V = end area × length
- Find the volume of cylinders, tanks and pipes
- Find the volume of cones, pyramids and spheres
- Change volumes into litres and US gallons
- Break a composite solid into basic solids

Key words

- volume
- cubic unit
- prism
- cylinder
- cone and pyramid
- sphere
- capacity







11.1 Volume and cubic units

Volume is the amount of space a solid takes up or holds. Length uses one direction and area uses two. Volume uses three: length, width and height.

Volume is counted in cubes. A **cubic unit** is a cube 1 unit long on each edge. A cube 1 cm on each edge is 1 cubic centimetre, written 1 cm³. The small 3 shows that three lengths were multiplied. The cubic units in this book are mm³, cm³ and m³, and in³ and ft³.

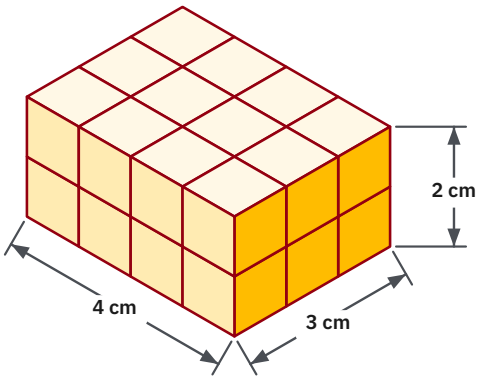


Figure 11.1 One layer holds 4 × 3 = 12 cubes of 1 cm³. Two layers hold 12 × 2 = 24 cm³.

Changing between cubic units uses the length factor three times. A cubic centimetre is 10 mm × 10 mm × 10 mm = 1,000 mm³. A cubic metre is 1,000 mm × 1,000 mm × 1,000 mm = 1,000,000,000 mm³. A cubic foot is 12 in × 12 in × 12 in = 1,728 in³.

THREE FACTORS, NOT ONE

To change mm³ to m³, divide by 1,000,000,000 (1,000 three times). Dividing by 1,000 only changes a length. Many people find it easier to change each length to metres first and then multiply.

TRY IT 11.1

1. A block of 1 cm cubes is 5 long, 2 wide and 3 high. How many cm³?
2. How many mm³ are in 1 cm³?
3. How many in³ are in 2 ft³?
4. Which is a volume: 8 m, 8 m² or 8 m³?

Answers are in the Answer Key.

11.2 Prisms

A **prism** is a solid with two identical, parallel ends joined by flat sides. Cut a slice anywhere along it and the slice is the same shape as the end. A box, a triangular trough and a length of angle bar are all prisms.

Think of the end as one layer. The volume is the area of that layer times how many layers there are, which is the length.

VOLUME OF A PRISM

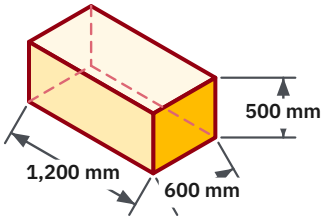
$$V = \text{end area} \times \text{length}$$

$$\text{Box: } V = l \times w \times h$$

Find the end area with the formulas from Chapter 10. All lengths in the same unit.

Example 11.1 A rectangular tank

A tank is 1,200 mm long, 600 mm wide and 500 mm high inside. Find its volume in mm³ and in m³.



SOLUTION

1. $V = l \times w \times h = 1{,}200 \times 600 \times 500 = 360{,}000{,}000$ mm³.
2. In metres the tank is 1.2 m by 0.6 m by 0.5 m.
3. $V = 1.2 \times 0.6 \times 0.5 = 0.36$ m³.

CALCULATOR

TI-30Xa

mm³

1200 × 600 × 500 =

Display 360000000.

m³

1.2 × 0.6 × 0.5 =

Display 0.36

Sharp EL-531XT

mm³

1200 × 600 × 500 =

Display 360’000’000.

m³

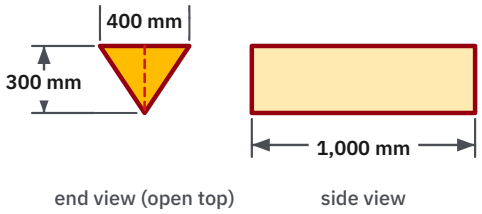
1.2 × 0.6 × 0.5 =

Display 0.36

Answer 360,000,000 mm³ = 0.36 m³

Example 11.2 A triangular trough

A trough is a triangular prism. Its end is a triangle with a base of 400 mm and a height of 300 mm. The trough is 1,000 mm long. Find the volume in m³.



SOLUTION

1. End area: 400 × 300 ÷ 2 = 60,000 mm².
2. Volume: 60,000 × 1,000 = 60,000,000 mm³.
3. In m³: 60,000,000 ÷ 1,000,000,000 = 0.06 m³.

CALCULATOR

TI-30Xa

400 × 300 ÷ 2 × 1000 ÷ 1000000000 =

Display 0.06

Sharp EL-531XT

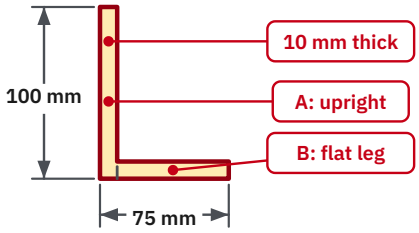
400 × 300 ÷ 2 × 1000 ÷ 1000000000 =

Display 0.06

Answer 0.06 m³

Example 11.3 Break down the end of an angle bar

An angle bar has legs of 100 mm and 75 mm, is 10 mm thick and 300 mm long. Find the volume of steel in cm³.



SOLUTION

1. Break the L-shaped end into two rectangles. Upright: 100 × 10 = 1,000 mm². Flat leg without the corner: (75 − 10) × 10 = 650 mm².
2. End area: 1,650 mm². Volume: 1,650 × 300 = 495,000 mm³.
3. In cm³ (1 cm³ = 1,000 mm³): 495,000 ÷ 1,000 = 495 cm³.

CALCULATOR

TI-30Xa

1650 × 300 ÷ 1000 =

Display 495.

Sharp EL-531XT

1650 × 300 ÷ 1000 =

Display 495.

Answer 495 cm³

TRY IT 11.2

1. Box 800 mm by 500 mm by 300 mm. Volume in m^3?
2. Box 24 in by 18 in by 12 in. Volume in in^3 and ft^3?
3. Triangular prism: end base 0.5 m, end height 0.4 m, length 3 m. Volume?
4. A flat bar 50 mm by 10 mm, 2,000 mm long. Volume in cm^3?

Answers are in the Answer Key.

11.3 Capacity: litres and US gallons

Capacity is how much a container holds. Liquids are measured in litres (L) or in gallons. The litre is tied to the cubic units: a cube 10 cm (100 mm) on each edge holds exactly 1 L.

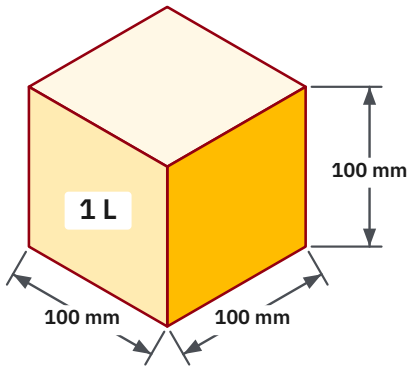


Figure 11.2 A cube 100 mm on each edge: 10 cm × 10 cm × 10 cm = 1,000 cm^3 = 1 L.

Volume	Capacity	Exact or rounded	Source
1,000 cm^3	1 L	exact	SI definition of the litre
1,000,000 mm^3	1 L	exact	100 mm × 100 mm × 100 mm
1 m^3	1,000 L	exact	SI definition of the litre
231 in^3	1 US gal	exact	NIST SP 1038
1 US gal	3.785411784 L	exact	231 × 16.387064 cm^3 (NIST)

Table 11.1 Capacity factors. The gallon factors are the NIST SP 1038 factors used in Chapter 7.

Example 11.4 Litres in a tank

How many litres does the 1,200 mm by 600 mm by 500 mm tank from Example 11.1 hold?

SOLUTION

1. Its volume is 360,000,000 mm³, which is 0.36 m³.
2. Method 1: 1 L = 1,000,000 mm³, so 360,000,000 ÷ 1,000,000 = 360 L.
3. Method 2: 1 m³ = 1,000 L, so 0.36 × 1,000 = 360 L.

CALCULATOR

TI-30Xa

360000000 ÷ 1000000 =

Display 360.

Sharp EL-531XT

360000000 ÷ 1000000 =

Display 360.

Answer 360 L

Example 11.5 A tank in US gallons and litres

A US-made tank is 24 in by 18 in by 12 in inside. Find its capacity in US gallons and in litres, to two decimal places.

SOLUTION

1. Volume: 24 × 18 × 12 = 5,184 in³.
2. US gallons: 1 US gal = 231 in³, so 5,184 ÷ 231 = 22.44 US gal.
3. Litres: multiply the full gallon value by 3.785411784: 84.95 L.

CALCULATOR

TI-30Xa

US gallons

24 × 18 × 12 ÷ 231 =

Display 22.44155844

Litres

24 × 18 × 12 ÷ 231 × 3.785411784 =

Display 84.95053978

Sharp EL-531XT

US gallons

24 × 18 × 12 ÷ 231 =

Display 22.44155844

Litres

24 × 18 × 12 ÷ 231 × 3.785411784 =

Display 84.95053978

Answer 22.44 US gal = 84.95 L

TWO DIFFERENT GALLONS

The US gallon is defined as 231 in³, which is 3.785411784 L (NIST SP 1038). The imperial gallon is defined as 4.54609 L (Weights and Measures Act). Check which gallon a label means. This book uses US gallons.

TRY IT 11.3

1. Write 2.5 m³ in litres.
2. A tank is 1,000 mm by 500 mm by 400 mm inside. Litres?
3. 50 US gal in litres, to two decimal places?
4. 200 L in US gallons, to two decimal places?
5. A box 12 in by 11 in by 7 in. US gallons?
6. How many litres in 1 ft³? Use 1 ft³ = 1,728 in³. Two decimal places.

Answers are in the Answer Key.

11.4 Cylinders: tanks and pipes

A **cylinder** has two equal circular ends joined by a curved side. It works like a prism: every slice is the same circle. So its volume is the circle area times the height (or length).

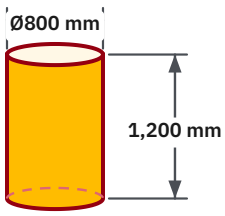
VOLUME OF A CYLINDER

$$V = \pi \times r^2 \times h$$

r = radius of the end, h = height or length. For a pipe, h is the length of the run.

Example 11.6 A vertical tank

A vertical tank is 800 mm across inside and 1,200 mm high. How many litres does it hold, to two decimal places?



SOLUTION

1. Radius: 800 ÷ 2 = 400 mm.
2. $V = \pi \times 400^2 \times 1{,}200 = 603{,}185{,}789.49$ mm³.
3. Litres: ÷ 1,000,000 gives 603.19 L.

CALCULATOR

TI-30Xa

π × 400 x² × 1200 ÷ 1000000 =

Display 603.1857895

Sharp EL-531XT

π × 400 x² × 1200 ÷ 1000000 =

Display 603.1857895

Answer 603.19 L

Example 11.7 Filling a pipe run

A pipe run is 30 m long and the pipe has an inside diameter of 102 mm. How many litres of water fill it, to two decimal places?

SOLUTION

1. Work in metres. Inside radius: 102 ÷ 2 = 51 mm = 0.051 m.
2. $V = \pi \times 0.051^2 \times 30 = 0.24514$ m³.
3. Litres: × 1,000 = 245.14 L.

CALCULATOR

TI-30Xa

π × 0.051 x² × 30 × 1000 =

Display 245.1384748

Sharp EL-531XT

π × 0.051 x² × 30 × 1000 =

Display 245.1384748

Answer 245.14 L

USE THE INSIDE DIAMETER

Capacity uses the inside diameter, because the water fills the inside. The outside diameter tells you how much space the pipe or tank takes up.

Inside diameter	Litres per metre	Compared with 25 mm
25 mm	0.49	1 time
50 mm	1.96	4 times
100 mm	7.85	16 times
200 mm	31.42	64 times

Table 11.2 Litres per metre of pipe, π × r² × 1 m × 1,000. Doubling the diameter holds 2 × 2 = 4 times as much.

TRY IT 11.4

1. Tank 600 mm across inside, 900 mm high. Litres to two decimal places?
2. Pipe, inside diameter 50 mm, 40 m long. Litres to two decimal places?
3. Tank 24 in across inside, 36 in tall. US gallons to two decimal places?
4. Pipe, inside diameter 150 mm. Litres per metre to two decimal places?

Answers are in the Answer Key.

11.5 Cones and pyramids

A **cone** has a circular base that narrows to a point. A **pyramid** has a flat base, often a square, and triangular sides that meet at a point. A cone holds one third of a cylinder with the same base and height, and a pyramid holds one third of a prism with the same base and height.



Figure 11.3 A cone and a square pyramid. Each height is the dashed line from the apex straight down to the centre of the base.

VOLUME OF A CONE AND A PYRAMID

$$V = \text{base area} \times h \div 3$$

$$\text{Cone: } V = \pi \times r^2 \times h \div 3$$

$$\text{Square pyramid: } V = b^2 \times h \div 3$$

h = height at right angles to the base, not the sloping side.

Example 11.8 A cone-shaped hopper

A cone-shaped hopper has a radius of 400 mm and a height of 600 mm. How many litres does it hold, to two decimal places?

SOLUTION

1. Base area: $\pi \times 400^2 = 502{,}654.82$ mm².
2. $V = 502{,}654.82 \times 600 \div 3 = 100{,}530{,}964.91$ mm³.
3. Litres: $\div 1{,}000{,}000 = 100.53$ L.

CALCULATOR

TI-30Xa

[π] [×] [400] [x²] [×] [600] [÷] [3] [÷] [1000000] [=]

Display 100.5309649

Sharp EL-531XT

[π] [×] [400] [x²] [×] [600] [÷] [3] [÷] [1000000] [=]

Display 100.5309649

Answer 100.53 L

TRY IT 11.5

1. Cone, radius 300 mm, height 400 mm. Litres to two decimal places?
2. Square pyramid, base 600 mm, height 400 mm. Litres?
3. Square pyramid, base 12 in, height 10 in. Volume in in³?
4. A cone has a radius of 300 mm, a height of 400 mm and a sloping side of 500 mm. Which length goes into the formula as h?

Answers are in the Answer Key.

11.6 Spheres

A **sphere** is a perfectly round ball. Every point on its surface is the same distance, the radius, from the centre. Half a sphere is a hemisphere.

VOLUME OF A SPHERE

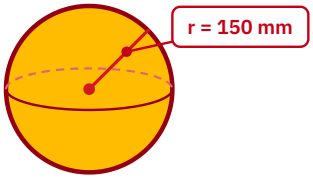
$$V = 4 \div 3 \times \pi \times r^3$$

Hemisphere: half of that

r = radius. r^3 means r × r × r. Use the x^3 key, or y^x 3.

Example 11.9 A spherical float

A spherical float has a radius of 150 mm. Find its volume in litres, to two decimal places.



SOLUTION

1. $r^3 = 150 \times 150 \times 150 = 3{,}375{,}000$.
2. $V = 4 \div 3 \times \pi \times 3{,}375{,}000 = 14{,}137{,}166.94\ \text{mm}^3$.
3. Litres: $\div 1{,}000{,}000 = 14.14$ L.

CALCULATOR

TI-30Xa

4 ÷ 3 × π × 150 2nd x³ ÷ 1000000 =

Display 14.13716694

Sharp EL-531XT

4 ÷ 3 × π × 150 x³ ÷ 1000000 =

Display 14.13716694

Answer **14.14 L**

TRY IT 11.6

1. Sphere, radius 100 mm. Litres to two decimal places?
2. Hemisphere, radius 300 mm. Litres to two decimal places?
3. Sphere, diameter 1 m. m^3 to three decimal places?
4. Sphere, radius 6 in. Volume in in^3 to two decimal places?

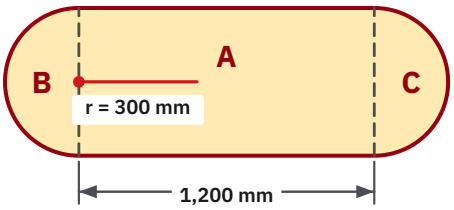
Answers are in the Answer Key.

11.7 Composite solids

Many tanks and hoppers are composite solids. Break the solid into basic solids, find each volume, and add. Take away the volume of any hollow part.

Example 11.10 A tank with round ends

A tank is a cylinder 1,200 mm long with a hemisphere on each end. The radius is 300 mm. How many litres does it hold, to two decimal places?



SOLUTION

1. The two hemispheres make one sphere. Cylinder: π × 300² × 1,200 = 339,292,006.59 mm³.
2. Sphere: 4 ÷ 3 × π × 300³ = 113,097,335.53 mm³.
3. Total: 452,389,342.12 mm³ = 452.39 L.

CALCULATOR

TI-30Xa

(π × 300 x^2 × 1200 + 4 ÷ 3 × π × 300 2nd x^3) ÷ 1000000 =

Display 452.3893421

Sharp EL-531XT

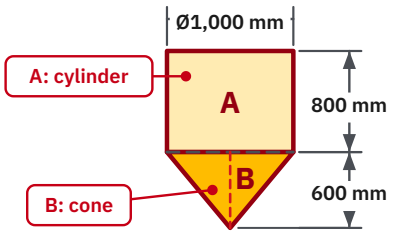
(π × 300 x^2 × 1200 + 4 ÷ 3 × π × 300 x^3) ÷ 1000000 =

Display 452.3893421

Answer 452.39 L

Example 11.11 A hopper

A hopper is a cylinder 1,000 mm across and 800 mm high, on top of a cone 600 mm high. Find the capacity in litres, to two decimal places.



SOLUTION

1. Radius: 1,000 ÷ 2 = 500 mm.
2. A, cylinder: π × 500² × 800 = 628,318,530.72 mm³.
3. B, cone: π × 500² × 600 ÷ 3 = 157,079,632.68 mm³.
4. A + B = 785,398,163.40 mm³ = 785.40 L.

CALCULATOR

TI-30Xa

(π × 500 x² × 800 + π × 500 x² × 600 ÷ 3) ÷ 1000000 =

Display 785.3981634

Sharp EL-531XT

(π × 500 x² × 800 + π × 500 x² × 600 ÷ 3) ÷ 1000000 =

Display 785.3981634

Answer **785.40 L**

CHANGE ONE VALUE

Double the radius of a cylinder and its volume becomes 2 × 2 = 4 times bigger, because r is squared. Double every length of a solid, and the volume becomes 2 × 2 × 2 = 8 times bigger.

TRY IT 11.7

1. Tube: outside diameter 60 mm, inside diameter 50 mm, 2,000 mm long. Volume of metal in cm³, to two decimal places?
2. A cylinder 400 mm across and 500 mm high sits on a cone 300 mm high with the same radius. Litres to two decimal places?
3. A rectangular tank 1,200 mm by 600 mm is filled to a depth of 350 mm. Litres?
4. A cylinder holds 20 L. The radius is doubled and the height kept. How many litres now?

Answers are in the Answer Key.

WORKSHEET 11A

Find the volume

Practice mode: one drawing hides a dimension behind a red ?

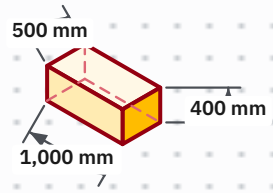
Name _____

Date _____

Write the formula and the numbers. Round to two decimal places where π is used.

1 Volume in m³? Capacity in litres?

SHAPE CANVAS READ THE DRAWING



WORKING

Answer _____

2 Volume of the cube in mm³ and in litres?

SHAPE CANVAS READ THE DRAWING

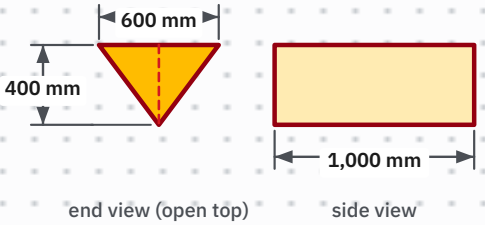


WORKING

Answer _____

3 Capacity of the trough in litres?

SHAPE CANVAS READ THE DRAWING

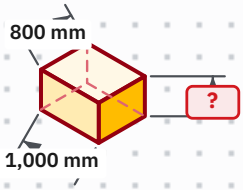


WORKING

Answer _____

4 The tank holds 480 L. Find the ?

SHAPE CANVAS PRACTICE: FIND THE ?

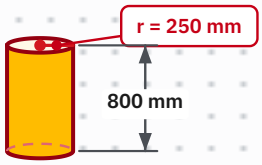


WORKING

Answer _____

5 Capacity of the cylinder in litres?

SHAPE CANVAS READ THE DRAWING

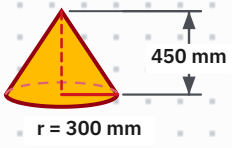


WORKING

Answer _____

6 Capacity of the cone in litres?

SHAPE CANVAS READ THE DRAWING



WORKING

Answer _____

WORKSHEET 11B

Tanks and hoppers

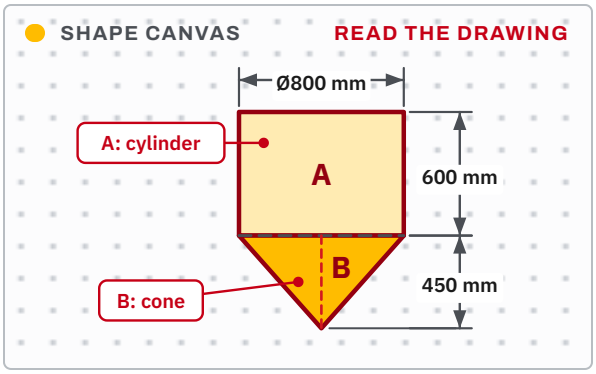
Break each solid into basic solids

Name _____

Date _____

Find each part, then add. Give answers to two decimal places.

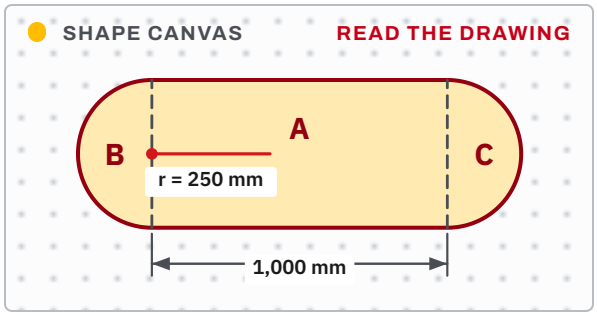
1 Find the capacity of the hopper in litres.



WORKING

Answer _____

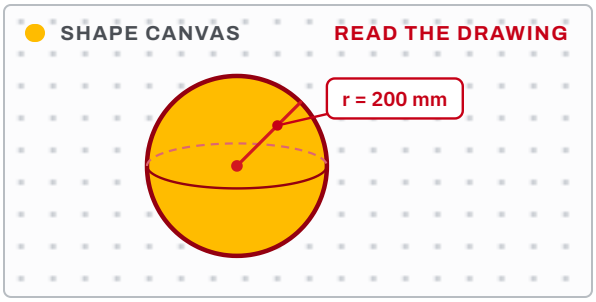
2 The tank has a hemisphere on each end. Find its capacity in litres.



WORKING

Answer _____

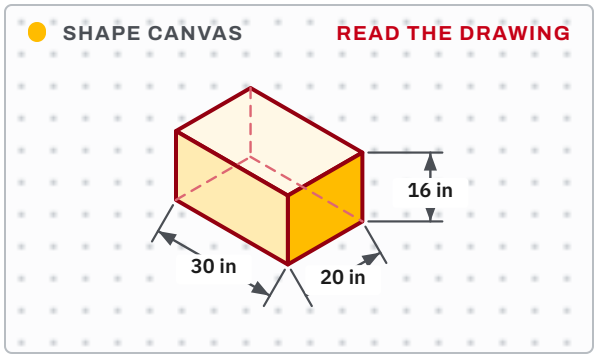
3 Find the capacity of the spherical tank in litres.



WORKING

Answer _____

4 A US-made tank is 30 in by 20 in by 16 in inside. Find its capacity in US gallons.



WORKING

Answer _____

WORKSHEET 11C

Change one value

Unfold it, fold it back, and see what changes

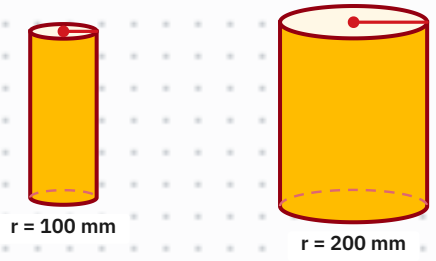
Name _____

Date _____

Give answers to two decimal places. For each pair, say how many times bigger the second answer is.

1 Both cylinders are 500 mm high. The radius doubles from 100 mm to 200 mm. Find both capacities in litres.

SHAPE CANVAS READ THE DRAWING



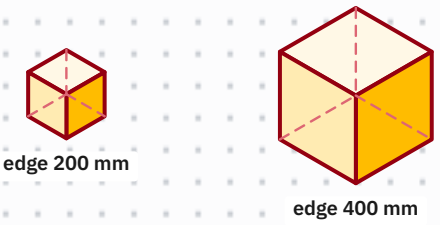
Both are 500 mm high.

WORKING

Answer _____

2 A cube edge doubles from 200 mm to 400 mm. Find both capacities in litres.

SHAPE CANVAS READ THE DRAWING



WORKING

Answer _____

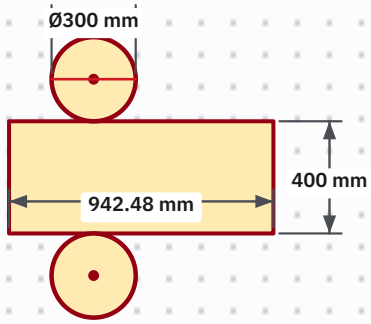
3 A 20 m pipe run has an inside diameter of 50 mm. A second run of the same length is 100 mm inside. How many litres fills each?

WORKING

Answer _____

4 This net folds up into a closed cylinder. Find the capacity of the cylinder in litres.

SHAPE CANVAS READ THE DRAWING



WORKING

Answer _____

CHAPTER

12

Angles and Right Triangles

Square corners hold a shop together. A frame that is out of square will not sit flat, a brace cut to the wrong length will not fit, and a bolt circle laid out at the wrong angle will not line up with its flange.

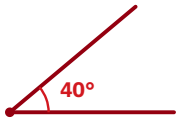
This chapter starts with angles: how they are measured in degrees and how they are named. Then it moves to the right triangle and the Pythagorean theorem, the rule that lets you find a missing side, check a corner for square with the 3-4-5 method and work out the diagonal of a frame.

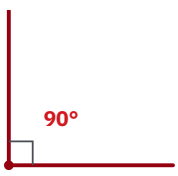
In this chapter you will

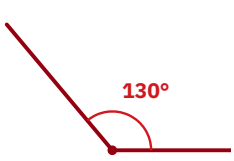
- Measure and name angles in degrees
- Tell acute, right, obtuse and straight angles apart
- Find a missing angle on a line, around a point and in a triangle
- Name the legs and the hypotenuse of a right triangle
- Use the Pythagorean theorem to find a missing side
- Check a corner for square with the 3-4-5 method and with diagonals

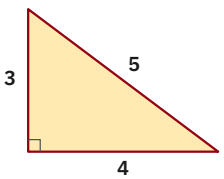
Key words

- angle
- degree
- right angle
- acute angle
- obtuse angle
- leg
- hypotenuse
- Pythagorean theorem









12.1 Angles and degrees

An **angle** is the amount of turn between two lines that meet at a point. The point is called the vertex, and the two lines are the arms of the angle.

Angles are measured in **degrees**. One full turn is 360 degrees, written 360°. Half a turn is 180°, and a quarter turn is 90°.

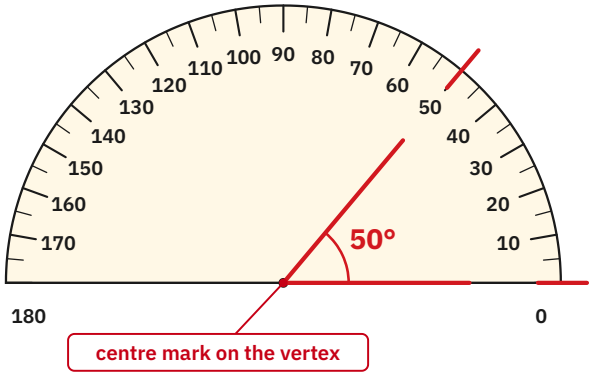


Figure 12.1 A protractor measures an angle. Put the centre mark on the vertex, line one arm up with 0 and read where the other arm crosses the scale. This angle is 50°.

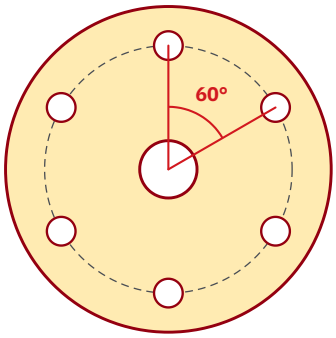
LONG ARMS DO NOT MAKE A BIGGER ANGLE

The size of an angle is the amount of turn, not the length of the arms. A 50° angle drawn with 20 mm arms is the same angle as one drawn with 2 m arms.

Equal parts of a turn come up in layout work all the time. Holes spaced evenly around a circle, such as the bolt holes in a flange, split the full 360° turn into equal angles.

Example 12.1 Angle between bolt holes

Six bolt holes are spaced evenly around a circle. What is the angle between two holes that are next to each other, measured at the centre?



SOLUTION

1. A full turn is 360°.
2. Six equal spaces share the turn: 360° ÷ 6 = 60°.

Answer **60° between holes.**

TRY IT 12.1

1. How many degrees are in half a turn?
2. How many degrees are in one eighth of a turn?
3. Eight bolt holes are evenly spaced on a circle. Angle between neighbours?
4. Five holes are evenly spaced on a circle. Angle between neighbours?
5. Twelve holes are evenly spaced on a circle. Angle between neighbours?
6. What fraction of a full turn is 270°?

Answers are in the Answer Key.

12.2 Types of angles

Angles are named by their size. The most important one in the shop is the **right angle**, exactly 90°. A right angle is a square corner, and drawings mark it with a small square instead of an arc.

An **acute angle** is less than 90°. An **obtuse angle** is more than 90° and less than 180°. A straight angle is exactly 180°: the two arms make a straight line.

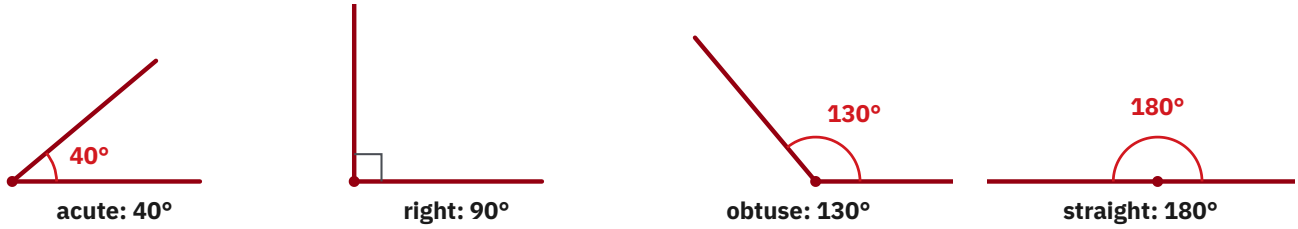
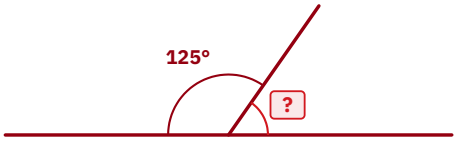


Figure 12.2 The four kinds of angle, drawn at their true size.

Two facts let you find a missing angle without a protractor. Angles that sit side by side on a straight line add to 180°. Angles that go all the way around a point add to 360°. For example, if three lines meet at a point and two of the angles are 110° and 95°, the third angle is 360° − 110° − 95° = 155°.

Example 12.2 A bent strap

A flat strap is bent so that one side of the bend makes an angle of 125° with the straight part, as shown. Find the angle on the other side of the bend.



SOLUTION

1. The two angles sit on a straight line, so they add to 180°.
2. Missing angle = 180° − 125° = 55°.

Answer **55°, an acute angle.**

TRY IT 12.2

1. Name each angle: 35°, 90°, 150°, 89°, 180°.
2. Two angles on a straight line. One is 72°. Find the other.
3. Two angles on a straight line. One is 135°. Find the other.
4. Three angles around a point: 120°, 135° and ?. Find the ?
5. Four equal angles go around a point. How big is each?
6. Is 91° acute, right or obtuse?

Answers are in the Answer Key.

12.3 Angles in a triangle

The three angles inside any triangle add to 180°. You can see why by tearing the three corners off a paper triangle and laying them side by side. They fit together into a straight line.



Figure 12.3 The corners of a 50°, 60°, 70° triangle (left) placed side by side make a straight angle: 50° + 60° + 70° = 180°.

ANGLES IN A TRIANGLE

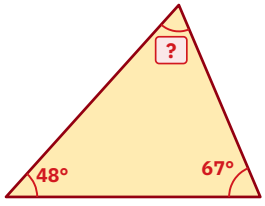
$$\angle A + \angle B + \angle C = 180°$$

In a right triangle: the two acute angles add to 90°

∠ means "angle". A right triangle already uses 90° of the 180°, so the other two angles share the other 90°.

Example 12.3 The third angle of a gusset

A triangular gusset has angles of 48° and 67°. Find the third angle.



SOLUTION

1. The three angles add to 180°.
2. Third angle = 180° − 48° − 67° = 65°.

Answer 65°

A triangle with two equal sides also has two equal angles. They are the angles at the ends of the third side. If the angle between the equal sides is 40°, the other two angles share 180° − 40° = 140°, so each one is 140° ÷ 2 = 70°.

TRY IT 12.3

1. A triangle has angles of 35° and 85°. Find the third angle.
2. A triangle has angles of 90° and 27°. Find the third angle.
3. One acute angle of a right triangle is 45°. Find the other.
4. Can a triangle have two right angles? Why?
5. A gusset has two equal sides with 50° between them. Find the other two angles.
6. All three angles of a triangle are equal. How big is each?

Answers are in the Answer Key.

12.4 Right triangles and the Pythagorean theorem

A right triangle has one right angle. Its sides have special names. The two sides that make the right angle are the **legs**. The side across from the right angle is the **hypotenuse**.

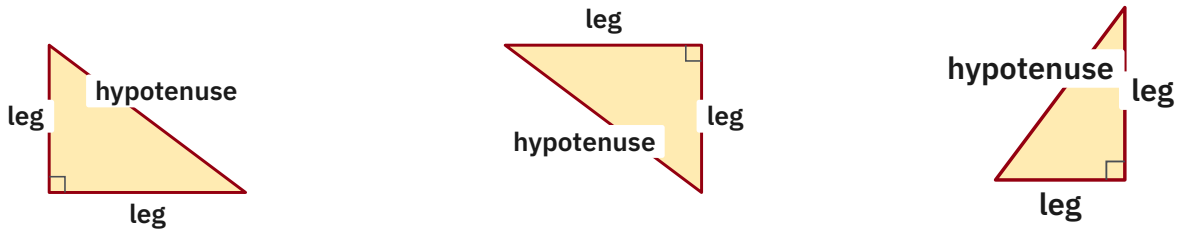


Figure 12.4 Three right triangles turned different ways. The hypotenuse is always across from the small square that marks the right angle, and it is always the longest side.

The **Pythagorean theorem** links the three sides of every right triangle. Square each leg and add the results. The total is the **square** of the hypotenuse.

Figure 12.5 shows it with real squares. A right triangle with legs of 3 and 4 units has a hypotenuse of 5 units. The squares on the legs hold 9 and 16 small squares. The square on the hypotenuse holds 25, and 9 + 16 = 25.

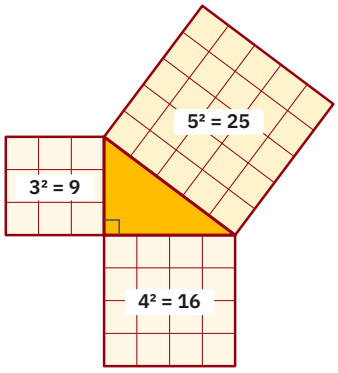


Figure 12.5 The squares on the sides of a 3-4-5 right triangle: 9 + 16 = 25.

PYTHAGOREAN THEOREM

$$a^2 + b^2 = c^2$$

Find the hypotenuse: $c = \sqrt{(a^2 + b^2)}$

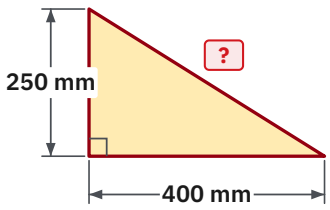
Find a leg: $a = \sqrt{(c^2 - b^2)}$

a and b are the legs, c is the hypotenuse. The second and third lines are the theorem **transposed** (Chapter 8) to make c or a the subject.

1. Sketch the triangle and mark the right angle.
2. Label the hypotenuse c and the legs a and b.
3. To find the hypotenuse, square both legs, add, then take the **square root**.
4. To find a leg, square the hypotenuse and the known leg, subtract, then take the square root.
5. Check: the hypotenuse must be the longest side.

Example 12.4 Length of a gusset edge

A triangular gusset has legs of 400 mm and 250 mm. Find the length of the sloped edge (the hypotenuse) to two decimal places.



SOLUTION

1. $c^2 = 400^2 + 250^2 = 160{,}000 + 62{,}500 = 222{,}500$.
2. $c = \sqrt{222{,}500}$.

CALCULATOR

TI-30Xa

(400 x^2 + 250 x^2) √x

Display 471.6990566

Sharp EL-531XT

√ (400 x^2 + 250 x^2) =

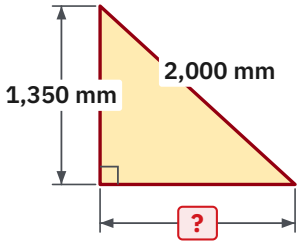
Display 471.6990566

The brackets make the calculator add the squares before it takes the square root. The Sharp puts the √ key first and the brackets after it.

Answer **471.70 mm**

Example 12.5 Finding a leg

A brace 2,000 mm long runs from the floor to a column. Its top end is 1,350 mm above the floor. How far from the column is its bottom end? Give the answer to two decimal places.



SOLUTION

1. The brace is the hypotenuse: c = 2,000. The known leg is a = 1,350.
2. $b^2 = 2{,}000^2 - 1{,}350^2 = 4{,}000{,}000 - 1{,}822{,}500 = 2{,}177{,}500$.
3. $b = \sqrt{2{,}177{,}500}$.

CALCULATOR

TI-30Xa

(2000 x^2 − 1350 x^2) √x

Display 1475.635456

Sharp EL-531XT

√ (2000 x^2 − 1350 x^2) =

Display 1’475.635456

Answer **1,475.64 mm. Check: it is shorter than the 2,000 mm brace.**

ADD TO FIND THE HYPOTENUSE, SUBTRACT TO FIND A LEG

If you add the squares when you are looking for a leg, your answer will be longer than the hypotenuse. That cannot happen. A leg is always shorter than the hypotenuse.

Example 12.6 Feet and inches

A right triangle has legs of 2′ 8″ and 1′ 8″. Find the hypotenuse to the nearest 1/16″, in feet and inches.

SOLUTION

1. Change the legs to inches: 2′ 8″ = 2 × 12 + 8 = 32″ and 1′ 8″ = 1 × 12 + 8 = 20″.
2. $c^2 = 32^2 + 20^2 = 1{,}024 + 400 = 1{,}424$.
3. $c = \sqrt{1{,}424} = 37.73592$″ (the display shows more digits).
4. Change the decimal part to sixteenths: 0.73592 × 16 = 11.77, which rounds to 12 sixteenths.
5. 12/16 = $\frac{3}{4}$ in lowest terms.

CALCULATOR

TI-30Xa

(32 x² + 20 x²) √x

Display 37.73592453

Sharp EL-531XT

√ (32 x² + 20 x²) =

Display 37.73592453

Answer $37\frac{3}{4}$″, which is 3′ $1\frac{3}{4}$″.

TRY IT 12.4

1. Legs 600 mm and 800 mm. Find the hypotenuse.
2. Legs 1,500 mm and 900 mm. Find the hypotenuse to two decimal places.
3. Hypotenuse 1,300 mm, one leg 500 mm. Find the other leg.
4. Hypotenuse 2.5 m, one leg 1.8 m. Find the other leg to two decimal places.
5. Legs 9″ and 12″. Find the hypotenuse.
6. Legs 2′ 0″ and 1′ 3″. Find the hypotenuse to the nearest 1/16″.

Answers are in the Answer Key.

12.5 The 3-4-5 check for square

The theorem also works backwards. If the sides of a triangle fit $a^2 + b^2 = c^2$, the corner between a and b is a right angle. The smallest whole numbers that fit are 3, 4 and 5: $3^2 + 4^2 = 9 + 16 = 25 = 5^2$.

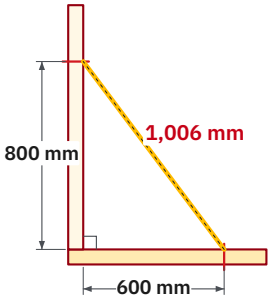
Any multiple of 3, 4 and 5 works too, in any unit. Pick the largest set that fits on the work. A reading error of 1 mm matters less on a 1,000 mm diagonal than on a 500 mm one.

1. From the corner, measure and mark 3 units along one edge.
2. From the same corner, measure and mark 4 units along the other edge.
3. Measure the straight distance between the two marks.

4 If it is exactly 5 units, the corner is square. If it is longer, the corner is open past 90°. If it is shorter, the corner is closed to less than 90°.

Example 12.7 Is the frame square?

You mark 600 mm along one side of a frame corner and 800 mm along the other. The tape reads 1,006 mm between the marks. Is the corner square?



SOLUTION

1. For a square corner: $600^2 + 800^2 = 360{,}000 + 640{,}000 = 1{,}000{,}000$, and $\sqrt{1{,}000{,}000} = 1{,}000$ mm.
2. The tape reads 1,006 mm, which is 6 mm too long.
3. A diagonal that is too long means the corner is opened past 90°.

Answer **No. Close the corner until the marks are 1,000 mm apart.**

Leg a	Leg b	Diagonal c	Check: $a^2 + b^2 = c^2$
300 mm	400 mm	500 mm	$90{,}000 + 160{,}000 = 250{,}000 = 500^2$
600 mm	800 mm	1,000 mm	$360{,}000 + 640{,}000 = 1{,}000{,}000 = 1{,}000^2$
3 ft	4 ft	5 ft	$9 + 16 = 25 = 5^2$
6 ft	8 ft	10 ft	$36 + 64 = 100 = 10^2$

Table 12.1 Multiples of 3-4-5 in metric and imperial units.

TRY IT 12.5

1. Marks at 300 mm and 400 mm. What should the diagonal be?
2. Marks at 9 ft and 12 ft. What should the diagonal be?
3. Marks at 900 mm and 1,200 mm. The diagonal reads 1,494 mm. Is the corner open or closed?
4. Marks at 1.2 m and 1.6 m. What should the diagonal be?
5. Marks at 15″ and 20″. What should the diagonal be?
6. Do 5, 12 and 13 make a right triangle? Show the check.

Answers are in the Answer Key.

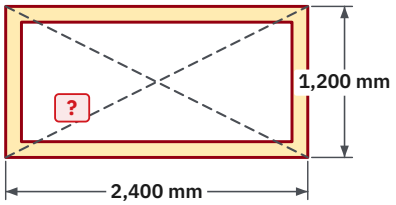
12.6 Diagonals of rectangles

A diagonal joins opposite corners of a rectangle. It cuts the rectangle into two right triangles, so its length comes from the Pythagorean theorem: $d = \sqrt{(l^2 + w^2)}$.

Diagonals are the fastest check for a whole frame. Measure both diagonals corner to corner. If the opposite sides are equal and the two diagonals are equal, every corner is 90°.

Example 12.8 Diagonal of a base frame

A base frame is 2,400 mm long and 1,200 mm wide. What should each diagonal measure, to the nearest millimetre?



SOLUTION

1. $d^2 = 2{,}400^2 + 1{,}200^2 =$
   $5{,}760{,}000 + 1{,}440{,}000 = 7{,}200{,}000.$
2. $d = \sqrt{7{,}200{,}000} = 2{,}683.28$ mm.

CALCULATOR

TI-30Xa

(2400 x² + 1200 x²) √x

Display 2683.281573

Sharp EL-531XT

√ (2400 x² + 1200 x²) =

Display 2′683.281573

Answer 2,683 mm

SQUARING A FRAME

When the two diagonals of a frame are different, the two corners at the ends of the long diagonal are less than 90°. Push those two corners toward each other and measure again until both diagonals match.

TRY IT 12.6

1. A rectangle is 1,600 mm by 1,200 mm. Find the diagonal.
2. A frame is 1,800 mm by 750 mm. Find the diagonal.
3. A frame is 3,000 mm by 1,500 mm. Find the diagonal to the nearest millimetre.
4. A square has 300 mm sides. Find the diagonal to two decimal places. Hint: both legs are 300 mm.
5. A frame measures 2,013 mm on one diagonal and 2,020 mm on the other. Is it square?
6. A door opening is 36″ by 80″. Find the diagonal to the nearest 1/16″.

Answers are in the Answer Key.

WORKSHEET 12A

Angles on the canvas

Each drawing shows the angles at their true size. Find every red ?

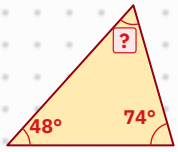
Name _____

Date _____

Use the angle facts from this chapter: a straight line is 180°, a full turn is 360° and the angles of a triangle add to 180°. Write the fact you used in the working box.

1 Find the third angle of the triangle.

SHAPE CANVAS PRACTICE: FIND THE ?

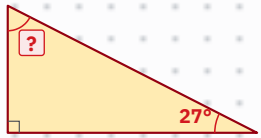


WORKING

Answer _____

2 Find the missing angle of the right triangle.

SHAPE CANVAS PRACTICE: FIND THE ?



WORKING

Answer _____

3 A strap is bent as shown. Find the ?

SHAPE CANVAS PRACTICE: FIND THE ?

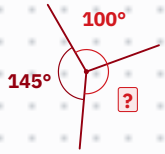


WORKING

Answer _____

4 Three lines meet at a point. Find the ?

SHAPE CANVAS PRACTICE: FIND THE ?

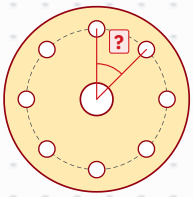


WORKING

Answer _____

5 Eight holes are evenly spaced. Find the angle between neighbours.

SHAPE CANVAS PRACTICE: FIND THE ?

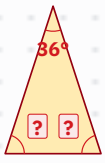


WORKING

Answer _____

6 The gusset has two equal sides. Find each ?

SHAPE CANVAS PRACTICE: FIND THE ?

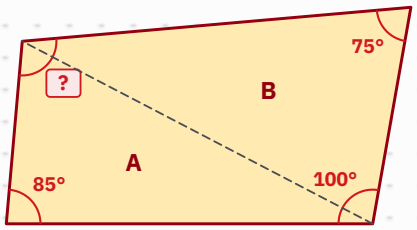


WORKING

Answer _____

7 Break it down. The dashed line splits the plate into triangles A and B. How many degrees do the four angles of the plate add to? Find the ?

SHAPE CANVAS — PRACTICE: FIND THE ?

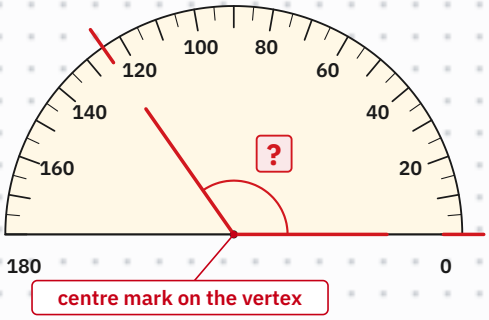


WORKING

Answer _____

8 Read the protractor. How big is the angle, and what type of angle is it?

SHAPE CANVAS — PRACTICE: FIND THE ?



WORKING

Answer _____

9 Change one value. The plate in problem 5 is made with 10 holes instead of 8, still evenly spaced. What is the angle between neighbours now? With 12 holes? Do more holes give a bigger or a smaller angle?

WORKING

Answer _____

WORKSHEET 12B

Pythagoras on the canvas

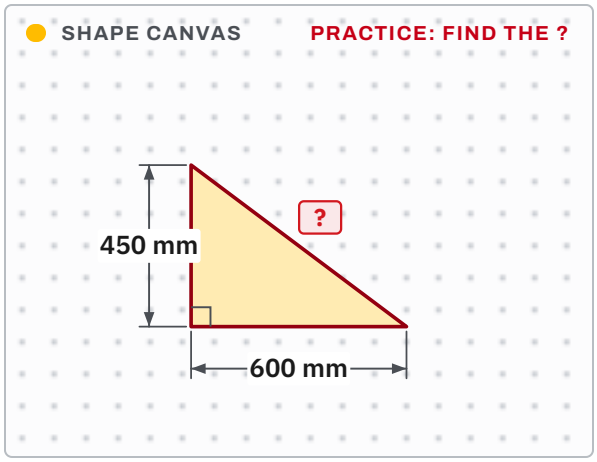
Every triangle has a right angle. Find every red ?

Name _____

Date _____

Label the hypotenuse c before you start. Add the squares to find the hypotenuse. Subtract to find a leg. Check that the hypotenuse is the longest side.

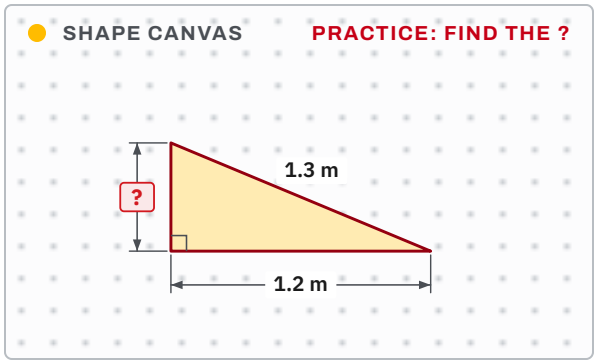
1 Find the hypotenuse.



WORKING

Answer _____

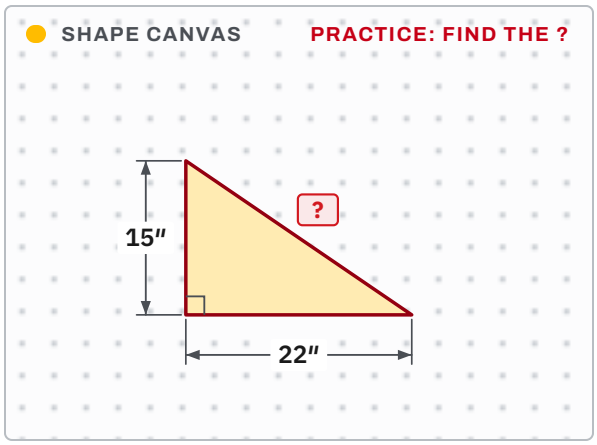
2 Find the missing leg.



WORKING

Answer _____

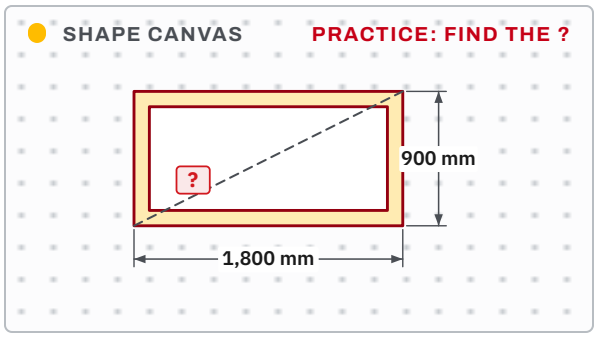
3 Find the hypotenuse to the nearest 1/16″.



WORKING

Answer _____

4 Find the diagonal of the frame to two decimal places.

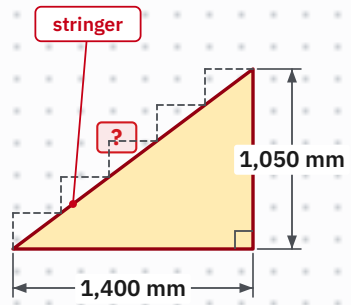


WORKING

Answer _____

5 A stair has a total rise of 1,050 mm and a total run of 1,400 mm. How long is the stringer from the bottom corner to the top corner?

SHAPE CANVAS — PRACTICE: FIND THE ?

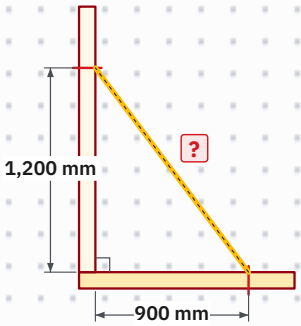


WORKING

Answer _____

6 You mark 900 mm and 1,200 mm from a corner. The tape reads 1,507 mm between the marks. What should it read? Is the corner open or closed?

SHAPE CANVAS — PRACTICE: FIND THE ?



WORKING

Answer _____

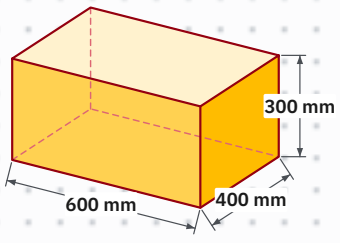
7 Change one value. A right triangle has legs of 300 mm and 400 mm. Find the hypotenuse. Then double both legs. What happens to the hypotenuse?

WORKING

Answer _____

8 Break it down. A box is 600 mm long, 400 mm wide and 300 mm high. First find the diagonal across the bottom. Then use it as a leg to find the longest straight rod that fits inside, corner to corner. Give the answer to two decimal places.

SHAPE CANVAS — READ THE DRAWING



WORKING

Answer _____

CHAPTER

13

Trigonometry

The Pythagorean theorem finds a side when you know two sides. On the job you often know one side and an angle instead: a brace set at 35°, a ramp that must not be steeper than a given slope, a pipe offset made with 45° fittings.

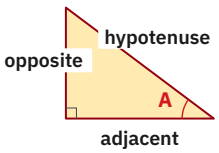
Trigonometry links the angles of a right triangle to its sides. With three ratios, sine, cosine and tangent, and the SIN, COS and TAN keys on your calculator, you can find any side or any angle of a right triangle.

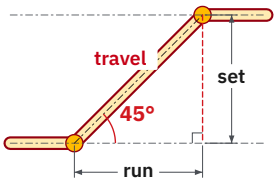
In this chapter you will

- Name the opposite side, adjacent side and hypotenuse from a given angle
- Write the sine, cosine and tangent ratios
- Use SIN, COS and TAN in DEG mode on both calculators
- Find a missing side from one side and one angle
- Find a missing angle with the inverse keys
- Solve slopes, ramps, pipe offsets, bevels and mitres

Key words

- trigonometry
- opposite side
- adjacent side
- sine
- cosine
- tangent
- inverse trig function
- offset





13.1 Naming the sides

Trigonometry is the math that links the angles of a right triangle to the lengths of its sides. It always starts from one of the two acute angles. From that angle, each side gets a name.

- The **hypotenuse** is across from the right angle. It is the longest side.
- The **opposite side** is across from the angle you are working with. It does not touch that angle.
- The **adjacent side** is next to the angle. It is the side that touches the angle and is not the hypotenuse.



Figure 13.1 The same triangle named from angle A (left) and from angle B (right). The opposite and adjacent sides swap. The hypotenuse stays the same.

MARK THE ANGLE FIRST

Put your finger on the angle you are working with. The side your finger cannot touch is the opposite side. Of the two sides that touch it, the longer one is the hypotenuse and the other one is the adjacent side.

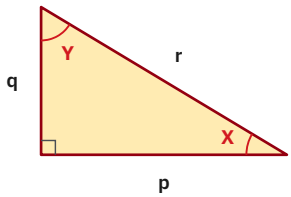


Figure 13.2 A right triangle with sides p, q and r and acute angles X and Y.

TRY IT 13.1

1. In Figure 13.2, which side is opposite angle X?
2. Which side is adjacent to angle X?
3. Which side is the hypotenuse?
4. Which side is opposite angle Y?
5. Which side is adjacent to angle Y?
6. Angle X is 31°. How big is angle Y?

Answers are in the Answer Key.

13.2 Sine, cosine and tangent

Draw any right triangle with a 30° angle, large or small. Divide the opposite side by the hypotenuse and you always get 0.5. The ratio of two sides depends only on the angle, not on the size of the triangle. That is why one angle and one side are enough to find the rest.

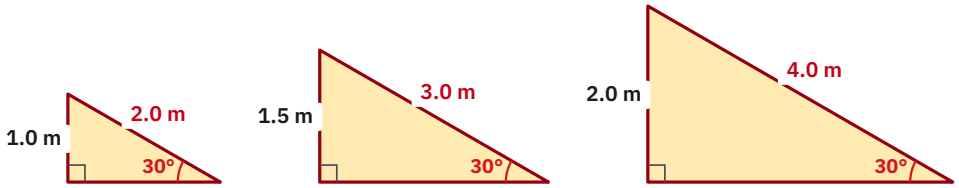


Figure 13.3 Three 30° triangles. In each one, opposite ÷ hypotenuse = 1 ÷ 2 = 1.5 ÷ 3 = 2 ÷ 4 = 0.5.

Each pair of sides gives a ratio with its own name. The **sine** (sin) of an angle is opposite ÷ hypotenuse. The **cosine** (cos) is adjacent ÷ hypotenuse. The **tangent** (tan) is opposite ÷ adjacent.

THE THREE TRIG RATIOS

$$\sin A = \text{opposite} \div \text{hypotenuse}$$

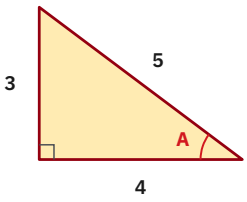
$$\cos A = \text{adjacent} \div \text{hypotenuse}$$

$$\tan A = \text{opposite} \div \text{adjacent}$$

A memory aid: SOH CAH TOA. S = O ÷ H, C = A ÷ H, T = O ÷ A.

Example 13.1 The ratios of a 3-4-5 triangle

A right triangle has sides 3, 4 and 5. Angle A is opposite the side of 3. Write sin A, cos A and tan A as fractions and as decimals.



SOLUTION

1. From A: opposite = 3, adjacent = 4, hypotenuse = 5.
2. sin A = 3 ÷ 5 = 0.6.
3. cos A = 4 ÷ 5 = 0.8.
4. tan A = 3 ÷ 4 = 0.75.

Answer **sin A = 3/5 = 0.6, cos A = 4/5 = 0.8, tan A = 3/4 = 0.75.**

The trig keys and DEG mode

Angles in this book are in degrees, so both calculators must be in degree mode. On the TI-30Xa, check that DEG shows in the display. If it does not, press DRG until it does. On the Sharp EL-531XT, DEG shows at the top of the display, and the DRG key steps through DEG, RAD and GRAD. Chapter 2 shows the keys.

The keys go in a different order. On the TI-30Xa, key the angle first, then SIN, COS or TAN. On the Sharp, press sin, cos or tan first, then the angle, then =.

CALCULATOR Check your mode: sin 30°

TI-30Xa

30 SIN

Display 0.5

Sharp EL-531XT

sin 30 =

Display 0.5

If sin 30° does not give 0.5, the calculator is not in degree mode. In radian mode the same keys give −0.988. Press DRG until DEG shows, then try again.

TRY IT 13.2

1. sin 40° to four decimal places
2. cos 40° to four decimal places
3. tan 40° to four decimal places
4. tan 45°
5. sin 60° and cos 30° to four decimal places. What do you notice?
6. A right triangle has opposite 5, adjacent 12, hypotenuse 13. Find sin A and tan A to four decimal places.

Answers are in the Answer Key.

13.3 Finding a side

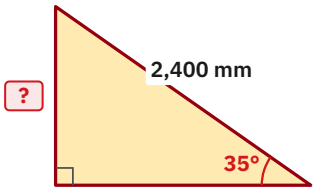
1. Mark the angle you know.
2. Name the side you know and the side you want: opposite, adjacent or hypotenuse.
3. Pick the ratio that uses those two sides: SOH, CAH or TOA.
4. Write the ratio with the numbers in it and **transpose** it for the side you want.
5. Check: the hypotenuse must be the longest side.

WHEN THE SIDE YOU WANT IS ON THE BOTTOM, DIVIDE

If the unknown side is the hypotenuse in SOH or CAH, or the adjacent side in TOA, it sits under the fraction bar. Then you divide the known side by sin, cos or tan instead of multiplying.

Example 13.2 Height of a brace (sine)

A brace 2,400 mm long is set at 35° to the floor. How high up the wall does it reach? Give the answer to two decimal places.



SOLUTION

1. From the 35° angle: the brace is the hypotenuse, the height is the opposite side.
2. Opposite and hypotenuse: SOH. sin 35° = opposite ÷ 2,400.
3. Multiply both sides by 2,400: opposite = 2,400 × sin 35°.

CALCULATOR

TI-30Xa

35 | SIN | × | 2400 | =

Display 1376.583447

Sharp EL-531XT

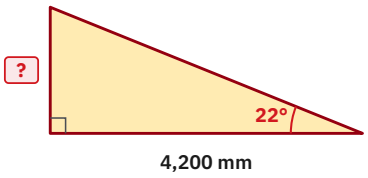
sin | 35 | × | 2400 | =

Display 1’376.583447

Answer 1,376.58 mm

Example 13.3 Rise of a truss (tangent)

The top chord of a truss rises at 22° over a half span of 4,200 mm. How high is the peak above the bottom chord? Give the answer to two decimal places.



SOLUTION

1. The half span is the adjacent side. The height is the opposite side.
2. Opposite and adjacent: TOA. opposite = 4,200 × tan 22°.

CALCULATOR

TI-30Xa

22 | TAN | × | 4200 | =

Display 1696.910149

Sharp EL-531XT

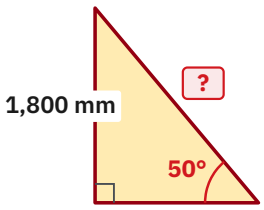
tan | 22 | × | 4200 | =

Display 1’696.910149

Answer 1,696.91 mm

Example 13.4 Length of a brace (divide)

A brace must reach 1,800 mm up a column and meet the floor at 50°. How long is the brace? Give the answer to two decimal places.



SOLUTION

1. The height is the opposite side. The brace is the hypotenuse.
2. SOH: sin 50° = 1,800 ÷ hypotenuse.
3. Transpose: hypotenuse = 1,800 ÷ sin 50°.

CALCULATOR

TI-30Xa

[1800] [÷] [50] [SIN] [=]

Display: 2349.733121

Sharp EL-531XT

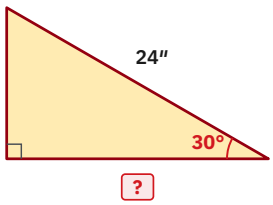
[1800] [÷] [sin] [50] [=]

Display: 2'349.733121

Answer **2,349.73 mm. Check: longer than the 1,800 mm height.**

Example 13.5 Inches to the nearest sixteenth

A 24″ gusset edge is set at 30° to the base. Find the base (the adjacent side) to the nearest 1/16″.



SOLUTION

1. CAH: adjacent = 24 × cos 30° = 20.78461″ (the display shows more digits).
2. Sixteenths: 0.78461 × 16 = 12.55, which rounds to 13.

CALCULATOR

TI-30Xa

[30] [COS] [×] [24] [=]

Display: 20.78460969

Sharp EL-531XT

[cos] [30] [×] [24] [=]

Display: 20.78460969

Answer $20\frac{13}{16}''$

TRY IT 13.3

1. Hypotenuse 800 mm, angle 25°. Find the opposite side to two decimal places.
2. Hypotenuse 1,500 mm, angle 40°. Find the adjacent side to two decimal places.
3. Adjacent 2,000 mm, angle 18°. Find the opposite side to two decimal places.
4. Opposite 600 mm, angle 35°. Find the hypotenuse to two decimal places.
5. Opposite 900 mm, angle 52°. Find the adjacent side to two decimal places.
6. Hypotenuse 30″, angle 40°. Find the opposite side to the nearest 1/16″.

Answers are in the Answer Key.

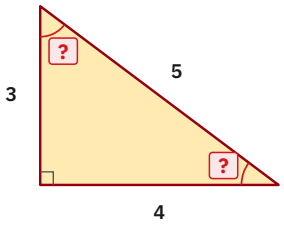
13.4 Finding an angle

When you know two sides, work out their ratio and then turn the ratio back into an angle. That is the job of an **inverse trig function**: SIN^{-1}, COS^{-1} and TAN^{-1}. They are the second functions of the SIN, COS and TAN keys.

On the TI-30Xa, key the ratio, then 2nd and SIN, COS or TAN. Put a division in brackets so it is worked out first. On the Sharp, press 2ndF and sin, cos or tan first, then the ratio, then =.

Example 13.6 The angles of a 3-4-5 triangle

Find both acute angles of a 3-4-5 triangle to two decimal places. Angle A is opposite the side of 3.



SOLUTION

1. From A, the opposite side is 3 and the adjacent side is 4: TOA.
2. $\tan A = 3 \div 4$, so $A = \tan^{-1}(3 \div 4) = 36.87°$.
3. The two acute angles add to 90°: $B = 90° - 36.87° = 53.13°$.

CALCULATOR

TI-30Xa

A from TOA

(3 ÷ 4) 2nd TAN⁻¹

Display 36.86989765

Check with SOH

(3 ÷ 5) 2nd SIN⁻¹

Display 36.86989765

Sharp EL-531XT

A from TOA

2ndF tan⁻¹ (3 ÷ 4) =

Display 36.86989765

Check with SOH

2ndF sin⁻¹ (3 ÷ 5) =

Display 36.86989765

Any ratio that uses two known sides gives the same angle. Here SOH, sin A = 3 ÷ 5, checks the answer.

Answer **A = 36.87°, B = 53.13°**

RATIO KEY OR ANGLE KEY?

SIN, COS and TAN turn an angle into a ratio. SIN^{-1}, COS^{-1} and TAN^{-1} turn a ratio into an angle. A sine or cosine ratio is always less than 1, because the hypotenuse is the longest side. If yours is more than 1, the sides are mixed up.

TRY IT 13.4

1. Opposite 500, adjacent 800. Find the angle to two decimal places.
2. Opposite 700, hypotenuse 1,000. Find the angle to two decimal places.
3. Adjacent 1,100, hypotenuse 1,400. Find the angle to two decimal places.
4. Opposite 9″, adjacent 12″. Find both acute angles to two decimal places.
5. sin A = 0.5. Find A.
6. tan A = 1. Find A.

Answers are in the Answer Key.

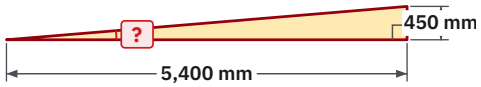
13.5 Slopes and ramps

A slope is a right triangle lying on its side. The rise is the opposite side, the run is the adjacent side and the length along the slope is the hypotenuse. The angle of the slope comes from the tangent: angle = $\tan^{-1}$(rise ÷ run).

Slopes are also given as a **percent** grade: rise ÷ run × 100. A slope written as 1 in 12 rises 1 unit for every 12 units of run.

Example 13.7 Angle and grade of a ramp

A ramp rises 450 mm over a run of 5,400 mm. Find its angle and its percent grade to two decimal places.



SOLUTION

1. Rise and run are opposite and adjacent: TOA.
2. Angle = $\tan^{-1}(450 \div 5{,}400)$.
3. Grade = $450 \div 5{,}400 \times 100 = 8.33\%$.

CALCULATOR

TI-30Xa

(450 ÷ 5400) 2nd TAN⁻¹

Display 4.763641691

Sharp EL-531XT

2ndF tan⁻¹ (450 ÷ 5400) =

Display 4.763641691

Answer **Angle 4.76°, grade 8.33%. The drawing is to scale: a slope this gentle looks almost flat.**

1 IN 12

450 ÷ 5,400 is the same ratio as 1 ÷ 12. So a slope of 1 in 12 has the same angle, 4.76°.

Example 13.8 Fall of a drain line

A drain line runs 12,000 mm at a grade of 2%. How far does it fall, and what is its angle to two decimal places?

SOLUTION

1. Grade = fall ÷ run × 100, so fall = run × grade ÷ 100 = 12,000 × 2 ÷ 100 = 240 mm.
2. Angle = $\tan^{-1}(2 \div 100) = \tan^{-1}(0.02) = 1.15°$. Key it the same way as the ramp angle.

Answer **Fall 240 mm, angle 1.15°.**

TRY IT 13.5

1. Rise 300 mm, run 4,000 mm. Find the angle and the grade to two decimal places.
2. A slope of 1 in 20. Find the angle to two decimal places.
3. A 1% grade over a 15,000 mm run. How far does it fall?
4. A ramp is 3,000 mm long, measured along the slope. It slopes at 4°. Find the rise to two decimal places.
5. A roof rises 6″ for every 12″ of run. Find its angle to two decimal places.
6. A ramp must rise 600 mm at an angle of 3°. How long is the run, to the nearest millimetre?

Answers are in the Answer Key.

13.6 Pipe offsets

An **offset** moves a pipe over to a new line that is parallel to the first one. In this book an offset has two fittings of the same angle, and the three measurements of an offset are the sides of a right triangle. In this book, set, run and travel are measured along the centreline of the pipe.

- **Set:** how far the pipe moves over, square to the two parallel lines. It is the side opposite the fitting angle.
- **Run:** the distance along the line of the pipe that the offset takes up. It is the side adjacent to the fitting angle.
- **Travel:** the length along the centreline of the angled part, from one fitting to the other. It is the hypotenuse.

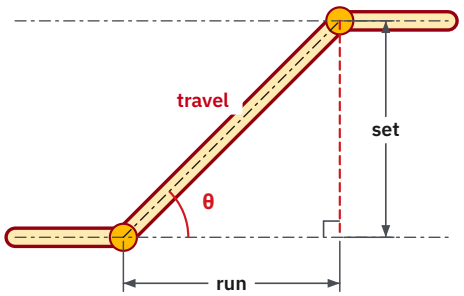


Figure 13.4 A 45° offset. Set, run and travel form a right triangle. The fitting angle θ (theta) is at the first fitting.

OFFSETS

$$\text{travel} = \text{set} \div \sin\theta$$

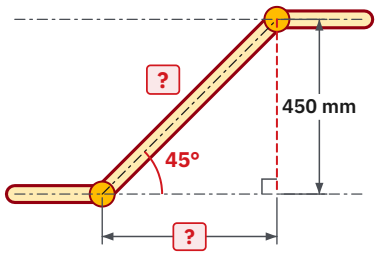
$$\text{run} = \text{set} \div \tan\theta$$

$$\text{set} = \text{travel} \times \sin\theta$$

θ (theta) is the fitting angle. These are SOH and TOA with the set as the opposite side.

Example 13.9 A 45° offset

A line must move over 450 mm using 45° fittings. Find the travel and the run, to two decimal places.



SOLUTION

1. Travel = 450 ÷ sin 45°.
2. Run = 450 ÷ tan 45°. Since tan 45° = 1, the run equals the set: 450.00 mm.

CALCULATOR

TI-30Xa

450 ÷ 45 SIN =

Display 636.3961031

Sharp EL-531XT

450 ÷ sin 45 =

Display 636.3961031

Answer **Travel 636.40 mm, run 450.00 mm.**

Example 13.10 A 60° offset in inches

A line must move over 14″ using 60° fittings. Find the travel and the run to the nearest 1/16″.

SOLUTION

1. Travel = 14 ÷ sin 60° = 16.166″.
2. Run = 14 ÷ tan 60° = 8.083″.
3. Change each decimal part to sixteenths and round.

CALCULATOR

TI-30Xa

Travel

14 ÷ 60 SIN =

Display 16.16580754

Run

14 ÷ 60 TAN =

Display 8.082903769

Sharp EL-531XT

Travel

14 ÷ sin 60 =

Display 16.16580754

Run

14 ÷ tan 60 =

Display 8.082903769

Answer **Travel $16\frac{3}{16}$″, run $8\frac{1}{16}$″.**

Where offset constants come from

Travel = set ÷ sin θ is the same as set × (1 ÷ sin θ). The number 1 ÷ sin θ depends only on the fitting angle, so it can be worked out once and kept in a table. The same goes for 1 ÷ tan θ for the run. Every one of these numbers comes from the calculator:

Fitting angle θ	Travel constant 1 ÷ sin θ	Run constant 1 ÷ tan θ
60°	1.155	0.577
45°	1.414	1.000
30°	2.000	1.732
22.5°	2.613	2.414
11.25°	5.126	5.027

Table 13.1 Offset constants worked out from sine and tangent, to three decimal places.

CALCULATOR The 45° travel constant: 1 ÷ sin 45°

TI-30Xa

45 SIN 1/x

Display 1.414213562

Sharp EL-531XT

(sin 45) 2ndF x^{-1} =

Display 1.414213562

The 1/x key (2ndF x^{-1} on the Sharp) divides 1 by the number. For the run constant, use TAN instead of SIN.

A ROUNDED CONSTANT GIVES A ROUNDED ANSWER

With the constant 1.414, the 45° travel above is 450 × 1.414 = 636.30 mm. Dividing by sin 45° on the calculator gives 636.40 mm. The difference grows with the set, so use the calculator value when you can. To find the fitting angle from a set and a run, use θ = $\tan^{-1}$(set ÷ run).

TRY IT 13.6

1. 45° fittings, set 600 mm. Find the travel to two decimal places.
2. 60° fittings, set 400 mm. Find the travel and run to two decimal places.
3. 22.5° fittings, set 250 mm. Find the travel to two decimal places.
4. 45° fittings, set 10″. Find the travel to the nearest 1/16″.
5. Travel 900 mm with 30° fittings. Find the set.
6. Set 350 mm, run 350 mm. What fitting angle?

Answers are in the Answer Key.

13.7 Bevel and mitre angles

A mitre joint joins two pieces at a corner. Each piece is cut at an angle so that the joint line splits the corner angle in half. For a square corner (90°), each piece is cut at 45° to its edge.

A frame with equal sides and equal corners turns 360° in all, so each corner turns 360° ÷ n, where n is the number of sides. The corner angle inside the frame is 180° − 360° ÷ n, and each piece is cut at half of that to its edge.

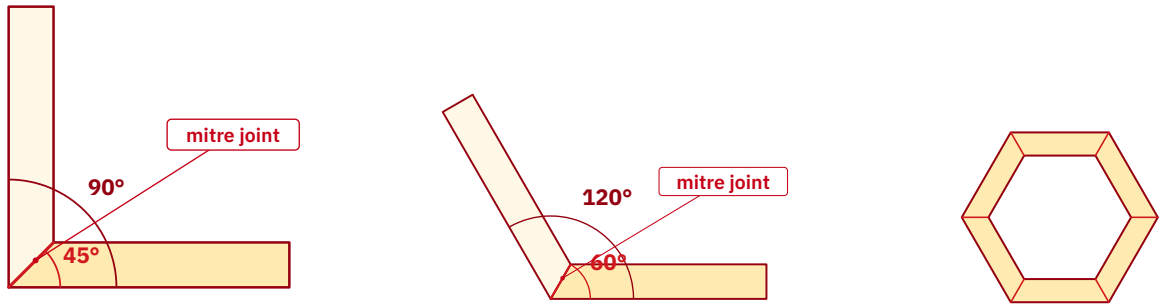


Figure 13.5 Mitre joints for a square corner (90°, cut at 45°) and for a six-sided frame (120°, cut at 60°).

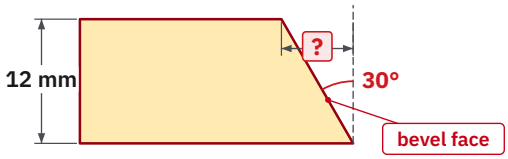
Sides n	Corner angle 180° − 360° ÷ n	Cut angle to the edge (half the corner)	Angle away from a square cut
4	90°	45°	45°
6	120°	60°	30°
8	135°	67.5°	22.5°
12	150°	75°	15°

Table 13.2 Mitre angles for frames with equal sides.

A bevel is an angled face cut along the edge of a plate. In this book the bevel angle is measured from the square edge. Through the full thickness t, the bevel is t × tan(angle) wide at the top. Two plates bevelled at the same angle and butted together open a groove of twice the bevel angle.

Example 13.11 Width of a bevel

A 12 mm plate is bevelled at 30° through its full thickness. How wide is the bevel at the top, to two decimal places?



SOLUTION

1. The thickness is the side adjacent to the bevel angle. The width is the opposite side: TOA.
2. Width = 12 × tan 30°.

CALCULATOR

TI-30Xa

12 × 30 TAN =

Display 6.92820323

Sharp EL-531XT

12 × tan 30 =

Display 6.92820323

Answer **6.93 mm. Two plates bevelled like this open a 60° groove. To find a bevel angle from the thickness and the width, use $\tan^{-1}$(width ÷ thickness).**

TRY IT 13.7

1. A frame has 5 equal sides. Find the corner angle and the cut angle.
2. Each corner of a frame is 150°. How many sides does it have?
3. A 20 mm plate is bevelled at 30° through its thickness. Bevel width to two decimal places?
4. A 16 mm plate has a 9 mm wide bevel. Find the bevel angle to two decimal places.
5. Two plates are each bevelled at 35°. What is the groove angle?
6. A square frame corner is cut at 45°. A cut of 46° is made by mistake on both pieces. What corner angle do the pieces make?

Answers are in the Answer Key.

WORKSHEET 13A

Find the side

Each triangle is drawn at its true angle. Find every red ?

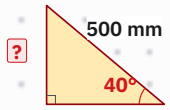
Name _____

Date _____

Mark the angle, name the sides, pick SOH, CAH or TOA, then multiply or divide. Give metric answers to two decimal places and inch answers to the nearest 1/16″.

1 Find the opposite side.

SHAPE CANVAS PRACTICE: FIND THE ?

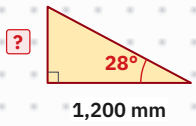


WORKING

Answer _____

2 Find the height.

SHAPE CANVAS PRACTICE: FIND THE ?

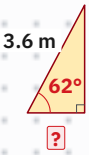


WORKING

Answer _____

3 Find the base.

SHAPE CANVAS PRACTICE: FIND THE ?

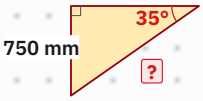


WORKING

Answer _____

4 Find the hypotenuse.

SHAPE CANVAS PRACTICE: FIND THE ?

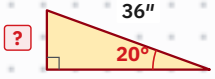


WORKING

Answer _____

5 Find the opposite side to the nearest 1/16″.

SHAPE CANVAS PRACTICE: FIND THE ?



WORKING

Answer _____

6 Find the hypotenuse to the nearest 1/16″.

SHAPE CANVAS PRACTICE: FIND THE ?



WORKING

Answer _____

WORKSHEET 13B

Find the angle

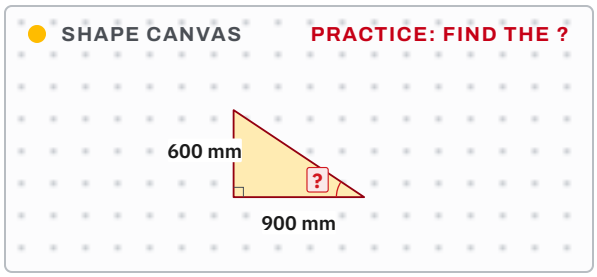
Use the inverse keys. Find every red ?

Name _____

Date _____

Name the two sides you know, pick the ratio, then use SIN^{-1}, COS^{-1} or TAN^{-1}. Give angles and percent grades to two decimal places. Check: the two acute angles add to 90°.

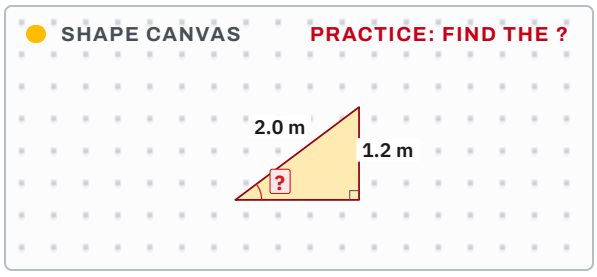
1 Find the angle.



WORKING

Answer _____

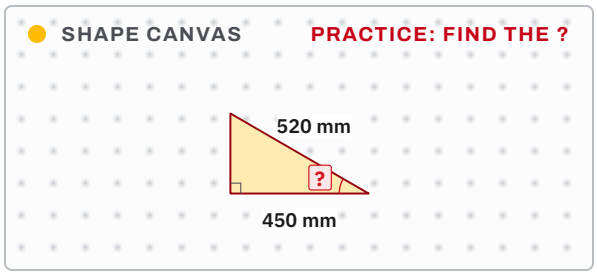
2 Find the angle.



WORKING

Answer _____

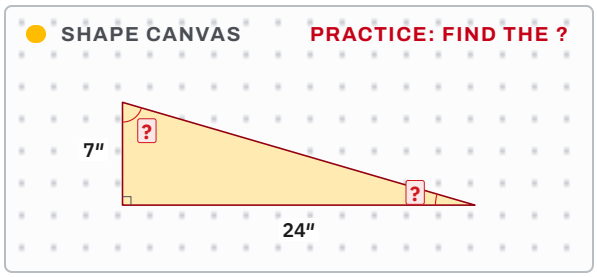
3 Find the angle.



WORKING

Answer _____

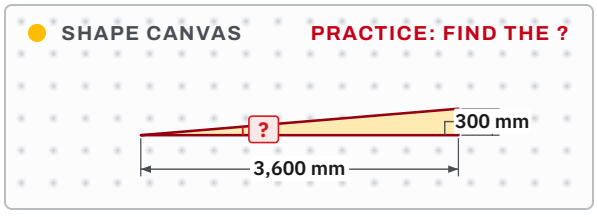
4 Find both acute angles.



WORKING

Answer _____

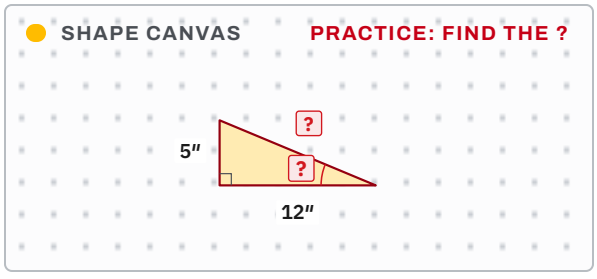
5 A ramp rises 300 mm over a run of 3,600 mm. Find its angle and its percent grade.



WORKING

Answer _____

6 A roof rises 5″ for every 12″ of run. Find the roof angle and the sloped length.



WORKING

Answer _____

WORKSHEET 13C

Shop triangles

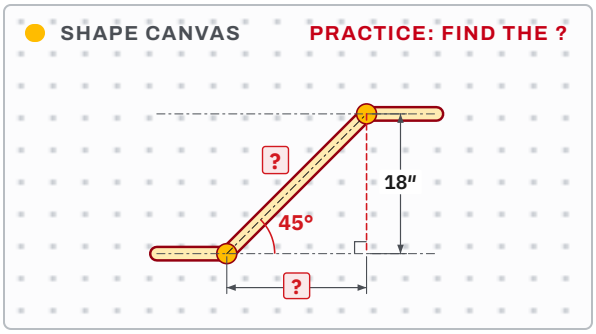
Offsets, bevels and mitres. Find every red ?

Name _____

Date _____

Find the right triangle in each job first. Set, run and travel are opposite, adjacent and hypotenuse from the fitting angle.

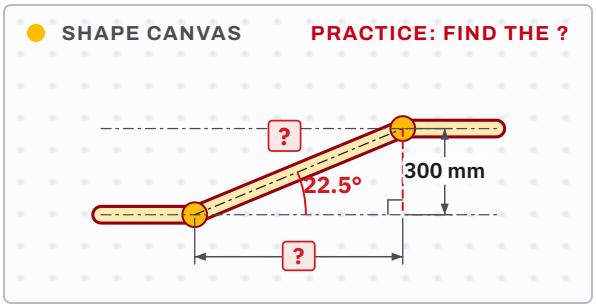
1 A 45° offset has a set of 18″. Find the travel to the nearest 1/16″ and the run.



WORKING

Answer _____

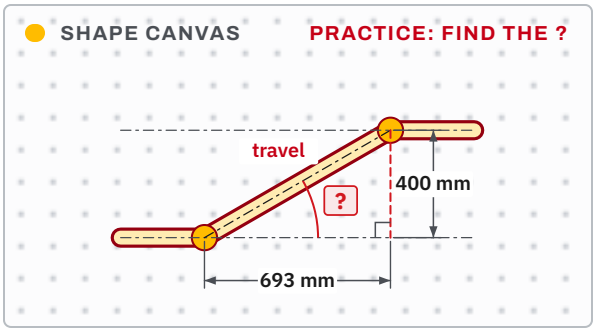
2 A 22.5° offset has a set of 300 mm. Find the travel and the run to two decimal places.



WORKING

Answer _____

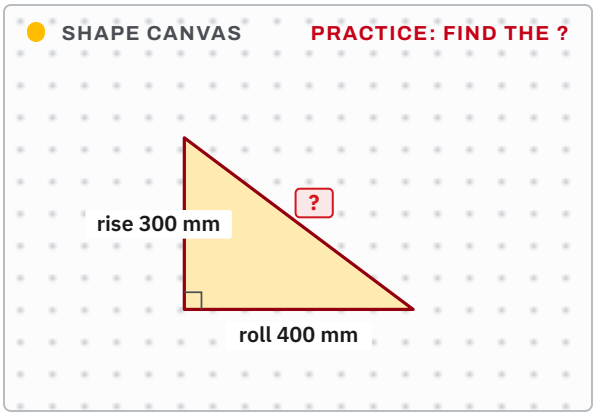
3 An offset has a set of 400 mm and a run of 693 mm. Find the fitting angle to two decimal places.



WORKING

Answer _____

4 Break it down. A rolling offset rises 300 mm and rolls 400 mm. First find the true offset (the set) with the Pythagorean theorem. Then find the travel with 45° fittings. Give both to two decimal places.

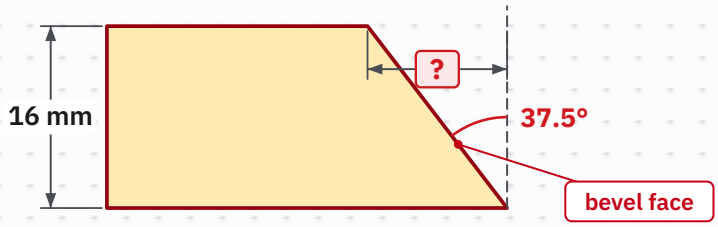


WORKING

Answer _____

5 A 16 mm plate is bevelled at 37.5° through its thickness. Find the bevel width to two decimal places.

SHAPE CANVAS — PRACTICE: FIND THE ?

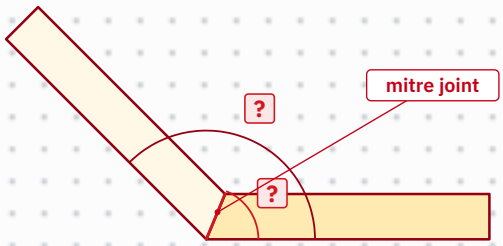


WORKING

Answer _____

6 An eight-sided frame. Find the corner angle and the cut angle of each piece.

SHAPE CANVAS — PRACTICE: FIND THE ?



WORKING

Answer _____

7 Change one value. A ramp is 3,000 mm long, measured along the slope, and slopes at 6°. Find its rise to two decimal places. Then double the angle to 12°. Does the rise double too?

WORKING

Answer _____

Formula sheet

Order of operations

Brackets, Exponents, Division and Multiplication (left to right), Addition and Subtraction (left to right)	**BEDMAS**

Perimeter and distance around

Rectangle	$P = 2 \times (l + w)$
Any shape with straight sides	**P = sum of the sides**
Circle	$C = \pi \times D$
Diameter and radius	$D = 2 \times r$

Area

Rectangle	$A = l \times w$
Square	$A = s^2$
Parallelogram	$A = b \times h$
Triangle	$A = b \times h \div 2$
Trapezoid	$A = (a + b) \times h \div 2$
Circle	$A = \pi \times r^2$
Ring (annulus)	$A = \pi \times (R^2 - r^2)$
Curved side of a cylinder	$A = \pi \times D \times h$

Volume

Rectangular prism	$V = l \times w \times h$
Any prism or cylinder	**V = base area × h**
Cylinder	$V = \pi \times r^2 \times h$
Cone	$V = \pi \times r^2 \times h \div 3$
Pyramid	**V = base area × h ÷ 3**
Sphere	$V = 4 \div 3 \times \pi \times r^3$

Right triangles

Pythagorean theorem	$c^2 = a^2 + b^2$
Angles of a triangle	$A + B + C = 180°$
Sine	$\sin \theta$ = **opposite ÷ hypotenuse**
Cosine	$\cos \theta$ = **adjacent ÷ hypotenuse**
Tangent	$\tan \theta$ = **opposite ÷ adjacent**

Percent and averages

Percent of a number	**part = percent ÷ 100 × whole**
Percent	**percent = part ÷ whole × 100**
Average	**sum ÷ how many**

Conversion factors

Exact factors are true by definition. Rounded factors are the exact value cut short. Keep the full exact factor on the calculator and round only the final answer.

From	To	Exact or rounded	Source
1 km	1,000 m	Exact	Weights and Measures Act
1 m	100 cm	Exact	Weights and Measures Act
1 m	1,000 mm	Exact	Weights and Measures Act
1 cm	10 mm	Exact	Weights and Measures Act
1 L	1,000 mL	Exact	Weights and Measures Act
1 kg	1,000 g	Exact	Weights and Measures Act
1 t (tonne)	1,000 kg	Exact	Weights and Measures Act
1 in	25.4 mm	Exact	NIST SP 811
1 ft	12 in = 304.8 mm = 0.3048 m	Exact	NIST SP 811
1 yd	0.9144 m	Exact	Weights and Measures Act
1/16 in	1.5875 mm	Exact	NIST SP 811
1 mm	0.03937 in	Rounded (0.039 370 078 7…)	NPC 2025 conversion tables
1 m	3.281 ft	Rounded (3.280 839 895…)	NPC 2025 conversion tables
1 m^2	1,000,000 mm^2	Exact	Weights and Measures Act
1 m^2	10,000 cm^2	Exact	Weights and Measures Act
1 ft^2	144 in^2	Exact	NIST SP 811
1 in^2	645.16 mm^2	Exact	NIST SP 811
1 ft^2	0.092 903 04 m^2	Exact	NIST SP 811
1 m^2	10.76 ft^2	Rounded (10.763 910 4…)	NPC 2025 conversion tables

1 m^3	1,000 L	Exact	Weights and Measures Act
1 L	1,000,000 mm^3	Exact	Weights and Measures Act
1 ft^3	1,728 in^3	Exact	NIST SP 811
1 ft^3	28.316 846 592 L	Exact	NIST SP 811
1 ft^3	7.48 US gal	Rounded (7.480 519 48… (1,728 ÷ 231))	NIST SP 1038
1 US gal	231 in^3 = 3.785 411 784 L	Exact	NIST SP 1038
1 imperial gal	4.546 09 L	Exact	Weights and Measures Act
1 imperial gal	1.201 US gal	Rounded (1.200 949 9…)	NIST SP 811
1 L	0.2200 imperial gal	Rounded (0.219 969 2…)	NPC 2025 conversion tables
1 psi	6.894 757 kPa	Rounded (6.894 757 293 168… (from the exact pound, inch and standard gravity))	NIST SP 811
1 kPa	0.1450 psi	Rounded (0.145 037 7…)	NPC 2025 conversion tables
1 bar	100 kPa	Exact	NIST SP 811
°F	1.8 × °C + 32	Exact	NIST SP 811
°C	(°F – 32) ÷ 1.8	Exact	NIST SP 811
1 lb	0.453 592 37 kg	Exact	Weights and Measures Act
1 kg	2.205 lb	Rounded (2.204 622 6…)	NPC 2025 conversion tables

Sources: Weights and Measures Act (Canada), Schedules I and II. NIST Special Publication 811, Guide for the Use of the International System of Units, Appendix B.8. NIST Special Publication 1038, The International System of Units (SI): Conversion Factors for General Use (2006), including the U.S. gallon of 231 cubic inches. National Plumbing Code of Canada 2025, conversion factor table (PDF page 245).

Glossary

acute angle
An angle less than 90°. Chapter 12

adjacent side
In a right triangle, the side next to the angle you are working with that is not the hypotenuse. Chapter 13

angle
The amount of turn between two lines that meet at a point. Angles are measured in degrees. Chapter 12

arc
A part of the circumference of a circle. Chapter 9

area
The amount of flat surface inside a shape, measured in square units. Chapter 10

average
The sum of a set of numbers divided by how many numbers there are. Also called the mean. Chapter 1

base
The side of a shape that a height is measured from. Any side can be used as the base. Chapter 10

BEDMAS
A memory aid for the order of operations: Brackets, Exponents, Division and Multiplication (left to right), Addition and Subtraction (left to right). Chapter 3

brackets
The symbols ( ) and [ ] that group part of a calculation so that it is worked out first. Chapter 3

capacity
How much a container holds, usually given in litres (L) or gallons. Chapter 11

circumference
The distance around a circle. $C = \pi \times D$. Chapter 9

common denominator
A denominator that two or more fractions share. For 3/8 and 5/16, 16 is a common denominator. Chapter 4

composite shape
A shape made of two or more basic shapes, such as rectangles, triangles and circles. Chapter 10

cone
A solid with a circular base that narrows to a point (the apex). Chapter 11

constant
A value in a formula that is always the same, such as π or the 2 in $P = 2 \times (l + w)$. Chapter 8

conversion factor
A number that changes a measurement from one unit to another, such as 25.4 to change inches to millimetres. Chapter 7

cosine
In a right triangle, the adjacent side divided by the hypotenuse. Written cos. Chapter 13

cubic unit
A cube that is 1 unit long on each edge, such as 1 mm^3, 1 cm^3, 1 m^3, 1 in^3 or 1 ft^3. Volume is counted in cubic units. Chapter 11

cylinder
A solid with two equal, parallel circular ends joined by a curved side. Chapter 11

D.A.L.
The way the Sharp EL-531XT takes keys: in the same order the calculation is written. The line is worked out when you press =. Chapter 2

decimal
A number written with a decimal point, such as 2.375. The digits after the point are tenths, hundredths, thousandths and so on. Chapter 5

decimal inch
An inch measurement written as a decimal, such as 0.375 in for 3/8 in. Chapter 5

decimal place
One position after the decimal point. 2.375 has three decimal places. Chapter 5

decimal point
The dot that separates the whole number part of a decimal from the part that is less than one. Chapter 5

degrees
The unit for measuring angles. One full turn is 360 degrees, written 360°. Chapter 12

denominator
The bottom number of a fraction. It tells how many equal parts the whole is split into. Chapter 4

diameter
The distance across a circle through its centre. D = 2 × r. Chapter 9

difference
The answer to a subtraction. Chapter 1

digits
The ten symbols 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9 used to write numbers. Chapter 1

display
The screen of the calculator. It shows the number you are keying in and the answer. Chapter 2

equivalent fractions
Fractions that are written differently but have the same value, such as 1/2, 2/4 and 8/16. Chapter 4

estimate
A quick, rounded answer used to check that an exact answer makes sense. Chapter 1

exponent
A small raised number that says how many of that number are multiplied together. In 5^3 the exponent is 3: 5^3 = 5 × 5 × 5 = 125. Chapter 3

factors
Numbers that are multiplied together. Chapter 1

formula
A rule written with letters and symbols that shows how to work out one quantity from others, such as A = l × w. Chapter 8

fraction
A number that names part of a whole, written as one whole number over another, such as 3/8. Chapter 4

height
The distance from the base to the opposite corner or side, measured at right angles (90°) to the base. Chapter 10

hypotenuse
The side of a right triangle across from the right angle. It is always the longest side. Chapter 12

imperial system
The system of units based on the inch, foot, yard, pound and gallon. Canada and the United States define these units from the metric units. Chapter 7

improper fraction
A fraction whose numerator is equal to or bigger than its denominator, such as 11/4. It is worth one whole or more. Chapter 4

inverse operation
An operation that undoes another. Subtracting undoes adding, dividing undoes multiplying, and a square root undoes squaring. Chapter 8

inverse trig function
A function that turns a trig ratio back into an angle: SIN^{-1}, COS^{-1} and TAN^{-1}. Chapter 13

kerf
The width of material removed by a saw cut. Chapter 1

key sequence
The list of keys you press, in order, to do a calculation. Chapter 2

legs
The two sides of a right triangle that form the right angle. Chapter 12

lowest terms
A fraction is in lowest terms when no whole number bigger than 1 divides evenly into both its numerator and its denominator. Chapter 4

memory
A place in the calculator that holds a number until you need it again. STO stores the number and RCL recalls it. Chapter 2

metric system
A system of units based on powers of ten, such as the millimetre, metre and kilometre. Chapter 7

mixed number
A whole number and a proper fraction written together, such as 2 3/4. Chapter 4

net
A flat pattern that folds or rolls up to make a solid. Each face of the solid is one piece of the net. Chapter 10

numerator
The top number of a fraction. It tells how many of the equal parts you have. Chapter 4

obtuse angle
An angle more than 90° and less than 180°. Chapter 12

offset
A change in the line of a pipe that moves it over to a new line parallel to the first. In this book an offset has two fittings of the same angle. Chapter 13

opposite side
In a right triangle, the side across from the angle you are working with. Chapter 13

order of operations
The agreed order for working out a calculation that has more than one operation in it. Chapter 3

percent
A number out of 100. 25% means 25 out of every 100, which is 0.25 or 1/4. Chapter 6

perimeter
The distance all the way around the outside of a flat shape. Chapter 9

pi
The number you get when you divide the circumference of any circle by its diameter, about 3.14159. Its symbol is π. Chapter 9

place value
The value a digit has because of its position in a number. In 12,450 the 2 is worth 2,000. Chapter 1

prefix
Letters in front of a unit name that multiply the unit by a power of ten, such as kilo (1,000) and milli (1/1,000). Chapter 7

prism
A solid with two identical, parallel ends joined by flat sides. Every slice parallel to the ends is the same shape. Chapter 11

product
The answer to a multiplication. Chapter 1

proper fraction
A fraction whose numerator is smaller than its denominator, such as 5/8. It is worth less than one whole. Chapter 4

proportion
A statement that two ratios are equal, such as 1/4 = 5/20. Chapter 6

pyramid
A solid with a flat base, often a square, and triangular sides that meet at a point (the apex). Chapter 11

Pythagorean theorem
In a right triangle, the square of the hypotenuse equals the sum of the squares of the two legs: $a^2 + b^2 = c^2$. Chapter 12

quotient
The answer to a division. Chapter 1

radius
The distance from the centre of a circle to its edge. Its letter is r. Chapter 9

rate
A ratio that compares two amounts in different units, such as litres per minute or dollars per metre. Chapter 6

ratio
A comparison of two amounts by division, written 3 : 1, 3 to 1 or 3/1. Chapter 6

reciprocal
The fraction turned upside down. The reciprocal of 3/4 is 4/3. A number times its reciprocal is 1. Chapter 4

remainder
The amount left over when a number does not divide evenly. Chapter 1

right angle
An angle of exactly 90°, a quarter turn. A square corner. Chapter 12

rounding
Replacing a number with a nearby number that is easier to work with, such as the nearest ten, hundred or thousand. Chapter 1

scale
The ratio of a length on a drawing to the real length. At 1 : 20, 1 mm on the drawing stands for 20 mm on the real part. Chapter 6

second function
The job printed above a key. Press 2nd (TI-30Xa) or 2ndF (Sharp) first, then the key. Chapter 2

SI
The International System of Units, the modern metric system. Its base units include the metre, the kilogram and the second. Chapter 7

sine
In a right triangle, the opposite side divided by the hypotenuse. Written sin. Chapter 13

sphere
A perfectly round ball. Every point on its surface is the same distance (the radius) from its centre. Chapter 11

square
A number multiplied by itself. The square of 5, written 5^2, is $5 \times 5 = 25$. Chapter 3

square root
The number that, multiplied by itself, gives the number you started with. $\sqrt{25} = 5$ because $5 \times 5 = 25$. Chapter 3

square unit
A square that is 1 unit long on each side, such as 1 mm^2, 1 m^2, 1 in^2 or 1 ft^2. Area is counted in square units. Chapter 10

stretch-out
The length of flat material that rolls or wraps into a round shape. For a cylinder it is $\pi \times D$. Chapter 9

subject of a formula
The letter on its own on one side of the equals sign. The formula tells you how to work it out. Chapter 8

substitute
To replace a letter in a formula with its value. Chapter 8

sum
The answer to an addition. Chapter 1

surface area
The total area of all the outside faces of a solid. Find it by adding the areas of the faces in its net. Chapter 10

tangent
In a right triangle, the opposite side divided by the adjacent side. Written tan. Chapter 13

transpose
To rearrange a formula so that a different letter becomes its subject. Chapter 8

trigonometry
The math that links the angles of a right triangle to the lengths of its sides. Chapter 13

unit
A fixed amount used to measure, such as the metre, the inch or the litre. A measurement is a number and a unit. Chapter 7

variable
A letter in a formula that stands for a quantity that can change, such as l for length. Chapter 8

volume
The amount of space a solid takes up or holds, measured in cubic units. Chapter 11

whole number
A number with no fraction or decimal part: 0, 1, 2, 3 and so on. Chapter 1